

Ratio & Proportion

Key - points:-

1) What are antecedent and consequent?

In the form $x : y$; x is called the *antecedent* and y is called *consequent*.

2) $x : y$ can also be written as $\frac{x}{y}$

3) Duplicate ratio (square) :-

Duplicate ratio of $x : y$ will be $x^2 : y^2$

4) Sub-duplicate Ratio (square root):-

Sub-duplicate Ratio of $x : y$ will be $\sqrt{x} : \sqrt{y}$

5) Triplicate Ratio (cube) :-

Triplicate ratio of $x : y$ will be $x^3 : y^3$

6) Sub-triplicate Ratio (cube root)

Sub-triplicate Ratio of $x : y$ will be $\sqrt[3]{x} : \sqrt[3]{y}$

7) Inverse ratio :- $x : y$ will be $y : x$

8) Compound Ratio : If two or more ratios are given, then the antecedent of one is multiplied with antecedent of the other and respective consequents are also multiplied.

If $a : b, c : d$ and $e : f$ are three ratios, then their compound ratio will be $\frac{ace}{bdf}$.

9) 3rd Proportional:- 3rd proportion of a and b i.e. $a : b :: b : x \Rightarrow$

$$\text{Then, } x = \frac{b^2}{a}$$

10) 4th Proportional: 4th proportional of a, b and c i.e. $a : b :: c : x$

$$\text{Then, } x = \frac{b \times c}{a}$$

11) Componendo and dividendo rule:

$$\text{If } a : b :: c : d, \text{ Then } \Rightarrow \frac{a+b}{a-b} = \frac{c+d}{c-d}$$

12) Mean Proportion : Mean proportion of two numbers, a and b , is given by $\sqrt{ab}$.

Partnership

Concept 1

The ratio of profits of two or more partners will be equal to the ratio of their respective equivalent investment ratio.

Equivalent investment = (Amount Invested) $\times$ (Time for which the amount is invested).

We will take some examples to understand the concepts of partnership.

Example 1: A invests Rs. 10,000 and B invests Rs. 20,000 for a period of 1 year each. The profit, after 1 year will be divided in the ratio:

$$\frac{\text{Profit of A}}{\text{Profit of B}} = \frac{10000 \times 12}{20000 \times 12} = \frac{1}{2}$$

Example 2: A invests Rs.100 in a business. After 6 months B joins with an investment of Rs. 200. At the end of 1 year C joins with an investment of Rs.100. What will be the ratio of profit after 3 years?

	A	B	C
Amount(Rs.)	100	200	100
Time(month)	36	30	24
Equivalent Investment	3600	6000	2400
Profit Ratio	3	5	2

Example 3: In the previous case if the total profit after 3 years is 10,000. What will be the share of A?

Share of A in total profit =

$$\frac{3}{3+5+2} \times 10,000 = \frac{3}{10} \times 10000 \\ \Rightarrow 3 \times 1000 = 3000$$

Note: In case someone invests in-between the total period of time, just take care for what time an amount is invested. Similarly there can be a case when someone opts out in that case also, the time for which the amount was invested will be taken into consideration and not the full period.

Concept 2

If $\frac{a}{b} = \frac{c}{d} = \frac{e}{f} = \frac{g}{g} = K$, where K is constant. Then, $K = \frac{a+c+e+f}{b+d+f+g}$

Example : $\frac{1+x}{2-x} = \frac{9-x}{3+x} = K$, Find the value of K ?

$$\text{Sol. :- } K = \frac{1+x+9-x}{2-x+3+x} = \frac{10}{5} = 2$$

Concept 3

If two or three ratios are given, and we have to find an individual ratio of all of them, then we take LCM of the common entities and then balancing the common entities, we get the required ratio.

Example :- $a : b = 3 : 4, b : c = 6 : 7$.
LCM of the common entities i.e $b = (4, 6) = 12$

$$a : b = 3 : 4 \Rightarrow 3 \times 3 = 9 : 12$$

$$b : c = 6 : 7 \Rightarrow 2 \times 6 = 12 : 14$$

$$\text{So, } a : b : c = 9 : 12 : 14$$

Note : In case there are more than three ratios given, then the LCM approach will be tedious in finding the individual ratio of the given entities.

Alternative method

$$\text{If } a : b = 2 : 3, \quad b : c = 4 : 5$$

$$c : d = 6 : 7, \quad d : e = 1 : 2$$

$$\begin{array}{ccccccc} a : b : c : d : e \\ \boxed{2 : 3} & \boxed{3} & \boxed{3} & \boxed{3} \\ \boxed{4} & \boxed{4 : 5} & \boxed{5} & \boxed{5} \\ \boxed{6} & \boxed{6} & \boxed{6 : 7} & \boxed{7} \\ \boxed{1} & \boxed{1} & \boxed{1} & \boxed{1 : 2} \end{array}$$

$$48 : 72 : 90 : 105 : 210$$

To understand this concept we will take some examples

Example 1 : ₹5600 is to be divided among A, B, C & D in such a way that the ratio of share of A : B is 1 : 2, B : C is 3 : 1, C : D is 2 : 3, find the share of A and B taken together?

Share of A : B = 1 : 2

Share of B : C = 3 : 1

Share of C : D = 2 : 3

A : B : C : D $\rightarrow$

$$1 \times 3 \times 2 : 2 \times 3 \times 2 : 2 \times 1 \times 2 : 2 \times 1 \times 3$$

$$= 6 : 12 : 4 : 6 \text{ or } 3 : 6 : 2 : 3$$

Sum of the share of all of them = $3 + 6 + 2 + 3 = 14$ unit

$$\text{Total share of (A + B)} = \frac{6+3}{14} \times 5600$$

$$= 9 \times 400 = ₹3600$$

Example 2 : Sum of ₹ x is divided among A, B, C such that the ratio of the shares of A and B is 6 : 7 and that of B and C is 3 : 2. If the difference of the share of A and C is ₹540. Find the value of x

Share of A and B = 6 : 7

Share of B and C = 3 : 2

Ratio of the shares of A, B and C = $6 \times 3 : 3 \times 7 : 7 \times 2 = 18 : 21 : 14$

Diff. of A and C = $18 - 14 = 4$ unit, which corresponds to ₹540

Total share of A, B and C = $18 + 21 + 14$

$$= 53 \text{ unit} = \frac{540}{4} \times 53 = ₹ 7155$$

Concept 4

If numerator and denominator of a ratio is multiplied and divided by same number, the ratio remains unchanged.

$$\text{E.g. :- } \frac{a}{b} = \frac{a \times k}{b \times k} = \frac{\frac{a}{k}}{\frac{b}{k}}$$

Ratio is independent of change in scale only and not of origin.

Concept 5

If there is 'n' number of unequal fractions

such as $\frac{a_1}{b_1}, \frac{a_2}{b_2}, \frac{a_3}{b_3}, \frac{a_4}{b_4}, \dots, \frac{a_n}{b_n}$ then

the ratio of $(a_1 + a_2 + a_3 + \dots + a_n)$ and $(b_1 + b_2 + b_3 + \dots + b_n)$ lies between lowest and highest values of these fractions.

Example:- three unequal fractions are $\frac{1}{2},$

$\frac{2}{3}, \frac{4}{7}$. So, the ratio of $(1 + 2 + 4) : (2 + 3 + 7) = 7 : 12$. Among these fractions,

lowest fraction is $\frac{1}{2}$ and the highest

fraction is $\frac{2}{3}$ so, the required value lies between $\frac{1}{2}$ and $\frac{2}{3}$

Concept 6:

If $aA = bB = cC$, where a, b, c are constants and A, B, C are variables. Then the ratio of $A : B : C = b \times c : c \times a : a \times b$.

Example: if $3A = 4B = 5C$, then find $A:B:C$
The ratio of $A : B : C = 4 \times 5 : 5 \times 3 : 3 \times 4 = 20 : 15 : 12$

Variety Questions

Q.1. The number of marbles in three bags is in the ratio of $2 : 3 : 4$. If 12 marbles are added to each bag, then the ratio changed to $8 : 11 : 14$. The total number of marbles in the beginning was:
SSC MTS 28/10/2024 (Morning)
(a) 216 (b) 162 (c) 200 (d) 126

Q.2. A function of two variable varies directly with x and inversely with y . Determine function when $x = 5$ and $y = 3$. Given that, for $x = 0$ and $y = 1$, $f = 15$ and for $x = 1$ and $y = 15$, $f = 2$.
SSC CGL 17/09/2024 (3rd Shift)
(a) 10 (b) 2 (c) 5 (d) 15

Q.3. A payment of ₹120 is made with ₹10, ₹5 and ₹2 coins. A total of 25 coins are used. Which of the following is the number of ₹10 coins used in the payment?
SSC CGL 12/09/2024 (1st Shift)
(a) 6 (b) 4 (c) 10 (d) 8

Q.4. What is the ratio of the mean proportional of 1.21 and 0.09 with the mean proportional of 0.16 and 0.36?
SSC CGL 11/09/2024 (3rd Shift)
(a) $11 : 8$ (b) $13 : 8$ (c) $13 : 12$ (d) $11 : 13$

Q.5. The mean proportional of $8a^2$ and $\frac{18}{a^4}$, where a is a positive number, is ____.
SSC CGL 09/09/2024 (3rd Shift)
(a) $\frac{12}{a^2}$ (b) $12a$ (c) $12a^2$ (d) $\frac{12}{a}$

Q.6. R pays ₹100 to P with ₹5, ₹2 and ₹1 coins. The total number of coins used for paying are 40. What is the number of coins of denomination ₹5 in the payment?
SSC CGL 09/09/2024 (1st Shift)
(a) 16 (b) 17 (c) 18 (d) 13

Q.7. Find the fourth proportion of the following numbers: $\frac{1}{3}$ -rd of 27, $\frac{4}{5}$ -th of

$30, \frac{3}{7}$ -th of 35.

Higher Secondary 24/06/2024 (Shift - 2)
(a) 35 (b) 40 (c) 45 (d) 38

Q.8. The concentration of three wines A, B and C are 10%, 20% and 30%, respectively. They are mixed in the ratio $2 : 3 : x$ resulting in a 23% concentration solution. Find the value of x .
SSC CHSL Tier II 02/11/2023
(a) 7 (b) 6 (c) 8 (d) 5

Q.9. In a school of 720 students, the ratio of boys to girls is $3 : 5$. Find how many new boys may be allowed to the school if 18 new girls are admitted, so that the ratio of boys to girls changes to $2 : 3$.
SSC CPO 05/10/2023 (3rd Shift)
(a) 42 (b) 50 (c) 44 (d) 38

Q.10. Ram has 12 hens and 5 cows, Shyam has 18 hens and 9 cows, Preetham has 4 hens and 8 cows, and Vinay has 10 hens and 7 cows. Who has a greater ratio of hens to cows?
SSC CPO 05/10/2023 (2nd Shift)
(a) Shyam (b) Preetham
(c) Vinay (d) Ram

Q.11. A company is manufacturing boxes of three types - plastic, wood and glass. The ratio of the numbers of plastic, wood and glass boxes is $2 : 3 : 5$. The difference between the numbers of wood and glass boxes is 144. If the number of plastic boxes is increased by 25%, then what will be the number of plastic boxes now?
SSC CPO 04/10/2023 (1st Shift)
(a) 144 (b) 216 (c) 360 (d) 180

Q.12. Find the fourth proportional of $12^{35}, 3^{12}$ and 4^{12} .
SSC MTS 14/09/2023 (3rd Shift)
(a) 12^{-23} (b) 3^{24} (c) 12^{-11} (d) 4^{24}

Q.13. Three numbers are in the ratio $1 : 2 : 3$. By adding 5 to each of the numbers then the ratio changes to $2 : 3 : 4$. The smallest number is:
SSC MTS 14/09/2023 (1st Shift)
(a) 5 (b) 15 (c) 12 (d) 20

Q.14. When $x = 1$ then y is directly proportional to $x + 3$ and $y = 8$. When $z = 6$ then x is inversely proportional to z and $x = 2$. Find the relation between y and z .
SSC MTS 12/09/2023 (2nd Shift)
(a) $(y - 6) / z = 24$ (b) $(y - 6) / z = 12$
(c) $z(y - 6) = 12$ (d) $z(y - 6) = 24$

Q.15. If $A : B = \frac{1}{3} : \frac{1}{5}$, $B : C = \frac{1}{7} : \frac{1}{6}$ and $C : D = \frac{1}{10} : \frac{1}{21}$, then $A : B : C : D$ is:

SSC MTS 06/09/2023 (3rd Shift)
(a) $40 : 24 : 21 : 10$ (b) $48 : 23 : 25 : 8$
(c) $8 : 3 : 7 : 5$ (d) $14 : 24 : 28 : 10$

Q.16. The average of the two numbers is 48. If 3 is added to a smaller number, then the ratio between them is $11 : 20$. Find the larger number. (Round up to two decimal places).
SSC MTS 05/09/2023 (3rd Shift)
(a) 40.87 (b) 3.87 (c) 63.87 (d) 70.87

Q.17. Arrange the ratios in ascending order $5 : 3\frac{1}{3}, 6\frac{1}{4} : 3\frac{1}{8}$ and $3.5 : 4.5$.
SSC MTS 05/09/2023 (1st Shift)

(a) $3.5 : 4.5 < 5 : 3\frac{1}{3} < 6\frac{1}{4} : 3\frac{1}{8}$
(b) $6\frac{1}{4} : 3\frac{1}{8} < 5 : 3\frac{1}{3} < 3.5 : 4.5$
(c) $5 : 3\frac{1}{3} < 3.5 : 4.5 < 6\frac{1}{4} : 3\frac{1}{8}$
(d) $5 : 3\frac{1}{3} < 6\frac{1}{4} : 3\frac{1}{8} < 3.5 : 4.5$

Q.18. The incomes of P, Q and R are in the ratio of $9 : 15 : 10$ and their savings are in the ratio is $10 : 15 : 12$. If R spends 60% of his income, what is the ratio of the expenditures of P, Q and R?
SSC MTS 05/09/2023 (1st Shift)
(a) $18 : 17 : 30$ (b) $17 : 30 : 18$
(c) $17 : 18 : 30$ (d) $18 : 30 : 17$

Q.19. Find the fourth proportional to $(a + b)^2 \times (a^3 - b^3), (a^3 + b^3), (a^2 - b^2)$.
SSC MTS 01/09/2023 (2nd Shift)
(a) $\frac{(a^2 - ab + b^2)}{(a + b)}$ (b) $\frac{(a^2 + ab + b^2)}{(a - b)}$
(c) $\frac{(a^2 - ab + b^2)}{(a^2 + ab + b^2)}$ (d) $\frac{(a^2 + ab + b^2)}{(a^2 - b^2)}$

Q.20. Two numbers A and B are such that the sum of 6% of A and 4% of B is $\frac{2}{5}$ of the sum of 7% of A and 11% of B, then the ratio A : B is:
SSC MTS 01/09/2023 (1st Shift)
(a) $3 : 7$ (b) $5 : 8$ (c) $1 : 8$ (d) $1 : 7$

Q.21. A certain sum was to be divided between A and B in the ratio $8 : 5$. However, by mistake, it was divided in the ratio $5 : 8$. Thus, A gets ₹300 less than his original share. Find the sum.
SSC CHSL 17/08/2023 (4th Shift)
(a) ₹2,600 (b) ₹1,500 (c) ₹1,300 (d) ₹600

Q.22. When P is subtracted from each of the numbers 8, 6, 2 and 9, the numbers so obtained in this order are in proportion. What is the mean proportional between $(3P - 6)$ and $(9P - 4)$?
SSC CHSL 10/08/2023 (4th Shift)
(a) 26 (b) 28 (c) 29 (d) 24

Q.23. A sum of money is divided among A, B, C in the ratio 3 : 2 : 5, respectively. If the share of C is ₹300 more than B. Then how much more money will A get than B?
SSC CHSL 03/08/2023 (1st Shift)
(a) ₹150 (b) ₹75 (c) ₹100 (d) ₹125

Q.24. The mean proportional of $(a + b)(a - b)^3, (a + b)^3(a - b)$ is:
SSC CGL 25/07/2023 (4th shift)
(a) $(a + b)^2(a - b)^2$ (b) $(a + b)^2(a - b)$
(c) $(a + b)(a - b)^2$ (d) $a^2 - b^2$

Q.25. The seats for Mathematics, English and Chemistry in a school are in the ratio of 6 : 4 : 7. If these seats are increased by 20%, 40% and 60%, respectively, the ratio of the increased seats is:
SSC CGL 19/07/2023 (2nd shift)
(a) 9 : 7 : 8 (b) 9 : 7 : 14
(c) 1 : 2 : 3 (d) 7 : 9 : 14

Q.26. A bag contains ₹550 in the form of 50p, 25p and 20p coins in the ratio 2 : 3 : 5. The difference between the amounts that are contributed by the 50 p and the 20 p coins is:
SSC CGL 17/07/2023 (3rd shift)
(a) ₹30 (b) ₹20 (c) ₹10 (d) ₹0

Q.27. In a party hall, there are people in blue and red dresses. The ratio of the number of men in blue to the number of women in red is 3 : 7. The ratio of men in red to the number of women in blue is 2 : 1. If the ratio of the number of people in blue to the number of people in red is 35 : 76, then what is the ratio of the number of men to the number of women?
SSC MTS 04/05/2023 (Morning)
(a) 47 : 51 (b) 40 : 43
(c) 62 : 57 (d) 52 : 59

Q.28. P's income is 250 percent of Q's income and P's expenditure is 180 percent of Q's expenditure. If P's income is 400 percent of Q's expenditure, then what is the respective ratio of P's savings and Q's savings?
SSC CHSL 17/03/2023 (3rd Shift)
(a) 8 : 3 (b) 11 : 3 (c) 11 : 5 (d) 5 : 2

Q.29. The number of units manufactured by company A was 12500 units in 2019 and 10625 units in 2020. While in company B, the production fell from 34000 units in 2019 to 30600 units in 2020. If X and Y are the percentage decrease in the number of units manufactured by company A and B respectively from 2019 to 2020, then what will be the ratio of X and Y?
SSC CGL 13/12/2022 (1st Shift)
(a) 5 : 3 (b) 3 : 4 (c) 3 : 2 (d) 8 : 5

Q.30. In an office with 40 officers, the average salaries of class-A, class-B and class-C officers are ₹600, ₹750 and ₹1,000 a day, respectively. The number of class-A, class-B and class-C officers in the office are in the ratio 5 : 4 : 1, respectively. Find the monthly average salary (in ₹) of an officers, all the 40 of them taken together. [Assume the number of days in a month to be 30.]
SSC MTS 08/07/2022 (Morning)
(a) 25,800 (b) 21,000
(c) 24,030 (d) 24,600

Q.31. The ratio of income of A to that of B is 4 : 3. The expenditure of A is $\frac{5}{12}$ of his income and the expenditure of B is $\frac{7}{15}$ of his income. The difference between their expenditure is ₹4,500. Find the income of B.
SSC CHSL 25/05/2022 (Evening)
(a) ₹49,625 (b) ₹48,625
(c) ₹51,625 (d) ₹50,625

Q.32. A sum of ₹46,800 is divided among A, B, C and D in such a way that the ratio of the combined share of A and D to the combined share of B and C is 8 : 5. The ratio of the share of B to that of C is 5 : 4. A receives ₹18,400. If x is the difference between the shares of A and B and y is the difference between the shares of C and D, then what is the value of (x - y) (in ₹)?
SSC CGL Tier II (29/01/2022)
(a) 6000 (b) 6500 (c) 5000 (d) 7000

Q.33. If a number of pens and pencils are bought in the ratio of 5 : 3, then Ashok has to pay ₹44. If the ratio is changed as 3 : 5, then he has to pay ₹36. Find the ratio of the price of one pen to one pencil.
SSC MTS 11/10/2021 (Evening)
(a) 5 : 3 (b) 7 : 3 (c) 8 : 5 (d) 7 : 4

Q.34. Atul purchased Bread costing Rs.20 and gave a 100 rupee note to the shopkeeper. The shopkeeper gave the balance money in coins of denomination 2, 5, and 10. If these coins are in the ratio 5 : 4 : 1, then how many 5 coins did the shopkeeper give?
SSC CGL 03/08/2021 (Evening)
(a) 5 (b) 6 (c) 8 (d) 4

Q.35. In a school, $\frac{3}{8}$ of the number of students are girls and the rest are boys, One - third of the number of boys are below 10 years and $\frac{2}{3}$ of the number of girls is also below 10 years. If the

number of students of age 10 or more years is 260, then the number of boys in the school is:
SSC CGL Tier II (15/11/2020)
(a) 280 (b) 300 (c) 234 (d) 312

Q.36. By adding 3 and 5 in numerator and denominator of a fraction it becomes $\frac{2}{3}$. If 1 and 3 are subtracted and added from numerator and denominator respectively, it becomes $\frac{2}{5}$. Find the fraction.
SSC CHSL 19/10/2020 (Morning shift)

(a) $\frac{5}{7}$ (b) $\frac{6}{7}$ (c) $\frac{7}{6}$ (d) $\frac{7}{5}$

Q.37. A, B and C spend 80%, 85% and 75% of their incomes, respectively. If their savings are in the ratio 8 : 9 : 20 and the difference between the incomes of A and C is Rs. 18,000, then the income of B is:
SSC CGL Tier II (13/09/2019)
(a) Rs. 24,000 (b) Rs. 27,000
(c) Rs. 30,000 (d) Rs. 36,000

Q.38. The ratio of income of A and B last year was 4 : 3, respectively. The ratios of their individual incomes of the last year and the present year are 3 : 4 and 5 : 6, respectively. If their total income for the present year is Rs. 8.04 lakh, then the income of B last year was :
SSC CGL Tier II (12/09/2019)
(a) Rs. 2.7 lakh (b) Rs. 3.6 lakh
(c) Rs. 2.4 lakh (d) Rs. 2.8 lakh

Q.39. The ratio of the income of A to that of B is 5 : 7. A and B save Rs. 4000 and Rs. 5000 respectively. If the expenditure of A is equal to $66\frac{2}{3}\%$ of the expenditure of B, then the total income of A and B is :
SSC CGL Tier II (11/09/2019)
(a) Rs. 25,200 (b) Rs. 24,000
(c) Rs. 26,400 (d) Rs. 28,800

Q.40. The ratio between the divisor and remainder is 3 : 2 and the ratio between the divisor and the quotient is 7 : 12. If the remainder is 14, then find the remainder when the dividend is divided by 9.
SSC MTS 21/08/2019 (Morning)
(a) 3 (b) 6 (c) 4 (d) 5

Q.41. Ravi's age is $\frac{3}{5}$ of Shyam's age. After x years the ratio of the ages of Ravi and Shyam becomes 5 : 7. If initially the sum of their ages is 32, then what is the value of x?
SSC MTS 08/08/2019 (Morning)
(a) 2 (b) 4 (c) 8 (d) 7

Q.42. Rs. 8000 is distributed among A, B and C such that they receive notes of Rs. 500, Rs. 200 and Rs. 100 respectively. The amounts received by them are in the ratio 15 : 2 : 3. What was the ratio of the numbers of notes of Rs. 500, Rs 200 and Rs 100 ?

SSC CHSL 10/07/2019 (Morning)

(a) 3:1:3 (b) 3:3:1 (c) 4:1:2 (d) 3:2:2

Q.43. A sum of Rs. 4360 was to be divided among A, B, C and D in the ratio of 3 : 4 : 5 : 8, but it was divided in the ratio of $\frac{1}{3} : \frac{1}{4} : \frac{1}{5} : \frac{1}{8}$ by mistake.

As a result:

SSC CHSL 01/07/2019 (Evening)

- (a) A received Rs. 956 more
(b) B received Rs. 318 more
(c) D received Rs. 1144 less
(d) C received Rs. 132 less

Q.44. A sum of Rs. x is divided among A, B and C such that the ratio of shares of A and B is 7 : 12 and that of B and C is 8 : 5. If the difference in the shares of A and C is Rs. 214, then the value of x is :

SSC CGL 07/06/2019 (Evening)

(a) 11556 (b) 11128 (c) 11770 (d) 11342

Q.45. The prices of two articles are in the ratio 4 : 5. If the price of the first article is increased by x% and that of the other is decreased by 30% , then the new prices of A and B will be in the ratio 10 : 7. The value of x is :

SSC CGL 07/06/2019 (Afternoon)

(a) 24.5 (b) 22.5 (c) 25 (d) 20

Practice Questions

SSC CHSL 2024 Tier - 2

Q.46. If a man's age is four times the sum of the ages of his three children, and after 6 years his age will be twice the sum of the ages of his three children, then what is the man's present age?

SSC CHSL Tier II (18/11/2024)

- (a) 60 years (b) 56 years
(c) 45 years (d) 30 years

SSC MTS 2024 Tier - 1

Q.47. In a student council election, there were three contestants. All the voters participated in the voting and 10% of votes were declared invalid. The contestants got the votes in the ratio 8 : 7 : 5. If the second candidate in the list got 4725 votes, the total number of valid votes is:

SSC MTS 07/10/2024 (Morning)

(a) 10,000 (b) 11,500 (c) 13,500 (d) 9,000

Q.48. 141 is divided into two parts in such a way that the one-eighth part of the first and one-ninth part of the second are in the ratio 5 : 6. Find the first part.

SSC MTS 07/10/2024 (Afternoon)

(a) 36 (b) 72 (c) 48 (d) 60

Q.49. The salaries of A and B are in a ratio of 2 : 3. Their savings are equal. If

A's expenditure is $\frac{1}{2}$ of the salary, then

what per cent of income does B spends ?

SSC MTS 07/10/2024 (Afternoon)

(a) 66.67% (b) 78.5% (c) 50% (d) 75%

Q.50. 360 toffees are distributed among Pushkal, Quadir and Rustam in the ratio 2 : 3 : 4. After distribution, Quadir gives 20% of his toffees to Pushkal and Rustam gives 25% of his toffees to Pushkal. What is the new ratio of toffees with Pushkal, Quadir and Rustam ?

SSC MTS 08/10/2024 (Evening)

- (a) 5 : 4 : 6 (b) 7 : 3 : 5
(c) 6 : 4 : 5 (d) 3 : 5 : 7

Q.51. Due to a fault in the voting machine, the invalid votes are counted in votes of the candidate B, who gets 5 times the invalid votes. A gets the votes 8 times the invalid votes and wins the election by 28. What is the total number of votes polled ?

SSC MTS 14/10/2024 (Afternoon)

(a) 168 (b) 184 (c) 196 (d) 176

Q.52. The mean proportional between the numbers p and q is 8. Which of the following pairs of numbers can be the values of p and q ?

SSC MTS 15/10/2024 (Afternoon)

- (a) 12 and 3 (b) 16 and 4
(c) 12 and 16 (d) 10 and 6

Q.53. The ratio of the incomes of Rama and Ganga is 4 : 3 and the ratio of their expenditures is 3 : 2. If each person saves ₹2,700, then find their expenditures, respectively.

SSC MTS 15/10/2024 (Evening)

- (a) ₹8,100 and ₹5,000
(b) ₹8,000 and ₹5,000
(c) ₹8,100 and ₹5,400
(d) ₹8,000 and ₹5,400

Q.54. Shankar comes to buy a T-shirt in a showroom. He purchases a shirt which costs ₹1,020 after a discount. He pays his bill in the form of 50 paise, 20 paise and 10 paise coins in the ratio 2 : 3 : 4 (assume for the sake of the problem). What is the number of 10 paise coins he used to pay his bill ?

SSC MTS 17/10/2024 (Afternoon)

(a) 1530 (b) 510 (c) 1020 (d) 2040

Q.55. What should be added to each of the four numbers 5, 19, 7 and 25 to make them proportional ?

SSC MTS 23/10/2024 (Morning)

(a) 2 (b) 3 (c) 1 (d) 4

Q.56. If $49 : x :: x : 81$ and $64 : y :: y : 169$, where x and y are both natural numbers, then find the value of $2x + 3y$.

SSC MTS 13/11/2024 (Evening)

(a) 312 (b) 126 (c) 348 (d) 438

Q.57. When x is added to each of 2, 3, 30 and 35, then the numbers obtained in this order are in proportion. What is the second largest part in the proportion ?

SSC MTS 14/11/2024 (Morning)

(a) 40 (b) 30 (c) 35 (d) 45

Q.58. The quantity of milk ordered on the first, second, third and fourth day of a four day ceremony is in the proportion of 30 litres, 24 litres, 35 litres and 'x' litres, respectively. Find the quantity of milk (in litres) ordered on the fourth day.

SSC MTS 14/11/2024 (Evening)

(a) 30 (b) 32 (c) 28 (d) 26

SSC CGL 2024 Tier - 1

Q.59. The average marks (out of 100) of boys and girls in an examination are 75 and 80, respectively. If the average marks of all the students in that examination are 78. Find the ratio of the number of boys to the number of girls.

SSC CGL 09/09/2024 (1st Shift)

(a) 2 : 3 (b) 3 : 4 (c) 1 : 3 (d) 1 : 2

Q.60. At present, A is thrice as old as B. C is 5 years elder than B. The sum of the ages of A, B, C is 75 years. The ratio of the age of B five years ago to the age of A three years from now is:

SSC CGL 09/09/2024 (2nd Shift)

(a) 2 : 3 (b) 1 : 4 (c) 4 : 7 (d) 1 : 5

Q.61. If x varies inversely as y and $x = 2$ when $y = 6$, then the value of y when $x = 3$ is:

SSC CGL 09/09/2024 (2nd Shift)

(a) 2 (b) 3 (c) 4 (d) 6

Q.62. The ages of two friends, Ram and Ravi, differ by 7 years. Ram's mother Mayadevi is three times as old as Ram, and Ravi is four times as old as his brother Soham. The ages of Mayadevi and Soham differ by 65 years. Find the age of Mayadevi (in years).

SSC CGL 10/09/2024 (1st Shift)

(a) 69 (b) 36 (c) 85 (d) 45

Q.63. What is the value of D, if $A : C = B : D$ in the given table?

A	B	C	D
25	40	30	?

SSC CGL 10/09/2024 (1st Shift)

(a) 56 (b) 48 (c) 45 (d) 42

Q.64. A money bag contains one rupee, 5 rupee and 10 rupee coins in the ratio of 3 : 5 : 7 respectively. If the total amount in the bag is ₹980, then find the number of coins of ₹10.

SSC CGL 10/09/2024 (1st Shift)

(a) 69 (b) 70 (c) 71 (d) 68

Q.65. A sum of money is divided among P, Q, R and S in the proportion of 4 : 9 : 2 : 3. If Q gets ₹560 more than R, then the share of S is:

SSC CGL 11/09/2024 (1st Shift)

(a) ₹140 (b) ₹340 (c) ₹240 (d) ₹440

Q.66. The incomes of P, Q and R are in the ratio 10 : 12 : 9 and their expenditures are in the ratio 12 : 15 : 8. If Q saves 25% of his income, then what is the ratio of the savings of P, Q and R?

SSC CGL 12/09/2024 (1st Shift)

(a) 15 : 14 : 21 (b) 14 : 15 : 21
(c) 21 : 15 : 14 (d) 21 : 14 : 15

Q.67. Raj divides ₹1,200 in the ratio 2 : 1 : 3 among three of his friends. The amount equal to the sum of three times the largest share and two times the smallest share is:

SSC CGL 12/09/2024 (1st Shift)

(a) ₹2,400 (b) ₹1,800 (c) ₹2,200 (d) ₹2,000

Q.68. The ratio between male and female members in a club is 2 : 3. If the number of male members is increased by 200, the ratio becomes 5 : 6. How many female members are there in the club?

SSC CGL 12/09/2024 (2nd Shift)

(a) 1200 (b) 900 (c) 1400 (d) 1000

Q.69. Find the mean proportional of 12.96 and 0.16.

SSC CGL 13/09/2024 (3rd Shift)

(a) 1.21 (b) 1.96 (c) 1.44 (d) 1.69

Q.70. In a collection of rare coins, there is one gold coin for every four non-gold coins. If 20 more gold coins are added to the collection, the ratio of the number of gold coins to that of non-gold coins will be 2:3. The total number of coins in the collection will now become _____.

SSC CGL 17/09/2024 (1st Shift)

(a) 80 (b) 60 (c) 100 (d) 48

Q.71. In a company, the ratio of men and women is 5 : 3, respectively. When 150 additional women are employed in the company, the ratio becomes 5 : 4. How many men are employed in the company?

SSC CGL 17/09/2024 (2nd Shift)

(a) 850 (b) 750 (c) 550 (d) 950

Q.72. There are y yellow marbles and b blue marbles in a box. 18% of the yellow marbles were added to 24% of the blue marbles, and the total was 30. Three times the number of yellow marbles exceeds two times the number of blue marbles by 200. What is the ratio of y^2 to b^2 ?

SSC CGL 17/09/2024 (3rd Shift)

(a) 6 : 5 (b) 4 : 1 (c) 2 : 1 (d) 5 : 2

Q.73. The mean proportional of $4a^2$ and x is $\frac{1}{5a}$, where a is a positive number, then the value of x is _____

SSC CGL 18/09/2024 (2nd Shift)

(a) $\frac{1}{100a^4}$ (b) $\frac{1}{100a^2}$ (c) $\frac{1}{25a^2}$ (d) $\frac{1}{25a^4}$

Q.74. The cost of a diamond varies directly as the square of its weight. Once, this diamond broke into four pieces with weights in the ratio of 4 : 3 : 2 : 1. When the pieces were sold, the merchant got ₹63,000 less than that of one piece. Find the original price of the diamond.

SSC CGL 19/09/2024 (1st Shift)

(a) ₹86,000 (b) ₹80,000
(c) ₹90,000 (d) ₹75,000

Q.75. The ratio of boys to girls initial in a college is 5 : 6. If 40 boys leave the college and 50 girls join the college, the ratio will become 8 : 11. Find the number of girls (initial) in the college.

SSC CGL 19/09/2024 (3rd Shift)

(a) 740 (b) 860 (c) 720 (d) 640

Q.76. 30 is the mean proportional of 18 and A. Find the value of A.

SSC CGL 23/09/2024 (1st Shift)

(a) 60 (b) 40 (c) 45 (d) 50

Q.77. The salaries of P and Q together amount to ₹9,60,000. P and Q save 64% and 44%, respectively of their individual salaries. If P's expenditure is 1.5 times Q's expenditure, then what is the ratio of P's salary to Q's salary?

SSC CGL 25/09/2024 (1st Shift)

(a) 2 : 3 (b) 7 : 3 (c) 3 : 7 (d) 3 : 2

Q.78. The ratio of circumference and diameter of a circle is 3 : 5. If the circumference is 6 cm, then the radius of the circle is:

SSC CGL 25/09/2024 (3rd Shift)

(a) 10 cm (b) 15 cm (c) 5 cm (d) 1.8 cm

Q.79. If $a : b = c : d = e : f = 5 : 7$, then what is the ratio $(3a + 5c + 11e) : (3b + 5d + 11f)$?

SSC CGL 26/09/2024 (2nd Shift)

(a) 7 : 11 (b) 3 : 7 (c) 5 : 7 (d) 11 : 7

Q.80. The monthly incomes of A, B and C are in the ratio 2 : 9 : 3 and their expenses are in the ratio 3 : 9 : 5. If A's saving is half of his total income, then savings of A, B and C are, respectively, in the ratio of:

SSC CGL 26/09/2024 (3rd Shift)

(a) 3 : 9 : 2 (b) 2 : 5 : 2
(c) 3 : 18 : 4 (d) 1 : 2 : 1

SSC CHSL 2024 Tier - 1

Q.81. Radhika requested the cashier of a bank to get her coins in lieu of her cheque worth ₹1,845. Cashier gave her a packet containing ₹5, ₹10 and ₹20 coins in the ratio of 3 : 5 : 7. What is the total worth of the smallest coin received by her?

SSC CHSL 01/07/2024 (1st Shift)

(a) ₹135 (b) ₹125 (c) ₹145 (d) ₹105

Q.82. The age of the older of two boys is thrice that of the younger. 9 years ago it was five times that of the younger. Find the present age of each.

SSC CHSL 01/07/2024 (2nd Shift)

(a) 15 years, 45 years
(b) 18 years, 54 years
(c) 17 years, 51 years
(d) 12 years, 36 years

Q.83. P pays Q a sum of ₹150 using coins of ₹2, ₹5 and ₹10. He uses a total of 50 coins. If the ratio of ₹2 and ₹5 coins used is 5 : 2, then how many coins of ₹10 are used in the payment?

SSC CHSL 01/07/2024 (3rd Shift)

(a) 2 (b) 1 (c) 4 (d) 5

Q.84. In what ratio must a grocer mix two varieties of pulses costing ₹60 and ₹75 per kg, respectively, so as to get a mixture worth ₹65 per kg?

SSC CHSL 01/07/2024 (3rd Shift)

(a) 3 : 4 (b) 2 : 1 (c) 2 : 3 (d) 1 : 2

Q.85. M varies inversely as $(N^2 + 3)$. If $M = 3$ when $N = 3$, then what will be the value of M when $N^2 = 69$?

SSC CHSL 01/07/2024 (4th Shift)

(a) 1 (b) $\frac{1}{4}$ (c) $\sqrt{69}$ (d) $\frac{1}{2}$

Q.86. The difference between two numbers is 48. The ratio of these two numbers is 7 : 3.

What is the sum of the two numbers?

SSC CHSL 01/07/2024 (4th Shift)

(a) 110 (b) 100 (c) 90 (d) 120

Q.87. Divide some money in the ratio Ravi, Reeta and Rahul that 5 (Part of Ravi) = 3 (Part of Reeta) = 11 (Part of

Rahul). The money ratio of Ravi : Reeta : Rahul is equal to :

SSC CHSL 02/07/2024 (4th Shift)

- (a) 11 : 5 : 3 (b) 11 : 33 : 15
(c) 5 : 11 : 3 (d) 33 : 55 : 15

Q.88. Among the ratios 3 : 4, 5 : 8, 1 : 2, 6 : 9 which is the greatest?

SSC CHSL 03/07/2024 (1st Shift)

- (a) 5 : 8 (b) 6 : 9 (c) 1 : 2 (d) 3 : 4

Q.89. If $a : b = 3 : 7$ and $b : c = 4 : 9$, then find $a : b : c$.

SSC CHSL 03/07/2024 (2nd Shift)

- (a) 12 : 28 : 63 (b) 3 : 28 : 63
(c) 12 : 28 : 9 (d) 3 : 28 : 9

Q.90. Which is the greatest ratio among the following?

$$\frac{8}{11}, \frac{4}{5}, \frac{1}{2} \text{ and } \frac{1}{11}$$

SSC CHSL 03/07/2024 (3rd Shift)

- (a) $\frac{4}{5}$ (b) $\frac{8}{11}$ (c) $\frac{1}{2}$ (d) $\frac{1}{11}$

Q.91. When 'X' is added to each of the numbers 7, 11, 18 and 23, then the number so obtained are in proportion. What is the mean proportional between the $(x - 1)$ and $(2x - 10)$?

SSC CHSL 03/07/2024 (4th Shift)

- (a) 78 (b) 98 (c) 48 (d) 68

Q.92. Convert 28% into the form of ratio.

SSC CHSL 04/07/2024 (1st Shift)

- (a) 25 : 7 (b) 7 : 25 (c) 7 : 18 (d) 18 : 7

Q.93. A box filled with gift articles weighs 25 kg. If the weight of the box and the gift articles, respectively, are in the ratio 1 : 9, then the weight of the articles (in grams) is:

SSC CHSL 04/07/2024 (1st Shift)

- (a) 22500 (b) 23500
(c) 21500 (d) 24500

Q.94. Find the third proportion to 36 and 48.

SSC CHSL 04/07/2024 (3rd Shift)

- (a) 36 (b) 54 (c) 48 (d) 64

Q.95. If $a : b = 3 : 4$, $b : c = 2 : 3$, and $c : d = 5 : 6$, then $a : b : c : d$ is:

SSC CHSL 05/07/2024 (1st Shift)

- (a) 10 : 20 : 30 : 36 (b) 15 : 25 : 30 : 36
(c) 15 : 20 : 30 : 36 (d) 15 : 20 : 35 : 36

Q.96. If $\frac{1}{M} : \frac{1}{N} : \frac{1}{O} = 3 : 4 : 5$, then $M : N : O$ is equal to :

SSC CHSL 05/07/2024 (2nd Shift)

- (a) 12 : 15 : 20 (b) 20 : 15 : 12
(c) 20 : 12 : 15 (d) 15 : 20 : 12

Q.97. The mean proportion of 0.14 and 2.24 is:

SSC CHSL 05/07/2024 (3rd Shift)

- (a) 0.56 (b) - 2.39 (c) 1.12 (d) 1.02

Q.98. Find the fourth proportion to 1.25, 5.75 and 3.5

SSC CHSL 05/07/2024 (4th Shift)

- (a) 4.5 (b) 16.1 (c) 12.5 (d) 6.5

Q.99. Two numbers are in the ratio 2 : 3. Their sum is 135. Find the numbers.

SSC CHSL 08/07/2024 (2nd Shift)

- (a) 54, 81 (b) 45, 90 (c) 50, 85 (d) 60, 75

Q.100. Find the mean proportional between 25 and 81.

SSC CHSL 08/07/2024 (3rd Shift)

- (a) 12 (b) 36 (c) 81 (d) 45

Q.101. Two numbers are, respectively, 30% and 60% more than a third number.

The ratio of the two numbers is:

SSC CHSL 09/07/2024 (1st Shift)

- (a) 13 : 16 (b) 12 : 13
(c) 11 : 15 (d) 13 : 17

Q.102. The strength of a school is 640, with the number of boys being 400. Find the ratio of girls to boys.

SSC CHSL 09/07/2024 (4th Shift)

- (a) 5 : 6 (b) 3 : 5 (c) 4 : 5 (d) 8 : 5

Q.103. Raju and Ramu have more land than Sonu by 15% and 25%, respectively. What is the ratio of land owned by Raju and Ramu?

SSC CHSL 10/07/2024 (1st Shift)

- (a) 56 : 25 (b) 21 : 23
(c) 23 : 25 (d) 25 : 23

Q.104. If $x : y = 5 : 6$, then $2x + 3y : 3x + 5y$ is:

SSC CHSL 10/07/2024 (2nd Shift)

- (a) 18 : 35 (b) 28 : 45
(c) 18 : 45 (d) 28 : 48

Q.105. Which of the following ratios is greater?

SSC CHSL 10/07/2024 (3rd Shift)

- (a) 7 : 4 (b) 8 : 3 (c) 2 : 3 (d) 1 : 5

Q.106. The ratio of red to green sweets in a bag is 3 : 4. How many red sweets are there in the bag if the number of green sweets is 120?

SSC CHSL 11/07/2024 (1st Shift)

- (a) 80 (b) 100 (c) 70 (d) 90

Q.107. Find the mean proportion to 0.72 and 2.85 (round up to one decimal place).

SSC CHSL 11/07/2024 (3rd Shift)

- (a) 1.4 (b) 0.7 (c) 2.8 (d) 0.4

Q.108. What should be added to each term of the ratio 5 : 11 so that the ratio becomes 3 : 5?

SSC CHSL 11/07/2024 (4th Shift)

- (a) 5 (b) 4 (c) 2 (d) 6

SSC CPO 2024 Tier - 1

Q.109. Two numbers are in the ratio of 3 : 7. If 11 is subtracted from each, the ratio between the numbers becomes 7 : 20. Find the smaller number.

SSC CPO 27/06/2024 (1st Shift)

- (a) 39 (b) 21 (c) 36 (d) 27

Q.110. If 40% of A = 60% of B = $\frac{1}{6}$ of C,

then find the ratio of A : B : C.

SSC CPO 27/06/2024 (3rd Shift)

- (a) 15:10:18 (b) 36:10:15
(c) 15:10:36 (d) 10:15:36

Q.111. The ratio of marks of A and B is 3 : 4 and the ratio of marks of B and C is 4 : 5. If the difference in the marks of C and A is 30, then how much marks does B get?

SSC CPO 27/06/2024 (3rd Shift)

- (a) 52 (b) 64 (c) 60 (d) 56

Q.112. A sum of ₹17,600 was divided between Jeevan, Praveen and Kanan, such that the ratio of the sums received by Jeevan and Praveen was 5 : 8, while the ratio of the sums received by Praveen and Kanan was 3 : 2. How much money did Jeevan receive as his share?

SSC CPO 28/06/2024 (1st Shift)

- (a) ₹4,900 (b) ₹4,750
(c) ₹4,850 (d) ₹4,800

Q.113. Find the natural number obtained when the number 13 is subtracted from the mean proportional of 75 and 12.

SSC CPO 28/06/2024 (1st Shift)

- (a) 14 (b) 30 (c) 27 (d) 17

Q.114. If four distinct positive numbers, a, b, c and d, in the order given are in proportion, then which of the following options is NOT correct?

SSC CPO 28/06/2024 (2nd Shift)

- (a) b, a, d and c, in the order given are in proportion.
(b) c, d, b and a, in the order given are in proportion.
(c) d, c, b and a, in the order given are in proportion.
(d) a, c, b and d, in the order given are in proportion.

Q.115. Let 'a' be a positive number and $b = \frac{4}{a}$. What is the mean proportional of a and b?

SSC CPO 28/06/2024 (3rd Shift)

- (a) 4 (b) 3 (c) 2 (d) a

Q.116. What is the ratio between S and T if S is 75% of T?

SSC CPO 28/06/2024 (3rd Shift)

- (a) 4 : 3 (b) 2 : 3 (c) 75 : 1 (d) 3 : 4

Q.117. In a school, the total number of

boys is 20% more than the total number of girls. What is the ratio of the number of boys to that of girls in the school?

SSC CPO 29/06/2024 (1st Shift)

(a) 6 : 5 (b) 5 : 6 (c) 5 : 3 (d) 4 : 5

Q.118. 42% of Ranita's weekly income is equal to 56% of Bhaskar's weekly income. If Bhaskar's weekly income was raised by ₹ 200, while Ranita's weekly income did not change, the ratio of the weekly incomes of Ranita and Bhaskar, respectively, would have been 5 : 4. What is Ranita's weekly income (in ₹)?

SSC CPO 29/06/2024 (1st Shift)

(a) 4200 (b) 3600 (c) 3000 (d) 4000

Q.119. Swati earns a sum of ₹40,000, out of which she saves 25%. She spends the remaining amount on food and education in the ratio of 3 : 2. What is the percentage of expenditure incurred on education to her total income?

SSC CPO 29/06/2024 (2nd Shift)

(a) 30% (b) 20% (c) 60% (d) 40%

SSC Selection Post (Phase - XII)

Q.120. In an educational institution, the ratio of the numbers of students in middle school and in high school, respectively, in a given year was given as 4 : 3. If there were 324 students in middle school in that year in that institution, what was the number of students in high school in the same year in that institution?

Matriculation Level 20/06/2024 (Shift - 1)

(a) 243 (b) 240 (c) 249 (d) 246

Q.121. If class A has 11 boys and 18 girls, class B has 13 boys and 17 girls, and class C has 20 boys and 9 girls, which class has the biggest ratio(s) of boys to girls?

Matriculation Level 20/06/2024 (Shift - 4)

(a) Class A (b) Both class A and class B (c) Class C (d) Class B

Q.122. A recipe is made using cheese and butter in the ratio 4 : 3. If Arohi uses 8 bowls of cheese, how many bowls of butter should she use?

Matriculation Level 21/06/2024 (Shift - 1)

(a) 4 (b) 6 (c) 8 (d) 3

Q.123. The ratio of the salaries of S to F is 2 : 3 and the ratio of their savings is 3 : 4. What is the ratio of their expenditures if their total savings are equal to the salary of S?

Matriculation Level 21/06/2024 (Shift - 1)

(a) 8 : 15 (b) 7 : 15 (c) 8 : 13 (d) 7 : 12

Q.124. S varies directly as (R + 7), and S

= 42 when R = 17. What is the value of S when R = 29?

Matriculation Level 24/06/2024 (Shift - 1)

(a) 51 (b) 63 (c) 58 (d) 66

Q.125. The price of a television set and a washing machine are in the ratio of 8 : 7. If the cost of the television set is ₹4,800 more than the cost of the washing machine, then find the price of the washing machine.

Matriculation Level 25/06/2024 (Shift - 1)

(a) ₹33,000 (b) ₹33,500

(c) ₹33,600 (d) ₹33,700

Q.126. Pooja, Chameli and Anju have a total of ₹450 in their purses. Pooja has ₹50 more than Chameli. Anju has two times what Pooja has. How much do Pooja, Chameli and Anju, respectively, have in their purses?

Higher Secondary 20/06/2024 (Shift - 2)

(a) ₹150, ₹75 and ₹225

(b) ₹100, ₹50 and ₹300

(c) ₹125, ₹75 and ₹250

(d) ₹125, ₹50 and ₹275

Q.127. Two numbers are in the ratio 4 : 5. If 3 is added to the smaller and 7 is subtracted from the larger number, they are in the ratio 1 : 1. Find the numbers.

Higher Secondary 20/06/2024 (Shift - 2)

(a) 60, 75 (b) 40, 50 (c) 55, 44 (d) 20, 25

Q.128. What is the mean proportional of $45x^4$ and $5y^2$?

Higher Secondary 21/06/2024 (Shift - 2)

(a) $5x^2y$ (b) $25x^4y$ (c) $15x^2y$ (d) $45x^2y^2$

Q.129. The ages of A and B are in the ratio 4 : 7, respectively. After 8 years, the ratio of their ages will be 2 : 3. What is the sum of the ages of A and B (in years)?

Higher Secondary 21/06/2024 (Shift - 2)

(a) 44 (b) 36 (c) 38 (d) 42

Q.130. Which number should be added to 3, 11, 4 and 14 to make them in proportion?

Higher Secondary 25/06/2024 (Shift - 2)

(a) 3 (b) 2 (c) 4 (d) 1

Q.131. Two numbers are in the ratio 4 : 5. If 17 is subtracted from each, the new numbers are in the ratio 11 : 14. If 20 is added to each of the original numbers, then the ratio becomes:

Higher Secondary 25/06/2024 (Shift - 4)

(a) 224 : 275 (b) 275 : 241

(c) 241 : 275 (d) 224 : 241

Q.132. A bag contains a total of 180 coins in the denominations of ₹5 and ₹1. Find the number of ₹5 coins in the bag, if the total value of the coins is ₹500.

Higher Secondary 26/06/2024 (Shift - 2)

(a) 60 (b) 120 (c) 80 (d) 100

Q.133. Find the third proportional of 9, 7, ?, 14.

Higher Secondary 26/06/2024 (Shift - 2)

(a) 21 (b) 16 (c) 24 (d) 18

Q.134. Find the ratio between the fourth proportional of 42, 56, 63 and the fourth proportional of 189, 273, and 153.

Graduate Level 21/06/2024 (Shift - 3)

(a) 33 : 221 (b) 217 : 42

(c) 33 : 84 (d) 84 : 221

Q.135. Rishi and Sarita have ₹4,760. If $\frac{9}{16}$ of Rishi's amount is equal to $\frac{3}{4}$ of Sarita's amount, how much does Sarita have?

Graduate Level 24/06/2024 (Shift - 3)

(a) ₹3,570 (b) ₹1,190

(c) ₹2,720 (d) ₹2,040

Q.136. A sum of money was divided between Cyrus, Rohan and Mishti. The ratio of the sums received by Cyrus and Rohan was 3 : 2, while the ratio of the sums received by Rohan and Mishti was 7 : 12. If the sum received by Cyrus was ₹1,800 less than the sum received by Mishti, how many rupees did Rohan receive as his share?

Graduate Level 24/06/2024 (Shift - 3)

(a) ₹9,240 (b) ₹7,560

(c) ₹8,400 (d) ₹6,300

Q.137. The third proportional to $(5^2 - 3^2)$ and $(5 - 3)$ is:

Graduate Level 25/06/2024 (Shift - 3)

(a) $\frac{1}{3}$ (b) $\frac{1}{8}$ (c) $\frac{1}{2}$ (d) $\frac{1}{4}$

Q.138. Four distinct positive numbers, 'a', 'b', 'c' and 'd', in the order given, are in proportion. 'b' is 35 more than 'a' and 'd' is 60 more than 'c'. The product of 'a' and 'c' is 5376. What is the sum of 'a', 'b', 'c' and 'd'?

Graduate Level 26/06/2024 (Shift - 3)

(a) 398 (b) 400 (c) 399 (d) 397

Q.139. A bag contains rupee, 50-paise and 25-paise coins in the ratio 3 : 4 : 6. If the total amount is ₹143, the number of 50-paise coins is:

Graduate Level 26/06/2024 (Shift - 4)

(a) 132 (b) 66 (c) 90 (d) 88

Q.140. The sum of the four numbers A, B, C and D is 875. If the ratio of A to B is 1 : 2, the ratio of B to C is 3 : 1 and the ratio of C to D is 2 : 3, find the value of C.

Graduate Level 26/06/2024 (Shift - 4)

(a) 120 (b) 130 (c) 125 (d) 135

SSC CHSL 2023 Tier - 2

Q.141. The ratio of the expenditure to the savings of Richa is 2 : 1. If her income and expenditure increase by 20% and 15%, respectively, then find the percentage increase in her savings.
SSC CHSL Tier II 10/01/2024
(a) 35% (b) 30% (c) 20% (d) 25%

Q.142. The ratio of prices of Coffee and Tea is 3 : 4 and the ratio of their quantities consumed by a family is 4 : 7. Find the ratio of expenditures on Coffee and Tea.
SSC CHSL Tier II 10/01/2024
(a) 3 : 7 (b) 7 : 3 (c) 7 : 4 (d) 4 : 7

Q.143. ₹7,786 is divided among A, B and C in such a manner that the ratio of the amount of A to that of B is 3 : 4 and the ratio of the amount of B to that of C is 12 : 13. The amount of money received by B is:
SSC CHSL Tier II 10/01/2024
(a) ₹7,138 (b) ₹2,977
(c) ₹2,748 (d) ₹5,038

SSC CPO 2023 Tier - 1

Q.144. The areas of the two triangles are in the ratio 4 : 3 and their heights are in the ratio 6 : 5. Find the ratio of their bases.
SSC CPO 03/10/2023 (1st Shift)
(a) 5 : 6 (b) 10 : 9 (c) 9 : 10 (d) 6 : 5

Q.145. If $\frac{x}{4y} = \frac{3}{4}$, then the value of $\frac{2x + 3y}{x - 2y}$ is
SSC CPO 05/10/2023 (2nd Shift)
(a) 8 (b) 6 (c) 9 (d) 7

SSC MTS 2023 Tier - 1

Q.146. P and Q are inversely proportional to each other and P is 10, when Q is 6. What will be the value of Q when P is 15?
SSC MTS 01/09/2023 (2nd Shift)
(a) 9 (b) 4 (c) 10 (d) 5

Q.147. The mean proportion of $(3 + 2\sqrt{2})^2$ and $(3 - 2\sqrt{2})^2$ is:
SSC MTS 04/09/2023 (1st Shift)
(a) $3 + 2\sqrt{2}$ (b) 5 (c) $2\sqrt{2}$ (d) 1

Q.148. Profit of ₹29,120 is divided among A, B and C so that A's share is $2\frac{1}{2}$ times B's and B's share is $1\frac{1}{5}$ times C. Find B's share.
SSC MTS 05/09/2023 (2nd Shift)
(a) ₹6,940 (b) ₹7,820 (c) ₹6,720 (d) ₹7,140

Q.149. Rupees 66,000 is to be divided amongst Asha, Bhupinder and Gargi in such a way that ratio of their shares is to be: Asha : Bhupinder = 9 : 5 and Bhupinder : Gargi = 2 : 1. How much should Asha get?
SSC MTS 05/09/2023 (3rd Shift)
(a) ₹38,000 (b) ₹36,000
(c) ₹32,000 (d) ₹34,000

Q.150. A certain sum is to be divided between A, B, and C in the ratio 2 : 5 : 7. But by mistake, it was divided in the ratio 5 : 2 : 7. Thus B loses by ₹150. Find the sum.
SSC MTS 06/09/2023 (1st Shift)
(a) ₹700 (b) ₹850 (c) ₹900 (d) ₹625

Q.151. If $p : q = 5 : 2$, then $p^2 - q^2 : p^2 + q^2 = ?$
SSC MTS 06/09/2023 (2nd Shift)
(a) 29 : 21 (b) 4 : 3 (c) 3 : 4 (d) 21 : 29

Q.152. The marks scored by a student are directly proportional to the time invested in studies. If the student gets 60 marks when he studies 4 hours a day, then how many hours he should use in his studies daily to get 75 marks?
SSC MTS 08/09/2023 (1st Shift)
(a) 4.5 (b) 5 (c) 5.5 (d) 6

Q.153. Three numbers are in the ratio 5 : 2 : 7. The sum of their squares is 3822. The biggest number is:
SSC MTS 08/09/2023 (2nd Shift)
(a) 35 (b) 38 (c) 78 (d) 49

Q.154. Among four numbers the sum of the smallest and largest is 72 more than the sum of middle two numbers. If the numbers are in the ratio 6 : 7 : 15 : 19, then the smallest number is:
SSC MTS 08/09/2023 (2nd Shift)
(a) 72 (b) 144 (c) 168 (d) 216

Q.155. If the income of T is 8 times his savings, then his expenditure is ___ times his savings.
SSC MTS 11/09/2023 (3rd Shift)
(a) 6 (b) 7 (c) 8 (d) 5

Q.156. The ratio of the number of boys to the number of girls in a school of 1650 students is 6 : 5. If 124 new boys are admitted and few new girls are admitted, the ratio changes to 4 : 3. What is the ratio of the newly admitted girls to the newly admitted boys?
SSC MTS 13/09/2023 (1st Shift)
(a) 3 : 4 (b) 62 : 9 (c) 4 : 3 (d) 9 : 62

Q.157. In a class of boys and girls, the number of girls is twice the number of boys. If 20 more boys join the class, the ratio of girls and boys is 1 : 1. How many

total students were there in the class initially?

SSC MTS 14/09/2023 (1st Shift)
(a) 60 (b) 75 (c) 45 (d) 90

Q.158. If $A : B : C = 3 : 4 : 5$, then $\sqrt{A^2 + B^2} : \sqrt{C^2 - B^2}$ is:
SSC MTS 14/09/2023 (3rd Shift)
(a) 3 : 2 (b) 4 : 5 (c) 5 : 3 (d) 3 : 5

SSC CHSL 2023 Tier - 1

Q.159. The cost of 3 cups and 5 plates is ₹1,080 and the cost of 2 cups and 4 plates is ₹840. Find the ratio of the cost of a cup to the cost of a plate.
SSC CHSL 03/08/2023 (3rd Shift)
(a) 1 : 3 (b) 2 : 3 (c) 1 : 1 (d) 1 : 2

Q.160. The compound ratio of 3 : 4 and 6 : 7 is 36 : x. Find x.
SSC CHSL 03/08/2023 (4th Shift)
(a) 64 (b) 75 (c) 56 (d) 48

Q.161. Find 2 numbers such that their mean proportional is 25 and their third proportional is 25.
SSC CHSL 07/08/2023 (1st Shift)
(a) 5 and 5 (b) 5 and 25
(c) 25 and 125 (d) 25 and 25

Q.162. The present age of a father is twice that of his son. 8.5 years hence the ratio would be 7 : 4. Find the son's present age.
SSC CHSL 07/08/2023 (2nd Shift)
(a) 25.5 years (b) 16.5 years
(c) 20.5 years (d) 24.6 years

Q.163. The sum of the amount that Sunaina and Ranjan have is ₹23 and the difference of their amount is ₹7. The ratio of the amount that Sunaina and Ranjan have is:—

SSC CHSL 07/08/2023 (3rd Shift)
(a) 5 : 3 (b) 7 : 15 (c) 15 : 8 (d) 2 : 17

Q.164. Two numbers are in the ratio of 1 : 2. If 7 is added to both, their ratio changes to 3 : 5. The greater number is:
SSC CHSL 07/08/2023 (3rd Shift)
(a) 30 (b) 18 (c) 14 (d) 28

Q.165. The difference between 24 and the mean proportional of 7 and 63 is:
SSC CHSL 07/08/2023 (4th Shift)
(a) 3 (b) 2 (c) 5 (d) 4

Q.166. If $a : b = 5 : 7$, then $(6a^2 - 2b^2) : (b^2 - a^2)$ will be:
SSC CHSL 08/08/2023 (4th Shift)
(a) 12 : 5 (b) 21 : 5 (c) 13 : 6 (d) 17 : 8

Q.167. What is the fourth proportional of $3\sqrt{5}$, $5\sqrt{8}$ and $3\sqrt{10}$?

SSC CHSL 09/08/2023 (4th Shift)

(a) $10\sqrt{5}$ (b) $40\sqrt{2}$ (c) 30 (d) 20

Q.168. The ratio of the father's age to his son's age is 5 : 3. The product of the numbers representing their ages is 960. The ratio of their ages after 6 years will be:

SSC CHSL 11/08/2023 (1st Shift)

(a) 23:15 (b) 23:17 (c) 21:15 (d) 21:17

Q.169. Find the third proportion of P and 12, when 8, P, 16 and 18 are in proportion.

SSC CHSL 11/08/2023 (2nd Shift)

(a) 16 (b) 8 (c) 18 (d) 12

Q.170. The total number of workers in a company is 960. If the number of female workers in the company is 420, then the respective ratio of the total number of male workers to the total number of female workers in the company is:

SSC CHSL 11/08/2023 (3rd Shift)

(a) 7 : 9 (b) 16 : 9 (c) 16 : 7 (d) 9 : 7

Q.171. Given that x varies directly as y and x = 45 when y = 9. What is the value of x when y = 50 ?

SSC CHSL 17/08/2023 (1st Shift)

(a) 225 (b) 190 (c) 175 (d) 250

Q.172. Divide ₹ 1488 among X, Y and Z. If the shares are in the ratio X : Y = 4 : 5, Y : Z = 7 : 6, then find out the share of Z.

SSC CHSL 17/08/2023 (3rd Shift)

(a) ₹ 460 (b) ₹ 450 (c) ₹ 470 (d) ₹ 480

SSC CGL 2023 Tier - 1

Q.173. The ratio of the number of boys to that of the girls in a school is 11 : 15. If there are 200 more girls than boys in the school, what is the number of boys in that school ?

SSC CGL 14/07/2023 (4th shift)

(a) 550 (b) 750 (c) 506 (d) 605

Q.174. The ratio between the monthly income and the expenditure of Vaidic is 8 : 5. If his income increases by 20% and his expenditure increases by 30%, then find the percentage increase or decrease in his monthly savings.

SSC CGL 14/07/2023 (4th shift)

(a) $5\frac{2}{3}\%$ increase (b) $5\frac{2}{3}\%$ decrease(c) $3\frac{1}{3}\%$ increase (d) $3\frac{1}{3}\%$ decrease

Q.175. In a factory, 48% of the number of male workers is equal to two-third of the number of female workers. What is the ratio of the number of males to that of the females in the factory ?

SSC CGL 18/07/2023 (2nd shift)

(a) 25 : 12 (b) 12 : 25 (c) 18 : 25 (d) 25 : 18

Q.176. Two numbers are, respectively, 17% and 50% more than a third number. The ratio of the two numbers is:

SSC CGL 19/07/2023 (4th shift)

(a) 27 : 25 (b) 39 : 50

(c) 19 : 11 (d) 29 : 25

Q.177. If $a : b :: b : c$, and $b = 96$, then which of the following can be a possible pair of values of a and c ?

SSC CGL 20/07/2023 (2nd shift)

(a) $a = 24$; $c = 374$ (b) $a = 32$; $c = 288$ (c) $a = 48$; $c = 168$ (d) $a = 16$; $c = 586$

Q.178. Three numbers are in the ratio $\frac{4}{5}$

 $:\frac{5}{6}:\frac{9}{10}$. The difference between the

smallest and the greatest numbers is 12. Find the number which is neither the smallest nor the greatest.

SSC CGL 24/07/2023 (2nd shift)

(a) 96 (b) 108 (c) 100 (d) 104

Q.179. A pole is 9030 cm long. It is colored saffron, white and green in the ratio of 6 : 7 : 8. Find the length of the green part of the pole.

SSC CGL 24/07/2023 (4th shift)

(a) 3440cm (b) 3010cm

(c) 3480cm (d) 2580cm

Q.180. If a 7-storey building has a 28 m long shadow, the number of storeys of the building whose shadow is 48 m long is:

SSC CGL 26/07/2023 (3rd shift)

(a) 14 (b) 24 (c) 16 (d) 12

Q.181. What smallest positive number should be subtracted from 2 and 3 so that 4 is the third proportion to them?

SSC CGL 26/07/2023 (4th shift)

(a) 1 (b) 2 (c) 4 (d) 3

Q.182. If the third proportional of $3x^2$ and $4xy$ is 48, then find the positive value of y.

SSC CGL 27/07/2023 (2nd shift)

(a) 3 (b) 5 (c) 2 (d) 4

SSC Selection Post (Phase - XI)

Q.183. The ratio of cheese to sauce for a single pizza is 1 cup to $\frac{1}{2}$ cup. If Rama used 15 cups of sauce to make a number of pizzas, then how many cups of cheese did she use on those pizzas ?

Graduate Level 27/06/2023 (Shift - 4)

(a) 20 cups (b) 10 cups

(c) 40 cups (d) 30 cups

Q.184. Out of the total income, a family saves 15% and the rest of the amount is

spent on three types of expenditure - health, education and food in the ratio of 1 : 2 : 2. The income is Rs.300 less than thrice the expenditure incurred on education. What is the difference between the savings and the expenditure incurred on health?

Graduate Level 27/06/2023 (Shift - 4)

(a) Rs.300 (b) Rs.350

(c) Rs.2,250 (d) Rs.500

Q.185. Let M denote the mean proportional between 16 and 9 and let N denote the third proportional of 9 and 15. Find the value of $5M + 3N$.

Graduate Level 28/06/2023 (Shift - 3)

(a) 300 (b) 125 (c) 135 (d) 225

Q.186. A company is manufacturing three different products - Glass, Plate and Bowl. The number of Plates is double the number of Glasses. The number of Glasses manufactured is twice the number of Bowls. What is the difference between the number of Plates and the number of Bowls if the total units manufactured by the company is 2380?

Graduate Level 28/06/2023 (Shift - 3)

(a) 1360 (b) 680 (c) 1120 (d) 1020

Q.187. Two numbers are in the ratio of 7 : 6. If the difference between their squares is 637, then the numbers are

Higher Secondary 30/06/2023 (Shift - 2)

(a) 49 and 42 (b) 49 and 12

(c) 43 and 36 (d) 14 and 12

Q.188. The students in three batches of a dance class are in ratio 2 : 3 : 5. If 20 students increase in each batch the ratio changes to 4 : 5 : 7. Find the total number of students in the three batches before the increase.

Matriculation Level 27/06/2023 (Shift - 1)

(a) 120 (b) 100 (c) 150 (d) 80

Q.189. In an examination, the ratio of practical to theory marks is 3 : 7. If 80% of the marks are scored in the practical and 60% of marks are scored in the theory, what is the ratio of marks scored in the theory and the practical ?

Matriculation Level 28/06/2023 (Shift - 1)

(a) 6 : 7 (b) 5 : 7 (c) 7 : 4 (d) 3 : 7

Q.190. If $a : b : c = 4 : 7 : 9$ and $b : c : d = 28 : 36 : 21$, then $a : b : c : d$ is:

Matriculation Level 28/06/2023 (Shift - 4)

(a) 12 : 21 : 27 : 17 (b) 16 : 28 : 36 : 21

(c) 8 : 14 : 18 : 21 (d) 4 : 7 : 9 : 11

Q.191. The ratio of the number of boys to that of girls in a village is 5 : 3. If 40% of the boys and 60% of the girls appeared in an examination, then the ratio of the number of boys and girls who appeared

in the examination to the number who did not appear in the same examination, is :-

Matriculation Level 30/06/2023 (Shift - 4)

- (a) 17 : 21 (b) 19 : 21
(c) 17 : 27 (d) 19 : 23

SSC CHSL 2022 Tier - 2

Q.192. The radii of two cylinders are in the ratio 2 : 1 and their heights are in the ratio 5 : 2. The ratio of their volumes is:

SSC CHSL Tier II (26/06/2023)

- (a) 20 : 1 (b) 15 : 1 (c) 10 : 1 (d) 5 : 1

SSC MTS 2022 Tier - 1

Q.193. The ratio of two numbers is 5 : 9. If 6 is added to the greater number and 6 is subtracted from the smaller number, then the greater number becomes 6 times the smaller number. What are the numbers ?

SSC MTS 02/05/2023 (Afternoon)

- (a) 15 and 27 (b) 10 and 18
(c) 25 and 45 (d) 20 and 36

Q.194. If $\frac{J}{4} = \frac{K}{5} = \frac{L}{3}$, then what is the value of $(L + J)^2 : (J + K)^2 : (K + L)^2$?

SSC MTS 04/05/2023 (Afternoon)

- (a) 25 : 16 : 9 (b) 64 : 25 : 81
(c) 49 : 81 : 62 (d) 49 : 81 : 64

Q.195. In three positive numbers, the ratio of first and second is 3 : 4 and the ratio of second and third is 3 : 5. The product of first and third is 4500. What is the sum of the three numbers?

SSC MTS 11/05/2023 (Morning)

- (a) 282 (b) 235 (c) 198 (d) 205

Q.196. When a sum of money was equally distributed among 50 children, each child received Rs. 80. If the same amount is equally distributed among another group of children such that each child get Rs. 50, then what is the number of children in the second group?

SSC MTS 15/05/2023 (Morning)

- (a) 50 (b) 80 (c) 60 (d) 40

Q.197. If $(x + y) : (y + z) : (z + x) = 3 : 5 : 7$ and $(x + y + z) = 45$, then what is the value of z ?

SSC MTS 16/05/2023 (Evening)

- (a) 18 (b) 30 (c) 27 (d) 20

Q.198. A sum of Rs. 120 is divided among p, q and r in such a way that p gets Rs. 9 more than q and q gets Rs. 6 more than r. What is the ratio of shares of p, q and r respectively ?

SSC MTS 17/05/2023 (Morning)

- (a) 18 : 15 : 13 (b) 15 : 12 : 10
(c) 16 : 13 : 11 (d) 11 : 8 : 10

SSC CHSL 2022 Tier - 1

Q.199. The expenditure and savings of a person are in a ratio of 4 : 3. His savings increase by $\frac{1}{5}$ of his income and income

remains the same. What is the new ratio of his expenditure and savings?

SSC CHSL 10/03/2023 (1st Shift)

- (a) 12 : 17 (b) 14 : 17
(c) 13 : 22 (d) 15 : 13

Q.200. Weight of Ayush and Abhishek are in the ratio of 8 : 5. Abhishek's weight increases by 40 percent and the total weight of Ayush and Abhishek both increases by 60 percent. If the total weight becomes 104 kg, then what is the weight of Ayush after the increment ?

SSC CHSL 10/03/2023 (3rd Shift)

- (a) 60 kg (b) 75 kg (c) 69 kg (d) 72 kg

Q.201. The ratio of income of Jitendra, and Narendra is 4 : 7 and the ratio of their expenditure is 6 : 5 respectively. If Jitendra saves Rs.400 out of Rs.2,800 income, then find the savings of Narendra.

SSC CHSL 13/03/2023 (2nd Shift)

- (a) Rs. 2,900 (b) Rs. 2,000
(c) Rs. 3,900 (d) Rs. 4,900

Q.202. The ratio of incomes of P and Q is 1 : 2. Ratio of income of Q and R is 3 : 2. If one-third of P's income is Rs. 4400 less than the half of P's income, then what is Q's income?

SSC CHSL 14/03/2023 (2nd Shift)

- (a) Rs. 56600 (b) Rs. 52800
(c) Rs. 46600 (d) Rs. 41200

Q.203. If the difference between the two numbers is 20 percent of their sum, then what is the ratio of larger number and smaller number ?

SSC CHSL 14/03/2023 (4th Shift)

- (a) 6 : 5 (b) 5 : 4 (c) 4 : 3 (d) 3 : 2

Q.204. The ratio of incomes of C and D is 3 : 4 respectively. The sum of their expenditure is Rs. 12000. Savings of C is the same as that of the expenditure of D. What is the sum of their savings?

SSC CHSL 16/03/2023 (2nd Shift)

- (a) Rs. 25000 (b) Rs. 15000
(c) Rs. 20000 (d) Rs. 16000

Q.205. The sum of the present ages of a father and his son is 94 years. 8 years ago their respective ages were in the ratio of 2 : 1. After 10 years what will be the ratio of ages of father and son ?

SSC CHSL 17/03/2023 (4th Shift)

- (a) 35 : 22 (b) 14 : 13 (c) 25 : 18 (d) 16 : 15

Q.206. Rs. 43,120 is divided among P, Q and R such that P's share is $\frac{4}{7}$ of total share of Q and R together and Q's share is $\frac{2}{5}$ of total share of P and R together.

What is the share of R ?

SSC CHSL 21/03/2023 (3rd Shift)

- (a) Rs.15120 (b) Rs.14220
(c) Rs.16400 (d) Rs.18050

SSC CGL 2022 Tier - 2

Q.207. The ratio of the incomes of two employees is 7 : 4, and the ratio of their expenditures is 3 : 1. If each of them manages to save ₹4,800 per month, find the sum of their monthly incomes (in ₹).

SSC CGL Tier II (02/03/2023)

- (a) 21120 (b) 20120 (c) 21150 (d) 18150

Q.208. The ratio of marks obtained by Rajesh, Rakesh and Ramesh in an exam is 2 : 4 : 9. What are the marks obtained by Rakesh and Ramesh, if Rajesh scored 30 marks in the exam?

SSC CGL Tier II (03/03/2023)

- (a) Rakesh = 40, Ramesh = 90
(b) Rakesh = 60, Ramesh = 135
(c) Rakesh = 120, Ramesh = 180
(d) Rakesh = 90, Ramesh = 40

SSC CGL 2022 Tier - 1

Q.209. By mistake, instead of dividing ₹702 among Ram, Ramesh, and Naresh in the ratio $\frac{1}{3} : \frac{1}{4} : \frac{1}{6}$, it was divided in the ratio of 3 : 4 : 6. Who gained the most and by how much?

SSC CGL 02/12/2022 (1st Shift)

- (a) Ram ₹158 (b) Ramesh ₹158
(c) Naresh ₹168 (d) Ram ₹168

Q.210. Which smallest positive number should be subtracted from each of 9 and 13 so that 18 is the third proportion to them ?

SSC CGL 03/12/2022 (3rd Shift)

- (a) 2 (b) 4 (c) 3 (d) 1

Q.211. A varies directly as the positive square root of B, and inversely as the cube of C. If A = 15, when B = 27 and C = 2, then find B when A = 9 and C = 2.

SSC CGL 05/12/2022 (2nd Shift)

- (a) $\frac{281}{42}$ (b) $\frac{243}{25}$ (c) $\frac{264}{37}$ (d) $\frac{275}{51}$

Q.212. The mean proportional of 6 and 54 is _____ more than 15.

SSC CGL 07/12/2022 (1st Shift)

(a) 3 (b) 6 (c) 4 (d) 5

SSC CPO 2022 Tier - 1

Q.213. Four numbers are in the ratio 1 : 2 : 3 : 5. Two times their sum is 44. Find the ratio of the numbers obtained by subtracting 3 from the cube of each number.

SSC CPO 09/11/2022 (Morning)

(a) 2 : 31 : 204 : 999

(b) 3 : 13 : 208 : 992

(c) 11 : 67 : 219 : 1003

(d) 5 : 61 : 213 : 997

Q.214. The salaries of A, B and C are in ratio of: $\frac{3}{2}, \frac{6}{5}, \frac{4}{3}$. The salary of A and B together is Rs.40,500. By what percentage is the salary of A more than that of C ?

SSC CPO 10/11/2022 (Evening)

(a) 12% (b) 13% (c) 11.5% (d) 12.5%

Q.215. The ratio of a father's age to his Mother's age is 3 : 2. The product of the numbers representing their age is 486. The ratio of their ages after 5 years will be: SSC CPO 10/11/2022 (Evening)

(a) 32:24 (b) 32:21 (c) 32:22 (d) 32:23

SSC CGL 2021 Tier - 2

Q.216. Salaries of B, C, D and E are in the ratio of 2 : 3 : 4 : 5 respectively. Their salaries are increased by 20 percent, 30 percent, 40 percent and 50 percent respectively. If the increased salary of D is ₹560, then what is the sum of the original salaries of B, C, D and E?

SSC CGL Tier II (08/08/2022)

(a) ₹1400 (b) ₹1560 (c) ₹1820 (d) ₹1260

Q.217. The cost of a diamond is directly proportional to the square of its weight. The cost of a 14 gm diamond is ₹2560. This diamond got broken down into two pieces in the ratio of 5 : 9. How much loss percent is incurred due to this breakage? (correct to two decimal places)

SSC CGL Tier II (08/08/2022)

(a) 53.47 percent (b) 49.71 percent

(c) 45.92 percent (d) 55.41 percent

Q.218. A sum of ₹ 1250 has to be distributed among A, B, C and D. Total share of B and D is equal to $(\frac{14}{11})$ of total share of A and C. Share of D is half of share of A. Share of C is 1.2 of share of A. What are the shares of A, B, C and D respectively?

SSC CGL Tier II (08/08/2022)

(a) ₹ 250, ₹ 575, ₹ 300, ₹ 125

(b) ₹ 350, ₹ 525, ₹ 300, ₹ 125

(c) ₹ 250, ₹ 525, ₹ 300, ₹ 125

(d) ₹ 250, ₹ 575, ₹ 300, ₹ 175

SSC Selection Post (Phase - X)

Q.219. If the mean proportional between 5 and 125 is x and the mean proportional between 4^4 and 6^2 is y, find the value of $2x : y$.

Graduate Level 04/08/2022 (Shift - 2)

(a) 25 : 48

(b) 25 : 128

(c) 25 : 64

(d) 25 : 96

Q.220. The mean proportional between 75 and x is five times the mean proportional between 15 and 26. The value of x is:

Higher Secondary 05/08/2022 (Shift - 2)

(a) 100 (b) 120 (c) 110 (d) 130

SSC MTS 2021 Tier - 1

Q.221. The ratio of the number of hens to the number of goats in farm A is 5 : 4 and that on farm B is 5 : 7. What could be the ratio of the total number of hens to the total number of goats in farms A and B taken together if total no. of animals in farm A and B are equal ?

SSC MTS 05/07/2022 (Morning)

(a) 35 : 37 (b) 1 : 2 (c) 9 : 11 (d) 7 : 11

Q.222. A person has some coins of ₹10, ₹5, and ₹2 denominations. The ratio of the products of the numbers of ₹10 and ₹5 coins, the numbers of ₹5 and ₹2 coins, and the numbers of ₹2 and ₹10 coins is 3 : 4 : 2 respectively. What could be the minimum amount of money this person has ?

SSC MTS 06/07/2022 (Morning)

(a) ₹52 (b) ₹88 (c) ₹68 (d) ₹74

Q.223. In a housing society, the ratio of two - wheelers and 4 - wheelers is 11 : 9. Among the four-wheelers, one-fourth are petrol vehicles and the remaining 81 are diesel vehicles. How many two - wheelers are there in this housing society?

SSC MTS 06/07/2022 (Evening)

(a) 88 (b) 132 (c) 121 (d) 99

Q.224. In a municipal election three candidates, Santosh, Ravi, and Vipin, contested and the valid votes they got were in the ratio of 11 : 4 : 9. If the winner defeated his nearest rival by 8350 votes, then how many valid votes were polled in that election ?

SSC MTS 07/07/2022 (Evening)

(a) 1,02,000

(b) 2,10,000

(c) 1,90,000

(d) 1,00,200

Q.225. A sum of money is distributed among A, B and C in the ratio 8 : 4 : 3. If A divides his part between D and E in the ratio 9 : 7, then what is the ratio of the share of C to that of D ?

SSC MTS 08/07/2022 (Afternoon)

(a) 4 : 5 (b) 3 : 4 (c) 2 : 3 (d) 6 : 7

Q.226. A number when subtracted from 3 times the second number, gives a result equal to $\frac{3}{4}$ of the first number.

What is the ratio of the first number to the second number ?

SSC MTS 11/07/2022 (Morning)

(a) 11 : 6 (b) 12 : 7 (c) 9 : 5 (d) 10 : 7

Q.227. In a bag there are coins of ₹5, ₹10 and ₹20 denominations. The total number of coins in the bag is 240. If the number of coins of ₹5, ₹10, ₹20 denominations are in the ratio of 2 : 3 : 5, then what is the total amount of money in the bag ?

SSC MTS 12/07/2022 (Afternoon)

(a) ₹3,540

(b) ₹4,600

(c) ₹4,620

(d) ₹3,360

Q.228. Three numbers a, b, and c are said to be in continued proportion if $a : b :: b : c$. What number must be added to each of the numbers 1, 7 and 25 so that the resulting sums, in that order, may be in continued proportion?

SSC MTS 14/07/2022 (Evening)

(a) 2 (b) 4 (c) 6 (d) 5

Q.229. A household of three humans has cats and dogs in the ratio 1:2. If the total number of legs in the household is 42, what is the total number of cats in the household ?

SSC MTS 14/07/2022 (Evening)

(a) 12 (b) 6 (c) 9 (d) 3

Q.230. When 8 is added to the numerator of a fraction and 12 is added to its denominator, the fraction becomes $\frac{1}{2}$.

When 2 is subtracted from its numerator and denominator, the fraction becomes $\frac{1}{8}$. Find the original fraction.

SSC MTS 15/07/2022 (Morning)

(a) $\frac{9}{19}$ (b) $\frac{3}{10}$ (c) $\frac{4}{7}$ (d) $\frac{5}{11}$

Q.231. If $a : b = b : c$, then what is the value of $a^6 : b^6$?

SSC MTS 15/07/2022 (Evening)

(a) $a^3 : b^3$ (b) $b^3 : ac$ (c) $a^3 : c^3$ (d) $a^2 : b^2$

Q.232. In a bag there are Rs 5 and Rs 10 coins such that the product of their numbers is 135 and the total amount in the bag is Rs 165, what is the amount

from all Rs 10 coins?

SSC MTS 18/07/2022 (Evening)

(a) 90 (b) 85 (c) 80 (d) 95

Q.233. There are a total of 60 students in a class and the ratio of the number of boys to that of the girls is 4 : 1, respectively. If 22 of the boys from this class are vegetarians, then find the number of non-vegetarian boys from the same class.

SSC MTS 19/07/2022 (Evening)

(a) 26 (b) 18 (c) 38 (d) 28

Q.234. The ratio of the quantities imported of steel and iron in 2018 was 3 : 2 and in 2020 it was 1 : 1. The ratio of the quantities imported of steel in 2018 and 2020 was 4 : 3. What was the ratio of the quantities imported of iron in 2018 and 2020 ?

SSC MTS 20/07/2022 (Morning)

(a) 11 : 12 (b) 7 : 5 (c) 8 : 9 (d) 7 : 9

Q.235. On a tree, there are some parrots and some pigeons in the ratio of 7:9, respectively. After an hour, 8 parrots fly away, and 6 pigeons and 10 sparrows come and sit on the tree. The ratio of the parrots and the pigeons on the tree now is 1 : 4. What is the ratio of the parrots and the sparrows that are now on the tree?

SSC MTS 20/07/2022 (Afternoon)

(a) 1 : 1 (b) 1 : 2 (c) 3 : 5 (d) 4 : 5

Q.236. 4 years ago, the age of Garvit was three times of Yuvansh, but one year ago the age of Garvit was two times that of Yuvansh. What is the difference between the ages of Garvit and Yuvansh?

SSC MTS 21/07/2022 (Morning)

(a) 13 years (b) 11 years
(c) 7 years (d) 6 years

Q.237. The ratio of the number of boys to the number of girls in a class is 2 : 5. If 10 boys leave the class and 20 new girls join the class, then the new ratio of boys and girls in the class becomes 4 : 25. What is the number of boys in the class now?

SSC MTS 22/07/2022 (Morning)

(a) 12 (b) 8 (c) 9 (d) 22

Q.238. Rs 21000 is to be distributed between Raj and Karan such that Raj gets Rs 9000 less than Karan. What is the ratio of the amount received by Raj to that received by Karan?

SSC MTS 22/07/2022 (Evening)

(a) 2 : 5 (b) 3 : 5 (c) 5 : 3 (d) 5 : 2

Q.239. Four years ago, Shyam's age was $\frac{3}{4}$ times that of Ram. Four years hence,

Shyam's age will be $\frac{5}{6}$ times that of Ram

.What is the present age of Shyam ?

SSC MTS 26/07/2022 (Morning)

(a) 24 years (b) 20 years
(c) 16 years (d) 15 years

SSC CHSL 2021 Tier - 1

Q.240. Soham's initial expenditure and savings were in the ratio 5 : 1. His income increases by 25%, while his savings increase by 20%. If Soham's current expenditure is ₹4,347, what was his initial income?

SSC CHSL 31/05/2022 (Morning)

(a) ₹4,130 (b) ₹4,140
(c) ₹4,150 (d) ₹4,160

Q.241. Two numbers, A and B, are such that the sum of 10% of A and 8% of B is $\frac{3}{5}$ th of the sum of 12% of A and 16% of

B. The ratio of A : B is :

SSC CHSL 31/05/2022 (Afternoon)

(a) 2 : 3 (b) 4 : 7 (c) 2 : 7 (d) 4 : 3

Q.242. Two numbers X and Y are such that the sum of 18% of X and 8% of Y is one-third of the sum of 22% of X and 36% of Y. Find the ratio of X and Y.

SSC CHSL 02/06/2022 (Afternoon)

(a) 2 : 5 (b) 3 : 11 (c) 3 : 8 (d) 8 : 3

Q.243. If 60% of a number is equal to three-fourth of another number, what is the ratio of the first number to the second number?

SSC CHSL 07/06/2022 (Evening)

(a) 3 : 7 (b) 2 : 5 (c) 7 : 3 (d) 5 : 4

Q.244. P, Q and R are batsmen. The ratio of the runs scored by them in a certain match was P : Q = 16 : 17 and Q : R = 15 : 16. At the end of the match, they scored a total of 956 runs. The number of runs scored by R is (nearest to an integer):

SSC CHSL 09/06/2022 (Afternoon)

(a) 440 (b) 335 (c) 339 (d) 430

Q.245. A varies jointly with B and C. A = 6 when B = 3 and C = 2. Find A when B = 5 and C = 7.

SSC CHSL 10/06/2022 (Evening)

(a) 17.5 (b) 35 (c) 105 (d) 70

SSC CGL 2021 Tier - 1

Q.246. The ratio of the monthly incomes of A and B is 11 : 13 and the ratio of their expenditure is 9 : 11. If both of them manage to save ₹ 4,000 per month, then find the difference in their incomes (in ₹).

SSC CGL 11/04/2022 (Morning)

(a) 2,500 (b) 3,200 (c) 4,000 (d) 3,000

Q.247. At present, A is younger than B by 8 years. If 4 years ago, their ages were in the ratio of 1 : 2, then what is the present age of B (in years)?

SSC CGL 12/04/2022 (Morning)

(a) 11 (b) 20 (c) 12 (d) 18

SSC CGL 2020 Tier - 2

Q.248. If a, b and c are positive numbers such that $(a^2 + b^2) : (b^2 + c^2) : (c^2 + a^2) = 34 : 61 : 45$, then $b - a : c - b : c - a =$ ____.

SSC CGL Tier II (29/01/2022)

(a) 1 : 2 : 3 (b) 3 : 1 : 2
(c) 2 : 1 : 3 (d) 3 : 2 : 1

Q.249. The income of A is $\frac{2}{3}$ of B's income and the expenditure of A is $\frac{3}{4}$ of

B's expenditure. If $\frac{1}{3}$ of the income of B is equal to the expenditure of A, then the ratio of the savings of A to those of B is:

SSC CGL Tier II (29/01/2022)

(a) 3 : 4 (b) 3 : 5 (c) 5 : 3 (d) 4 : 3

Q.250. Alloy A contains metals x and y only in the ratio 5 : 2, while alloy B contains them in the ratio 3 : 4. Alloy C is prepared by mixing alloys A and B in the ratio 4 : 5. The percentage of x in alloy C is:

SSC CGL Tier II (29/01/2022)

(a) $55\frac{4}{9}$ (b) $55\frac{2}{9}$ (c) $55\frac{1}{9}$ (d) $55\frac{5}{9}$

Q.251. One cup has juice and water in the ratio 5 : 2, while another cup of the same capacity has them in the ratio 7 : 4, respectively. If the contents of both the cups (when full) are poured in a vessel, then what will be the final ratio of water to juice in the vessel?

SSC CGL Tier II (29/01/2022)

(a) 25:52 (b) 52:25 (c) 25:26 (d) 26:25

Q.252. The ratio of the incomes of A and B in the last year was 4 : 3. The ratios of their individual incomes in the last year and the present year are 3 : 4 and 5 : 6, respectively. If their total income in the present year is ₹24.12 lakhs, then the sum of the income (in ₹ lakhs) of A in the last year and that of B in the present year is:

SSC CGL Tier II (29/01/2022)

(a) 20.52 (b) 10.98 (c) 22.17 (d) 21.28

Q.253. Let x, y, z be fractions such that $x < y < z$. If z is divided by x, the result is $\frac{5}{2}$, which exceeds y by $\frac{7}{4}$. If $x + y + z = 1\frac{11}{12}$ then the ratio of $(z - x) : (y - x)$ is:

SSC CGL Tier II (03/02/2022)

(a) 6 : 5 (b) 9 : 5 (c) 5 : 6 (d) 5 : 9

Q.254. In an examination, the number of students who passed and the number of students who failed were in the ratio 25 : 4. If one more student had appeared and passed and the number of failed students was 3 less than earlier, the ratio of passed students to failed students would have become 22 : 3. What is the difference between the number of students who, initially, passed the examination and the number of students who failed the examination?

SSC CGL Tier II (03/02/2022)

(a) 132 (b) 126 (c) 174 (d) 150

SSC MTS 2020 Tier - 1

Q.255. Suman divided a certain sum between her three daughters in the ratio 2 : 3 : 4. Had she divided the sum in the ratio $\frac{1}{2} : \frac{1}{3} : \frac{1}{4}$, the daughter who got the least share earlier, would have got ₹3,500 more. The sum was:

SSC MTS 05/10/2021(Afternoon)

(a) ₹14,000 (b) ₹14,500
(c) ₹13,750 (d) ₹14,625

Q.256. In a library, the ratio of the number of mathematics books to that of physics books, is the same as the ratio of the number of physics books to that of chemistry books. If there are 144 mathematics books and 100 chemistry books, then the ratio of the total number of mathematics and physics books to the total number of chemistry and physics books is :-

SSC MTS 05/10/2021 (Evening)

(a) 3 : 2 (b) 6 : 5 (c) 5 : 4 (d) 4 : 3

Q.257. 70 males and 100 females are working in a firm. During one mass recruitment drive, an equal number of males and females are recruited and their ratio becomes 3 : 4. The total number of people working in the firm after recruitment is:

SSC MTS 06/10/2021 (Morning)

(a) 240 (b) 200 (c) 210 (d) 220

Q.258. Between two consecutive years, my income is in the ratio 2 : 5 and expenses are in the ratio 4 : 7. If my income in the second year is ₹75,000 and my expenses in the first year are ₹20,000, then my total savings in the two years together are:

SSC MTS 06/10/2021 (Morning)

(a) ₹55,000 (b) ₹60,000
(c) ₹50,000 (d) ₹40,000

Q.259. A sum of ₹3,780 is divided between A, B and C such that if their shares are decreased by ₹130, ₹150 and ₹200, respectively, then they are in the ratio of 5 : 2 : 4. What is the original share of C?

SSC MTS 06/10/2021 (Evening)

(a) ₹1,350 (b) ₹1,330 (c) ₹1,400 (d) ₹1,430

Q.260. A, B and C are three boxes containing marbles in the ratio 3 : 5 : 7, and the total number of marbles is 75. If 3 marbles are transferred from B to A, and 5 marbles are transferred from C to B, then the new ratio of the marbles is:

SSC MTS 07/10/2021 (Morning)

(a) 6 : 8 : 11 (b) 5 : 6 : 7
(c) 7 : 9 : 10 (d) 6 : 9 : 10

Q.261. If $(a + b) : (b + c) : (c + a) = 7 : 4 : 5$, and $a + b + c = 16$, then $(a^2 + b^2 + c^2) : (ab + bc + ca)$ is equal to :

SSC MTS 08/10/2021 (Evening)

(a) 24 : 17 (b) 26 : 19
(c) 27 : 20 (d) 23 : 15

Q.262. Renuka got $1\frac{1}{2}$ times as many marks in Mathematics as in English. Her total marks in Mathematics, English and Science are 190. If the ratio of the marks in Mathematics and Science is 2 : 3, find her Science marks.

SSC MTS 11/10/2021 (Evening)

(a) 92 (b) 90 (c) 85 (d) 88

Q.263. A man ordered 10 physics books and some chemistry books. The price of a chemistry book is twice the price of a Physics book. While preparing the bill the clerk interchanged the number of physics and chemistry books by mistake, which decreased the bill by $12\frac{1}{2}\%$. The ratio of

the number of physics books to the number of chemistry books in the original order is:

SSC MTS 12/10/2021 (Evening)

(a) 3 : 5 (b) 3 : 4 (c) 4 : 5 (d) 2 : 3

Q.264. The ratio of the monthly income and expenditure of Dinesh is 17 : 14, and his monthly savings are ₹12,000. If his monthly income is increased by ₹10,000 and expenditure is decreased by ₹2,000, then the new ratio of his income and expenditure is:

SSC MTS 12/10/2021 (Evening)

(a) 13 : 8 (b) 13 : 9 (c) 11 : 9 (d) 11 : 7

Q.265. A certain sum is divided among A, B, C and D such that B's share is $\frac{1}{3}$ of A's share, C's share is 40% of B's share, and D's Share is 50% of C's share. If the

difference between the shares of B and D is ₹1,600, then the sum is:

SSC MTS 13/10/2021 (Morning)

(a) ₹9,600 (b) ₹9,400 (c) ₹9,000 (d) ₹9,200

Q.266. A, B and C are boxes with marbles in the ratio 2 : 3 : 4. The total number of marbles in the boxes is 360. The above ratio can be changed to 3 : 8 : 7 by which of the following?

(i) By transferring 20 marbles from A to B and 20 marbles from C to B.

(ii) By transferring 20 marbles from A to B and 10 marbles from C to B.

(iii) By transferring 40 marbles from A to B and 20 marbles from C to B

(iv) By transferring 30 marbles from A to B and 30 marbles from C to B.

SSC MTS 18/10/2021 (Morning)

(a) (ii) (b) (iv) (c) (i) (d) (iii)

Q.267. The number of students in three sections of Grade 10 in a school are in the proportion 3 : 5 : 8. If 15, 30 and 15 more students are admitted in the three sections, respectively, the new proportion becomes 4 : 7 : 9. The total number of students before the new admissions is:

SSC MTS 18/10/2021 (Afternoon)

(a) 320 (b) 160 (c) 240 (d) 400

Q.268. The price of a semi-precious stone weighing 48 g, is ₹2,30,400. It is broken into two pieces whose weights are in the ratio 5 : 7. If the price is proportional to the square of the weight, then the loss (in ₹) incurred, is:

SSC MTS 18/10/2021 (Afternoon)

(a) 1,12,000 (b) 40,000
(c) 1,18,400 (d) 78,400

Q.269. x and y are two numbers having opposite signs such that $x^2 : y^2 = 49 : 64$. What is the value of $(5x - 6y) : (6x - 7y)$?

SSC MTS 18/10/2021 (Evening)

(a) 44 : 65 (b) 13 : 14
(c) 94 : 117 (d) 83 : 98

Q.270. In an examination, the ratio of the number of candidates who passed to those who failed was 5 : 2. If the number of failed candidates had been 14 more, then the ratio of the number of passed candidates to those who failed would have been 4 : 3. The number of candidates who appeared in the examination was:

SSC MTS 20/10/2021 (Afternoon)

(a) 126 (b) 98 (c) 70 (d) 100

Q.271. The ratio of the number of students in Science to that in Humanities, in Grade 12 of 360 students, is 2 : 3. When some new students are admitted, the number of Science

students increases by 18 and the ratio of the Science students to Humanities students changes to 3 : 5. What is the number of new Humanities students admitted?

SSC MTS 20/10/2021 (Evening)

(a) 54 (b) 45 (c) 63 (d) 72

Q.272. The ratio of the present incomes of X and Y is 5 : 8 and the ratios of their individual incomes, in the last year and in the present year, are 2 : 3 and 3 : 4, respectively. If their total income for the last year was ₹56,000, then what is the present income of Y?

SSC MTS 26/10/2021 (Afternoon)

(a) ₹48,000 (b) ₹36,000
(c) ₹40,000 (d) ₹30,000

Q.273. A shopkeeper placed a supply order for jeans, shirts and t-shirts at the rates of ₹500, ₹350 and ₹200 per piece respectively. He buys articles of each type in the ratio 1 : 2 : 3 for ₹46,800. For how many t-shirts did he place the order?

SSC MTS 27/10/2021 (Morning)

(a) 104 (b) 26 (c) 78 (d) 52

Q.274. The present ages of a mother and her son are in the ratio of 6 : 1. When the son becomes as old as his mother now, then the ratio of his father's age to his mother's age is 13 : 11. When the son becomes as old as his father now, then the sum of the ages of the father and son will be 115 years. The present age of the father is:

SSC MTS 27/10/2021 (Evening)

(a) 40 years (b) 45 years
(c) 42 years (d) 44 years

Q.275. In an examination, the ratio of the number of candidates who passed to those who failed was 5 : 2. If the number of failed candidates had been 14 more, then the ratio of the number of passed candidates to those who failed would have been 4 : 3. How many candidates passed the examination?

SSC MTS 02/11/2021 (Afternoon)

(a) 126 (b) 70 (c) 100 (d) 98

Q.276. A batter missed her century by four runs in an innings, in which the scoring shots in the form of sixes, fours, threes, two's and singles, were in the proportion 1 : 3 : 2 : 7 : 10. How many runs did she score in singles?

SSC MTS 02/11/2021 (Evening)

(a) 20 (b) 16 (c) 10 (d) 12

SSC CGL 2020 Tier - 1

Q.277. A, B and C divide a certain sum of money among themselves. The average of the amounts with them is 4520. Share

of A is $10\frac{2}{3}\%$ more than share of B and

$33\frac{1}{3}\%$ less than share of C. What is the

share of B (in rupee)?

SSC CGL 13/08/2021 (Evening)

(a) 3500 (b) 5976 (c) 3600 (d) 3984

Q.278. Alloy A contains metal x and y in the ratio 5 : 2 and alloy B contains these metals in the ratio 3 : 4. Alloy C is prepared by mixing A and B in the ratio 4 : 5. The percentage of y in alloy C is:

SSC CGL 18/08/2021 (Evening)

(a) $44\frac{4}{9}\%$ (b) $33\frac{4}{9}\%$ (c) $66\frac{4}{9}\%$ (d) $55\frac{5}{9}\%$

Q.279. Monthly salaries of Anil and Kumud are in the ratio 19 : 17. If Anil and Kumud get salary hike of Rs. 2000 and Rs. 1000 respectively, then the ratio in their salaries becomes 8 : 7. What is the present salary of Kumud (in Rs) ?

SSC CGL 24/08/2021 (Afternoon)

(a) 18000 (b) 38000 (c) 34000 (d) 35000

SSC CHSL 2020 Tier - 1

Q.280. A bag contains coins of denominations Rs.1, Rs.2 and Rs.5 in the ratio of 4 : 5 : 8. If the total value of these coins is Rs.432, then what is the number of Rs.2 coins ?

SSC CHSL 05/08/2021 (Morning)

(a) 40 (b) 30 (c) 50 (d) 60

Q.281. The ratio of the incomes of A and B is 7 : 5. A and B save Rs. 4,000 and Rs. 3,500, respectively. If the expenditure of B is half the expenditure of A, then the total income of A and B (in Rs.) will be:

SSC CHSL 05/08/2021 (Afternoon)

(a) 10,000 (b) 13,500 (c) 12,000 (d) 15,000

Q.282. In a proportion, the product of the first and fourth terms is 70 and that of the second and third items is $3.5y$. The value of y is.

SSC CHSL 09/08/2021 (Morning)

(a) 17 (b) 20 (c) 15 (d) 22

SSC CPO 2020 Tier - 1

Q.283. Seven years ago, the ratio of the ages of A and B was 4 : 5. Eight years hence, the ratio of the ages of A and B will be 9 : 10. What is the sum of their present ages in years?

SSC CPO 23/11/2020 (Morning)

(a) 41 (b) 82 (c) 32 (d) 56

Q.284. In a school, $\frac{5}{12}$ of the number of students are girls and the rest are boys, $\frac{4}{7}$ of the number of boys are below 14

years of age, and $\frac{2}{5}$ of the number of

girls are 14 years or above 14 years of age. If the number of students below 14 years of age is 1120, then the total number of students in the school is:

SSC CPO 25/11/2020 (Evening)

(a) 1820 (b) 1900 (c) 1920 (d) 1290

Answer Key :-

1.(b)	2.(a)	3.(d)	4.(a)
5.(d)	6.(d)	7.(b)	8.(d)
9.(a)	10.(d)	11.(d)	12.(a)
13.(a)	14.(d)	15.(a)	16.(c)
17.(a)	18.(b)	19.(c)	20.(c)
21.(c)	22.(b)	23.(c)	24.(a)
25.(b)	26.(d)	27.(d)	28.(b)
29.(c)	30.(b)	31.(d)	32.(a)
33.(b)	34.(c)	35.(b)	36.(a)
37.(b)	38.(a)	39.(b)	40.(d)
41.(c)	42.(a)	43.(c)	44.(d)
45.(c)	46.(a)	47.(c)	48.(d)
49.(a)	50.(c)	51.(c)	52.(b)
53.(c)	54.(d)	55.(a)	56.(d)
57.(c)	58.(c)	59.(a)	60.(d)
61.(c)	62.(a)	63.(b)	64.(b)
65.(c)	66.(b)	67.(c)	68.(a)
69.(c)	70.(a)	71.(b)	72.(b)
73.(a)	74.(c)	75.(c)	76.(d)
77.(b)	78.(c)	79.(c)	80.(c)
81.(a)	82.(b)	83.(b)	84.(b)
85.(d)	86.(d)	87.(d)	88.(d)
89.(a)	90.(a)	91.(c)	92.(b)
93.(a)	94.(d)	95.(c)	96.(b)
97.(a)	98.(b)	99.(a)	100.(d)
101.(a)	102.(b)	103.(c)	104.(b)
105.(b)	106.(d)	107.(a)	108.(b)
109.(a)	110.(c)	111.(c)	112.(d)
113.(d)	114.(b)	115.(c)	116.(d)
117.(a)	118.(d)	119.(a)	120.(a)
121.(c)	122.(b)	123.(c)	124.(b)
125.(c)	126.(c)	127.(b)	128.(c)
129.(a)	130.(d)	131.(a)	132.(c)
133.(d)	134.(d)	135.(d)	136.(c)
137.(d)	138.(c)	139.(d)	140.(c)
141.(b)	142.(a)	143.(c)	144.(b)
145.(c)	146.(b)	147.(d)	148.(c)
149.(b)	150.(a)	151.(d)	152.(b)
153.(d)	154.(b)	155.(b)	156.(d)

157.(a)	158.(c)	159.(a)	160.(c)
161.(d)	162.(a)	163.(c)	164.(d)
165.(a)	166.(c)	167.(d)	168.(a)
169.(a)	170.(d)	171.(d)	172.(d)
173.(a)	174.(c)	175.(d)	176.(b)
177.(b)	178.(c)	179.(a)	180.(d)
181.(a)	182.(a)	183.(d)	184.(a)
185.(c)	186.(d)	187.(a)	188.(b)
189.(c)	190.(b)	191.(b)	192.(c)
193.(b)	194.(d)	195.(d)	196.(b)
197.(c)	198.(c)	199.(c)	200.(c)
201.(a)	202.(b)	203.(d)	204.(d)
205.(a)	206.(a)	207.(a)	208.(b)
209.(c)	210.(d)	211.(b)	212.(a)
213.(d)	214.(d)	215.(d)	216.(a)
217.(c)	218.(a)	219.(a)	220.(d)
221.(a)	222.(c)	223.(b)	224.(d)
225.(c)	226.(b)	227.(d)	228.(a)
229.(d)	230.(b)	231.(c)	232.(a)
233.(a)	234.(c)	235.(c)	236.(d)
237.(a)	238.(a)	239.(c)	240.(b)
241.(b)	242.(c)	243.(d)	244.(c)
245.(b)	246.(c)	247.(b)	248.(c)
249.(b)	250.(d)	251.(a)	252.(a)
253.(a)	254.(b)	255.(d)	256.(b)
257.(c)	258.(c)	259.(c)	260.(d)
261.(b)	262.(b)	263.(d)	264.(b)
265.(d)	266.(c)	267.(c)	268.(a)
269.(d)	270.(b)	271.(a)	272.(a)
273.(c)	274.(a)	275.(b)	276.(a)
277.(c)	278.(a)	279.(c)	280.(a)
281.(c)	282.(b)	283.(a)	284.(c)

Solutions :-

Sol.1.(b) Let marbles initially in each bag = $2x, 3x, 4x$

After adding 12 marbles to each bag

Marbles in each bag = $8y, 11y, 14y$

According to the question,

$$2x + 12 = 8y$$

$$8y - 2x = 12 \text{-----(i)}$$

$$\text{And, } 3x + 12 = 11y$$

$$11y - 3x = 12 \text{-----(ii)}$$

By the equation (i) $\times 3$ - (ii) $\times 2$ we get;

$$24y - 6x - (22y - 6x) = 36 - 24$$

$$2y = 12 \Rightarrow y = 6$$

Putting the value of y in equation (i) we get;

$$8 \times (6) - 2x = 12$$

$$2x = 48 - 12 = 36 \Rightarrow x = 18$$

Now, total marbles initially has

$$= (2x + 3x + 4x) = 9x \Rightarrow 9 \times 18 = 162$$

Sol.2.(a) Here two values of function is given hence it is of two variable.

According to the question,

$$f \propto x \text{ and } f \propto \frac{1}{y}$$

$$\Rightarrow f = k_1 x + k_2 \times \frac{1}{y} \dots (i)$$

Where k_1 and k_2 are constant

On putting $x = 0, y = 1, f = 15$ in (i) we get

$$f = k_1 x + k_2 \times \frac{1}{y}$$

$$k_2 = 15 \dots (ii)$$

On putting $x = 1, y = 15, f = 2$ in (i) we

$$\text{get, } f = k_1 x + k_2 \times \frac{1}{y}$$

$$2 = k_1 + 15 \times \frac{1}{15} \dots [\text{from (ii)}]$$

$$k_1 = 1. \text{ Now,}$$

when $x = 5$ and $y = 3$

$$\Rightarrow f = 1 \times 5 + 15 \times \frac{1}{3}$$

$$\Rightarrow f = 5 + 5 = 10$$

Sol.3.(d) Let ₹10, ₹5, ₹2 coins be x, y, z respectively.

According to question,

$$x + y + z = 25 \dots (i)$$

$$10x + 5y + 2z = 120 \dots (ii)$$

Multiply by 2 in equation (i) and subtracting equation (i) from (ii) we get

$$8x + 3y = 70$$

Now on satisfying the options we find that only option (d) gives positive value of y and no. of coins must be positive.

Hence, no. of ₹10 coins = 8

Sol.4.(a) Mean proportional of 1.21 and

$$0.09 = \sqrt{1.21 \times 0.09} = 0.33$$

Mean proportional of 0.16 and 0.36

$$= \sqrt{0.16 \times 0.36} = 0.24$$

$$\text{Required ratio} = 0.33 : 0.24 = 11 : 8$$

Sol.5.(d) Mean proportion

$$= \sqrt{8a^2 \times \frac{18}{a^4}} = \sqrt{\frac{144}{a^2}} = \frac{12}{a}$$

Sol.6.(d) Let ₹5, ₹2, ₹1 coins be x, y, z respectively.

According to question,

$$x + y + z = 40 \dots (i)$$

$$5x + 2y + z = 100 \dots (ii)$$

On subtracting equation (i) from (ii)

$$4x + y = 60$$

Now on satisfying the options we find

that only option (d) gives positive value of y and no. of coins must be positive.

Hence, no. of ₹5 coins = 13

Sol.7.(b) For the fourth proportion

$$\frac{\frac{1}{3} \times 27}{\frac{4}{5} \times 30} = \frac{\frac{3}{7} \times 35}{x} \Rightarrow \frac{9}{24} = \frac{15}{x}$$

$$\Rightarrow x = 40$$

Sol.8.(d)

Ratio of their concentration = $10 : 20 : 30$

Ratio of their quantity = $2 : 3 : x$

$\therefore$ Concentration of mixed solution

$$= \frac{10 \times 2 + 20 \times 3 + 30 \times x}{5 + x} = 23$$

$$20 + 60 + 30x = 115 + 23x \Rightarrow 7x = 35, x = 5$$

Sol.9.(a) Total students = 720

$$\text{No. of Boys} = \frac{3}{8} \times 720 = 270$$

$$\text{No. of Girls} = \frac{5}{8} \times 720 = 450$$

According to question

$$\frac{270 + x}{450 + 18} = \frac{2}{3}$$

$$= 810 + 3x = 2 \times 468 \Rightarrow 3x = 936 - 810$$

$$x = \frac{126}{3} = 42$$

Sol.10.(d) According to the question,

$$\text{Ram} = \frac{12}{5} = 2.4, \text{ Shyam} = \frac{18}{9} = 2$$

$$\text{Preetham} = \frac{4}{8} = 0.5, \text{ Vinay} = \frac{10}{7} = 1.42$$

Sol.11.(d) Let the number of plastic, wood and glass = $2x, 3x$ and $5x$

According to question,

$$= 5x - 3x = 144 \Rightarrow x = 72$$

$$\text{Number of plastic woods} = 2x = 144$$

Now,

$$\text{After increasing } 25\% = 144 \times \frac{5}{4} = 180$$

Sol.12.(a) Fourth proportional

$$= \frac{3^{12} \times 4^{12}}{12^{35}} \Rightarrow \frac{3^{12} \times 4^{12}}{3^{35} \times 4^{35}} = 3^{-23} \times 4^{-23} = 12^{-23}$$

Sol.13.(a) According to the question,

$$\frac{x + 5}{2x + 5} = \frac{2}{3} \Rightarrow x = 5$$

Smallest number (x) = 5

Sol.14.(d) Given, $y \propto (x + 3)$

$$y = k_1(x + 3) \dots (I)$$

When $y = 8$, and $x = 1$

$$\Rightarrow 8 = k_1 \times 4 \Rightarrow k_1 = 2$$

$$x \propto \frac{1}{z}$$

$$x = k_2 \times \frac{1}{z} \dots (II)$$

$$x = 2 \text{ and } z = 6.$$

$$\Rightarrow 2 = k_2 \times \frac{1}{6} \Rightarrow k_2 = 12$$

at $k_2 = 12$ equation (II) become

$$x = \frac{12}{z}, \text{ put this value in eq. (I)}$$

$$y = 2\left(\frac{12}{z} + 3\right) \Rightarrow y = 2\left(\frac{12 + 3z}{z}\right)$$

$$\Rightarrow y \times z = 24 + 6z \Rightarrow z(y - 6) = 24$$

$$\text{Sol.15.(a)} A : B = \frac{1}{3} : \frac{1}{5} = 5 : 3$$

$$B : C = \frac{1}{7} : \frac{1}{8} = 8 : 7$$

$$C : D = \frac{1}{10} : \frac{1}{21} = 21 : 10$$

$$\begin{array}{l} A : B : C : D \\ 5 : 3 : 3 : 3 \\ 8 : 8 : 7 : 7 \\ 21 : 21 : 21 : 10 \end{array}$$

$$40 : 24 : 21 : 10$$

Sol.16.(c) According to the question,
 $x + y = 96$ {let $x < y$ }

$$\text{So, } \frac{x+3}{96-x} = \frac{11}{20} \Rightarrow 20x + 60 = 96 \times 11 - 11x$$

$$31x = 996 \Rightarrow x = \frac{996}{31} = 32.12$$

$$\text{Largest number (y)} \\ = 96 - x = 96 - 32.12 \approx 63.87$$

Sol.17.(a) Given :-

$$5 : 3\frac{1}{3}, 6\frac{1}{4} : 3\frac{1}{8} \text{ and } 3.5 : 4.5$$

$$5 : 3\frac{1}{3} = 5 : \frac{10}{3} \Rightarrow 3 : 2 = 1.5$$

$$6\frac{1}{4} : 3\frac{1}{8} = \frac{25}{4} : \frac{25}{8} \Rightarrow 2 : 1 = 2$$

$$3.5 : 4.5 = 35 : 45 \Rightarrow 7 : 9 = 0.78$$

$$\text{Required ascending order} = 0.78 < 1.5 < 2$$

$$\text{OR } 3.5 : 4.5 < 5 : 3\frac{1}{3} < 6\frac{1}{4} : 3\frac{1}{8}$$

Sol.18.(b)

Let income of P, Q, and R is $9x$, $15x$ and $10x$
 and savings of P, Q, and R is $10y$, $15y$, $12y$
 According to the question,

$$10x - 12y = 10x \times \frac{60}{100}$$

$$10x - 12y = 6x \Rightarrow \frac{x}{y} = \frac{3}{1}$$

Ratio of expenditure of P, Q, and R

$$= (9x - 10y) : (15x - 15y) : (10x - 12y)$$

$$= (27 - 10) : (45 - 15) : (30 - 12)$$

$$= 17 : 30 : 18$$

Sol.19.(c) Fourth proportional

$$= \frac{(a^3 + b^3) \times (a^2 - b^2)}{(a + b)^2 \times (a^3 - b^3)}$$

$$= \frac{(a + b)(a^2 - ab + b^2) \times (a + b)(a - b)}{(a + b)^2(a - b)(a^2 + ab + b^2)}$$

$$= \frac{(a^2 - ab + b^2)}{(a^2 + ab + b^2)}$$

Sol.20.(c) According to the question,
 $(A \times 6\% + B \times 4\%)$

$$= (A \times 7\% + B \times 11\%) \times \frac{2}{5}$$

$$A \times 30\% + B \times 20\% = A \times 14\% + B \times 22\%$$

$$A \times 16\% = B \times 2\% \Rightarrow A : B = 1 : 8$$

Sol.21.(c)

$$\begin{array}{l} \text{Ratio} \rightarrow A : B \\ \text{Initial} \rightarrow 8x : 5x = 13 \\ \text{Final} \rightarrow 5x : 8x = 13 \end{array}$$

According to the question,
 Difference in A's share $8x - 5x = 300$

$$x = \frac{300}{3} = 100$$

$$\text{Total sum} = (8x + 5x) = 13x = ₹1300$$

Sol.22.(b)

$$\begin{aligned} \text{Required number} &= \frac{ad - bc}{(a + d) - (b + c)} \\ &= \frac{8 \times 9 - 6 \times 2}{(8 + 9) - (6 + 2)} = \frac{72 - 12}{9} = \frac{20}{3} \end{aligned}$$

$$\text{Now, } (3P - 6) = 14 \text{ and } (9P - 4) = 56$$

$$\text{Mean proportional} = \sqrt{14 \times 56} = 28$$

Sol.23.(c) $A : B : C = 3 : 2 : 5$

Difference between share of C and B

$$= 3 \text{ units} \Rightarrow 3 \text{ units} = ₹ 300$$

$$\text{Share of A more than B} = 1 \text{ units} = ₹ 100$$

Sol.24.(a) Mean proportional

$$\begin{aligned} &= \sqrt{(a + b)(a - b)^3 \times (a - b)(a + b)^3} \\ &= \sqrt{(a - b)^4 \times (a + b)^4} \\ &= (a - b)^2 \times (a + b)^2 \end{aligned}$$

Sol.25.(b)

Ratio - Maths : English : Chemistry

$$\text{Initial} - 6 : 4 : 7$$

$$\text{Final} - 7.2 : 5.6 : 11.2$$

$$\begin{aligned} \text{Required ratio} &= 7.2 : 5.6 : 11.2 \\ &= 9 : 7 : 14 \end{aligned}$$

Sol.26.(d)

$$\text{Ratio} - 0.5 ₹ : 0.25 ₹ : 0.20 ₹$$

$$\text{Coins} - 2 : 3 : 5$$

$$\text{Amount} - 1 : 0.75 : 1$$

$$\text{Required difference} = (1 - 1) = ₹ 0$$

Sol.27.(d)

$$\text{Blue}_{(\text{men})} : \text{Red}_{(\text{women})} = 3x : 7x ;$$

$$\text{Red}_{(\text{men})} : \text{Blue}_{(\text{women})} = 2y : y ;$$

$$\frac{3x + y}{7x + 2y} = \frac{35}{76}$$

$$\Rightarrow 228x + 76y = 245x + 70y \Rightarrow \frac{x}{y} = \frac{6}{17}$$

$$\text{Number of men} = 3x + 2y = 52$$

$$\text{Number of women} = 7x + y = 59$$

$$\text{Required ratio} = 52 : 59$$

Sol.28.(b)

$$\text{Ratio} \rightarrow P : Q$$

$$\text{Income} \rightarrow 40 : 16$$

$$\text{Expenditure} \rightarrow 18 : 10$$

$$\text{Savings} \rightarrow 11 : 3$$

Therefore, the ratio of the savings of P
 And Q = 11 : 3

Sol.29.(c) % decrease (X) in the number
 of units manufactured by company A

$$= \frac{12500 - 10625}{12500} \times 100$$

$$= \frac{1,875}{12500} \times 100 = 15\%$$

% decrease (Y) in the number of units
 manufactured by company

$$B = \frac{34,000 - 30,600}{34,000} \times 100$$

$$= \frac{3,400}{34,000} \times 100 = 10\%$$

$$\text{Required ratio} = 15 : 10 = 3 : 2$$

Sol.30.(b)

$$\text{Ratio} \rightarrow A : B : C$$

$$\text{No of officers} \rightarrow 5 : 4 : 1$$

$$\text{Average} \rightarrow (600) : (600+150) : (600+400)$$

So, the average of 40 officers $\rightarrow$

$$= 600 + \frac{(150 \times 4) + (400 \times 1)}{(5 + 4 + 1)}$$

$$= 600 + \frac{1000}{10} = 700$$

So, the monthly salary of an officer

$$= 700 \times 30 = ₹21,000$$

Sol.31.(d)

$$\text{Ratio} \rightarrow \text{Income} = \text{exp.} + \text{Savings}$$

$$A \Rightarrow 12 = 5 + 7 \Rightarrow 4 \times 5$$

$$B \Rightarrow 15 = 7 + 8 \Rightarrow 3 \times 4$$

Difference b/w their expenditure $\rightarrow$

$$(4 \times 5 \times 5) - (3 \times 4 \times 7) \text{ units} = 4500 \text{ Rs.}$$

$$16 \text{ units} = 4500 \text{ Rs.}$$

Income of B ($3 \times 4 \times 15$) units

$$= \frac{4500}{16} \times 180 = 50625 \text{ Rs.}$$

Sol.32.(a)

$$\text{Share of A} = \frac{18400}{46800} = \frac{46}{117} \text{ of the sum.}$$

Let total sum be $117a$, share of A = $46a$

$$\text{Share of A + D} = \frac{8}{13} \times 117a = 72a$$

$$\text{Share of D} = 72a - 46a = 26a$$

$$\text{Share of B + C} = \frac{5}{13} \times 117a = 45a$$

Given, share of B : share of C = 5 : 4

So, share of B = $25a$

and share of C = $20a$

x = difference of share of A and B

$$= 46a - 25a = 21a.$$

y = Difference of share of C and D

$$= 26a - 20a = 6a$$

$$x - y = 21a - 6a = 15a$$

$$117a = 46,800, a = 400$$

$$\text{So, } 15a = 15 \times 400 = 6000$$

Sol.33.(b) Let price of one pen = x

and price of one pencil = y

$$\text{So, } 5x + 3y = 44 \dots\dots(1)$$

$$3x + 5y = 36 \dots\dots(2)$$

On solving (1) and (2), we get

$$x = 7 \text{ and } y = 3$$

$$\text{So required ratio} = 7 : 3$$

Sol.34.(c)

$$\text{Denomination} \rightarrow ₹ 2 : ₹ 5 : ₹ 10$$

$$\text{Number of coins} \rightarrow 5 : 4 : 1$$

$$\text{Amount Ratio} \rightarrow 10 : 20 : 10$$

$$\text{Total Amount} = 40 \text{ unit} = 80 \text{ Rs}$$

$$1 \text{ unit} = 2 \text{ Rs.}$$

Number of 5 rupees coin = $4 \times 2 = 8$

Sol.35.(b) Number of boys = $\frac{5}{8}$

Boys Below 10 years = $\frac{5}{8} \times \frac{1}{3} = \frac{5}{24}$

Girls Below 10 years = $\frac{3}{8} \times \frac{2}{3} = \frac{6}{24}$

Students Below 10 years = $\frac{11}{24}$

Students Above 10 years = $\frac{13}{24}$

Total students = $260 \times \frac{24}{13} = 480$

Number of boys = $480 \times \frac{5}{8} = 300$

Sol.36.(a) Go through option (a)

$\frac{5+3}{7+5} = \frac{8}{12} = \frac{2}{3}$ (which satisfy)

$\frac{5-1}{7+3} = \frac{4}{10} = \frac{2}{5}$ (which satisfy)

Sol.37.(b)

Ratio $\Rightarrow$ Income = expenditure + saving

A (80%) $\Rightarrow 5 = 4 + 1 \times 8$

B (85%) $\Rightarrow 20 = 17 + 3 \times 3$

C (75%) $\Rightarrow 4 = 3 + 1 \times 20$

According to question, Difference of income of A and C (80 - 40) units

$\Rightarrow$ (40 units) = 18000 Rs.

Then, Income of B (60 units) = 27000 Rs

Sol.38. (a)

Ratio $\rightarrow$ A : B

Last year Income $\rightarrow$ 4 : 3

Now,

Last Year : Present Year

A $\rightarrow$ (3 : 4) $\times 4 \times 5$

B $\rightarrow$ (5 : 6) $\times 3 \times 3$

Last Year : Present Year

A $\rightarrow$ 60 : 80

B $\rightarrow$ 45 : 54

According to the question,

(80 + 54) units = 8.04 lakh

1 unit = 0.06 lakh

Last year income of B (45 units)

= $45 \times 0.06 = 2.7$ lakh

Sol.39.(b)

Fractional Value $\rightarrow 66\frac{2}{3}\% = \frac{2}{3}$

Ratio $\rightarrow$ A : B

Income $\rightarrow$ 5 : 7

Expenditure $\rightarrow$ 2 : 3

Savings $\rightarrow$ 4000 : 5000

Now,

$(5 \times 3) - (7 \times 2)$ units

= $(3 \times 4000) - (2 \times 5000)$

$\Rightarrow$ 1 unit = 2000 Rs.

Total income of A and B $\rightarrow$ (5 + 7) units

12 units = $12 \times 2000 = 24000$ Rs.

Sol.40. (d)

Ratio $\rightarrow$ Divisor : Quotient : Remainder

7 : 12
3 : 2

21 : 36 : 14

According to the question

14 unit = 14 $\Rightarrow$ 1 unit = 1

21 unit (Divisor) = 21

Required remainder = $\frac{21 \times 36 + 14}{9}$
= 5 (Rem.)

Sol.41.(c)

Let the age of shyam = 5 units and

age of Ravi = 3 units

According to the question

(5 + 3) unit = 32 years $\Rightarrow$ 1 unit = 4 years

Age of Shyam = $5 \times 4 = 20$ years and

Age of Ravi = $3 \times 4 = 12$ years

Now, $\frac{20+x}{12+x} = \frac{7}{5}$

$100 + 5x = 84 + 7x \Rightarrow x = 8$ years

Short Trick :-

Ratio $\Rightarrow$ Ravi : Shyam

Present $\Rightarrow$ 3 : 5

After $\Rightarrow$ 5 : 7

According to question,

Sum of present age (3 + 5) units = 32 yrs.

Then (2 units) = 8 yrs.

Sol.42.(a)

notes worth $\rightarrow$ 5 : 2 : 1

Amount $\rightarrow$ 15 : 2 : 3

No. of notes = $\frac{\text{Total amount}}{\text{note's worth}}$

Ratio of number of notes

= $\frac{15}{5} : \frac{2}{2} : \frac{3}{1} = 3 : 1 : 3$

Sol.43. (c) Actual ratio = 3 : 4 : 5 : 8

$\Rightarrow$ (3 + 4 + 5 + 8) unit = 4360

$\Rightarrow$ 20 unit = 4360 $\Rightarrow$ 1 unit = 218

A's share (3 unit) = $3 \times 218 = 654$

B's share (4 unit) = $4 \times 218 = 872$

C's share (5 unit) = $5 \times 218 = 1090$

D's share (8 unit) = $8 \times 218 = 1744$

Mistaken ratio = $\frac{1}{3} : \frac{1}{4} : \frac{1}{5} : \frac{1}{8}$

= 40 : 30 : 24 : 15

= (40 + 30 + 24 + 15) unit = 4360

= 1 unit = 40

A's share (40 unit) = $40 \times 40 = 1600$

B's share (30 unit) = $30 \times 40 = 1200$

C's share (24 unit) = $24 \times 40 = 960$

D's share (15 unit) = $15 \times 40 = 600$

Clearly D get 1744 - 600 = Rs.1144 less

Sol.44.(d)

Ratio $\rightarrow$ A : B : C

7 : 12

8 : 5

14 : 24 : 15

According to the question

(15 - 14) = 1 unit = 214

Sum (x) 53 unit = $(14 + 24 + 15) \times 214$
= 11342

Sol.45.(c) Let the price of the articles be 400 and 500.

According to the question,

$\frac{400 \times \frac{100+x}{100}}{500 \times \frac{100-30}{100}} = \frac{10}{7}$

$\Rightarrow \frac{4(100+x)}{5 \times 70} = \frac{10}{7}$

$\Rightarrow 400 + 4x = 500 \Rightarrow x = 25\%$

Sol.46.(a) Let age of three children are p year, q year, r year

According to question,

Man's age (M) = $4(p + q + r)$... (i)

After 6 Years

$\Rightarrow M + 6 = 2(p + 6 + q + 6 + r + 6)$

$\Rightarrow M + 6 = 2(p + q + r) + 36$

From eq (i)

$\Rightarrow 4(p + q + r) + 6 = 2(p + q + r) + 36$

$\Rightarrow 2(p + q + r) = 30 \Rightarrow (p + q + r) = 15$

$\therefore$ Man's Present age = $4 \times 15 = 60$ years

Sol.47.(c) Let number of votes got the contestants are $8x$, $7x$, $5x$

According to question,

$\Rightarrow 7x = 4725 \Rightarrow x = 675$

$\therefore$ Total number of valid votes (20x)

= $20 \times 675 = 13,500$

Sol.48.(d) Let the first part and second part be x and y respectively,

According to question,

$\Rightarrow \frac{141 \times \frac{x}{8}}{141 \times \frac{y}{9}} = \frac{5}{6}$

$\Rightarrow \frac{9x}{8y} = \frac{5}{6} \Rightarrow \frac{x}{y} = \frac{20}{27}$

Now, 47 units = 141

$\Rightarrow 20 \text{ units} = \frac{141}{47} \times 20 = 60$

Sol.49.(a)

Ratio of A's and B's salaries = 2 : 3

According to question,

A's expenditure = $\frac{1}{2} \times$ A's salary

$\Rightarrow (2x - \text{saving of A}) = \frac{1}{2} \times 2x$

$\Rightarrow$ Saving of A = x = Saving of B (given)

Expenditure of B = $3x - x = 2x$

Required percentage = $\frac{2x}{3x} \times 100$

= 66.67%

Sol.50.(c) Let the distribution of toffee among Pushkal, Quadir and Rustam be 2

x, 3x and 4x respectively.

According to question,

$\Rightarrow 9x = 360 \Rightarrow x = 40$

Pushkal = 80, Quadir = 120 and Rustam = 160

Then, Quadir = $120 \times \frac{20}{100} = 24$

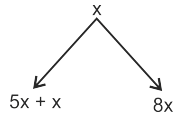
Remaining toffees of Quadir
 $= 120 - 24 = 96$

$$\text{Rustam} = 160 \times \frac{25}{100} = 40$$

Remaining toffees of Rustam
 $= 160 - 40 = 120$

Pushkal has toffees $= 80 + 24 + 40 = 144$
 Now, Pushkal : Quadir : Rustam
 $= 144 : 96 : 120$ or $6 : 4 : 5$

Sol.51.(c) Let no of invalid votes are $= x$
 According to question,



$$B = 6x, A = 8x$$

$$\Rightarrow (8x - 6x)\text{unit} = 28 \Rightarrow x = 14$$

$$\therefore \text{Total no of votes polled (14x)}$$

$$= 14 \times 14 = 196$$

Sol.52.(b) Mean proportional between p and $q = \sqrt{p \times q} = 8$
 Squaring on both sides $= p \times q = 64$
 By checking options, option (b) satisfies i.e. 16 and 4

Sol.53.(c)

Ratio	-	Rama	:	Ganga
Income	-	$4x$	:	$3x$
Expendi.	-	$3y$	:	$2y$

Savings - $4x - 3y$: $3x - 2y$

According to the question,

$$4x - 3y = 3x - 2y$$

$$x = y$$

savings of Rama $(4x - 3y)$

$$= 4x - 3x = x = ₹2700$$

So, expenditure of Rama $= 3x = ₹8100$

And expenditure of Ganga $= 2x = ₹5400$

Sol.54.(d)

Denomination $\rightarrow ₹0.5 : ₹0.2 : ₹0.1$

Number of coins $\rightarrow 2 : 3 : 4$

Amount Ratio $\rightarrow 1 : 0.6 : 0.4$

Total amount $= 1 + 0.6 + 0.4 = 2$ unit

According to the question,

$$2 \text{ unit} = ₹1020$$

So, number of 10 paise coins (4 unit)

$$= ₹1020 \times \frac{4}{2} = ₹2040$$

Sol.55.(a) Let x be the number added to each the four numbers 5, 19, 7 and 25

$$\text{Formula :- } x = \left| \frac{ad - bc}{(a + d) - (b + c)} \right|$$

$$\Rightarrow x = \frac{19 \times 7 - 5 \times 25}{(5 + 25) - (19 + 7)}$$

$$= \frac{133 - 125}{30 - 26} = \frac{8}{4} \Rightarrow x = 2$$

Sol.56.(d) In $49 : x :: x : 81$

$$x^2 = \sqrt{49 \times 81} \Rightarrow x = 63$$

In $64 : y :: y : 169$

$$y^2 = \sqrt{64 \times 169} \Rightarrow y = 104$$

Now

$$2x + 3y = 2 \times 63 + 3 \times 104$$

$$= 126 + 312 = 438$$

Sol.57.(c)

$$\text{Formula :- } x = \left| \frac{ad - bc}{(a + d) - (b + c)} \right|$$

$$\Rightarrow x = \left| \frac{2 \times 35 - 3 \times 30}{(2 + 35) - (3 + 30)} \right|$$

$$= \left| \frac{-20}{4} \right| = |-5| = 5$$

Now, second largest part of the proportion $= (30 + 5) = 35$

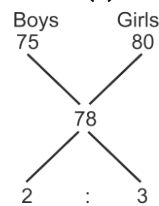
Sol.58.(c)

Required proportion $= 30 : 24 :: 35 : x$

$$30 \times x = 24 \times 35$$

$$x = \frac{24 \times 35}{30} = 28 \text{ litres}$$

Sol.59.(a)



Required ratio $= 2 : 3$

Sol.60.(d) According to the question,

$$\{A = 3B, C = B + 5\} \text{ ----- (i)}$$

$$A + B + C = 75 \text{ ----- (ii)}$$

By putting the value of A and C from eq

(i) to (ii) we get,

$$3B + B + (B + 5) = 75$$

$$5B = 70 \Rightarrow B = 14 \Rightarrow A = 3B = 42$$

$$\text{Hence, } (B - 5) : (A + 3) = 9 : 45 = 1 : 5$$

Sol.61.(c)

According to the question, $x \propto \frac{1}{y}$

$$x = k \times \frac{1}{y} \quad \{\text{where } k \text{ is constant}\}$$

Given that $x = 2$ and $y = 6$ we get;

$$2 = k \times \frac{1}{6} \Rightarrow k = 12$$

Now, for $x = 3$

$$3 = 12 \times \frac{1}{y} \Rightarrow y = 4$$

Sol.62.(a) Let the age of Soham be x

Then, age of Ravi $= 4x$

Age of Mayadevi $= 65 + x$

$$\text{Age of Ram} = \frac{1}{3}(65 + x)$$

According to question,

$$\frac{1}{3}(65 + x) - 4x = 7$$

$$\frac{65}{3} + \frac{x}{3} - 4x = 7 \Rightarrow -\frac{11x}{3} = 7 - \frac{65}{3}$$

$$\frac{11x}{3} = \frac{44}{3} \Rightarrow x = 4$$

So, the age of Mayadevi $= 65 + 4 = 69$ yrs

Sol.63.(b) According to question,

$$\frac{A}{C} = \frac{B}{D} \Rightarrow \frac{25}{30} = \frac{40}{D} \Rightarrow D = \frac{30 \times 40}{25} = 48$$

Sol.64.(b)

Denomination $\rightarrow ₹1 : ₹5 : ₹10$

No of coins $\rightarrow 3x : 5x : 7x$

Total amount in the bag

$$= 3x + 5 \times 5x + 10 \times 7x$$

$$= 3x + 25x + 70x = 98x$$

$$\text{Now, } 98x = 980 \Rightarrow x = 10$$

No of coins of ₹10 $= 7 \times 10 = 70$

Sol.65.(c) According to the question,

$(9 - 2)$ unit $= ₹560$

$$7 \text{ unit} = ₹560 \Rightarrow 1 \text{ unit} = ₹80$$

$$\text{Share of S} = 80 \times 3 = ₹240$$

Sol.66.(b) Let P's income $= 1000$

Q's income $= 1200$

R's income $= 900$

$$\text{Q's savings} = 1200 \times \frac{1}{4} = 300$$

$$\Rightarrow \text{Q's expenditure} = 1200 - 300 = 900$$

Ratio of expenditure

$$P : Q : R$$

$$12 : 15 : 8$$

$$\Rightarrow 720 : 900 : 480$$

Hence, P's savings $= 280$

Q's savings $= 300$

R's savings $= 420$

Ratio of savings $= 280 : 300 : 420$

$$= 14 : 15 : 21$$

Sol.67.(c) $2x + x + 3x = 1200$

$$\Rightarrow 6x = 1200 \Rightarrow x = 200$$

Now,

$$3(3x) + 2(x) = 11x = 11 \times 200 = 2200$$

Sol.68.(a) Let male $= 2x$

Female $= 3x$

New ratio

$$\frac{2x + 200}{3x} = \frac{5}{6} \Rightarrow 4x + 400 = 5x$$

$$\Rightarrow x = 400 \Rightarrow \text{Female} = 3 \times 400 = 1200$$

Sol.69.(c) Mean proportional of 12.96

$$\text{and } 0.16 = \sqrt{12.96 \times 0.16}$$

$$= \sqrt{3.6 \times 3.6 \times 0.4 \times 0.4}$$

$$= 3.6 \times 0.4 = 1.44$$

Sol.70.(a) Let the number of gold coins

and non gold coins be G and N respectively,

$$\Rightarrow G = \frac{1}{4} \times N \Rightarrow \frac{G}{N} = \frac{1}{4}$$

$G = x$ and $N = 4x$

According to the question,

$$\Rightarrow \frac{x + 20}{4x} = \frac{2}{3} \Rightarrow 3x + 60 = 8x \Rightarrow x = 12$$

Gold coins $= 12$

Non gold coins $= 4 \times 12 = 48$

Total number of coins in the collection

$$= 12 + 48 + 20 = 80$$

Sol.71.(b)

Ratio → Men : Women

$$\begin{array}{ccc} 5 & : & 3 \\ 5 & : & 4 \end{array} \quad \left. \begin{array}{c} \\ \\ \end{array} \right\} 1$$

According to question

⇒ 1 unit = 150

No of men employee in company (5 unit)
= 5 × 150 = 750**Sol.72.(b)** According to the question,

$$y \times \frac{18}{100} + b \times \frac{24}{100} = 30$$

$$\Rightarrow 3y + 4b = 500 \dots\dots(i)$$

$$\text{And, } 3y = 2b + 200$$

$$\Rightarrow 3y - 2b = 200 \dots\dots(ii)$$

Subtract equation (i) from equation (ii), we get

$$6b = 300 \Rightarrow b = 50$$

Put the value of b in equation (i)

$$3y + 200 = 500 \Rightarrow 3y = 300 \Rightarrow y = 100$$

Required ratio

$$y^2 : b^2 = 100 \times 100 : 50 \times 50 = 4 : 1$$

Sol.73.(a) Mean proportion = $\sqrt{pq}$

$$\Rightarrow \frac{1}{5a} = \sqrt{4a^2 \times x}$$

Taking square both side,

$$\Rightarrow \frac{1}{25a^2} = 4a^2 \times x \Rightarrow x = \frac{1}{100a^4}$$

Sol.74.(c) Given, Price $\propto$ (Weight)²

Let the weights of four pieces be 4x, 3x, 2x, 1x

∴ Total Price of four pieces

$$= 16x^2 + 9x^2 + 4x^2 + 1x^2 = 30x^2$$

Original weight of unbroken diamond

$$= 4x + 3x + 2x + 1x = 10x$$

Original price of unbroken diamond

$$= (10x)^2 = 100x^2$$

Difference between prices

$$= 100x^2 - 30x^2 = 70x^2 \Rightarrow 70x^2 = 63000$$

$$\Rightarrow 1x^2 = 900 \Rightarrow 100x^2 = ₹90,000$$

Sol.75.(c) According to the question,

$$\frac{5x - 40}{6x + 50} = \frac{8}{11} \Rightarrow 55x - 440 = 48x + 400$$

$$\Rightarrow 7x = 840 \Rightarrow x = 120$$

Hence, the number of girls

$$= 6 \times 120 = 720$$

Sol.76.(d) Mean proportional = $\sqrt{ab}$

$$\Rightarrow 30 = \sqrt{18 \times A}$$

Taking square both side,

$$\Rightarrow 900 = 18 \times A \Rightarrow A = 50$$

Sol.77.(b)

Ratio → P : Q

Expenditure → 36% of P : 56% of Q

$$\frac{\text{Expenditure of P}}{\text{Expenditure of Q}} = \frac{\frac{36}{100} \times \text{salary of p}}{\frac{56}{100} \times \text{salary of q}}$$

$$\frac{1.5}{1} = \frac{\frac{36}{100} \times \text{salary of p}}{\frac{56}{100} \times \text{salary of q}}$$

$$\frac{1.5}{1} = \frac{36 \times \text{salary of p}}{56 \times \text{salary of q}}$$

$$\frac{1.5 \times 56}{36} = \frac{\text{salary of p}}{\text{salary of q}}$$

$$\frac{\text{salary of p}}{\text{salary of q}} = \frac{7}{3} \text{ or } 7 : 3$$

$$\text{Sol.78.(c)} \frac{\text{Circumference}}{\text{diameter}} = \frac{3}{5}$$

$$\frac{6}{\text{diameter}} = \frac{3}{5} \Rightarrow \text{Diameter} = 10 \text{ cm}$$

Radius = 5 cm

Sol.79.(c) Given : a : b : c : d : e : f = 5 : 7

Now, (3a + 5c + 11e) : (3b + 5d + 11f)

$$(3 \times 5 + 5 \times 5 + 11 \times 5) : (3 \times 7 + 5 \times 7 + 11 \times 7)$$

$$(15 + 25 + 55) : (21 + 35 + 77)$$

$$95 : 133 \Rightarrow 5 : 7$$

Sol.80.(c)

Ratio → A : B : C

Income → 2 : 9 : 3

Expense → 3 : 9 : 5

Given condition, A's saving is half of his total income

To satisfy this condition, multiply income ratio by 3

Income → 6 : 27 : 9

Expense → 3 : 9 : 5

Saving → 3 : 18 : 4

Sol.81.(a)

Ratio - 5 : 10 : 20

$$3 : 5 : 7$$

Final - 15 : 50 : 140

Total amount = 15 + 50 + 140 = 205 units

205 units = ₹ 1845

(required worth) 15 units

$$= \frac{1845}{205} \times 15 = ₹ 135$$

Sol.82.(b)

Let the age of younger boy be x year

Then, age of older boy = 3x

Now, according to the question,

$$(3x - 9) = 5(x - 9)$$

$$3x - 9 = 5x - 45 \Rightarrow 2x = 36 \Rightarrow x = 18$$

So, the present age of each is 18 yr and 54 yr.

Sol.83.(b)

Ratio - ₹2 : ₹5 : ₹10

Number of coins - 5x : 2x : 50 - 7x

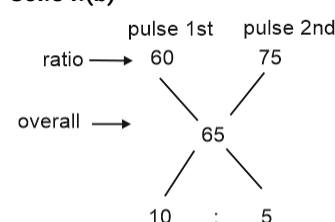
Total amount

$$= 10x + 10x + 500 - 70x = 150$$

$$50x = 350, x = 7$$

So the number of 10 rupees coins

$$= 50 - (7 \times 7) = 1$$

Sol.84.(b)

Required ratio = 2 : 1

Sol.85.(d) According to the question,

$$M \propto \frac{1}{(N^2 + 3)}$$

$$M = k \times \frac{1}{(N^2 + 3)} \dots\dots(i)$$

{where k is constant}

Putting M = 3 and N = 3 in equation (i)

$$3 = k \times \frac{1}{(3^2 + 3)} \Rightarrow k = 36$$

Now, value of M when given that N² = 69

$$M = 36 \times \frac{1}{(69 + 3)} \Rightarrow M = \frac{36}{72} = \frac{1}{2}$$

Sol.86.(d) Let numbers be 7x and 3x respectively,

According to the question,

$$7x - 3x = 48 \Rightarrow x = 12$$

Sum of numbers = 10x = 120

Sol.87.(d) 5 (Part of Ravi)

= 3 (Part of Reeta) = 11 (Part of Rahul)

$$\frac{\text{ravi}}{\text{reeta}} = \frac{3}{5}, \frac{\text{reeta}}{\text{rahul}} = \frac{11}{3}$$

Ratio - ravi : reeta : rahul

$$3 : 5 : 5$$

$$11 : 11 : 3$$

Final - 33 : 55 : 15

Sol.88.(d) According to the question,

$$\frac{3}{4} = 0.75, \frac{5}{8} = 0.625, \frac{1}{2} = 0.5,$$

$$\frac{6}{9} = 0.666$$

We can clearly see that the greatest ratio is 3 : 4

Sol.89.(a)

Ratio - a : b : c

$$3 : 7 : 7$$

$$4 : 4 : 9$$

Final - 12 : 28 : 63

Sol.90.(a) $\frac{8}{11}, \frac{4}{5}, \frac{1}{2}$ and $\frac{1}{11}$

Denominators of given fractions are 11, 5, 2 and 11

LCM (11, 5, 2, 11) = 110

Multiply each fraction by 110 we get fractions

$$\frac{8}{11} \times 110, \frac{4}{5} \times 110, \frac{1}{2} \times 110, \frac{1}{11} \times 110$$

$$= 80, 88, 55 \text{ and } 10$$

Here, 88 is the largest among all

Hence, required fraction = $\frac{4}{5}$ **Sol.91.(c)** Value of x = $\left| \frac{ad - bc}{(a + d) - (b + c)} \right|$

$$= \left| \frac{7 \times 23 - 11 \times 18}{(7 + 23) - (11 + 18)} \right|$$

$$= \left| \frac{161 - 198}{(30) - (29)} \right| = \left| \frac{-37}{1} \right| = 37$$

Then, mean proportion of $(x - 1)$ and $(2x - 10) = \sqrt{(x - 1)(2x - 10)}$

$$= \sqrt{(37 - 1)(74 - 10)}$$

$$= \sqrt{36 \times 64} = 6 \times 8 = 48$$

Sol.92.(b) Required ratio = $\frac{28}{100} = 7 : 25$

Sol.93.(a) According to the question,
Let the weight of the box and the gift articles is x and $9x$ gram respectively.

$$x + 9x = 25 \times 1000$$

$$10x = 25000 \Rightarrow x = 2500$$

So the weight of the articles will be
 $9 \times 2500 = 22500$ gram

Sol.94.(d) Third proportion

$$= \frac{(48)^2}{36} = 8 \times 8 = 64$$

Sol.95.(c)

Ratio $\rightarrow a : b : c : d$

$$3 : 4 : 4 : 4$$

$$2 : 2 : 3 : 3$$

$$5 : 5 : 5 : 6$$

$$30 : 40 : 60 : 72$$

Required ratio $\rightarrow 15 : 20 : 30 : 36$

Sol.96.(b) According to the question,

$$M : N : O = \left(\frac{1}{3} : \frac{1}{4} : \frac{1}{5}\right) \times 60$$

$$M : N : O = 20 : 15 : 12$$

Sol.97.(a) Mean proportion

$$= \sqrt{0.14 \times 2.24} = \sqrt{\frac{14}{100} \times \frac{224}{100}}$$

$$= \sqrt{\frac{14}{100} \times \frac{14 \times 16}{100}} = \frac{56}{100} = 0.56$$

Sol.98.(b) Let the fourth proportion be x

$$\Rightarrow \frac{1.25}{5.75} = \frac{3.5}{x} \Rightarrow x = \frac{3.5 \times 5.75}{1.25} = 16.1$$

Sol.99.(a) Sum of numbers = 5 unit = 135
1 unit = 27

$$\text{So, numbers} = 2 \times 27 = 54, 3 \times 27 = 81$$

Sol.100.(d) Mean proportional = $\sqrt{ab}$

Mean proportional of 25 and 81

$$= \sqrt{25 \times 81} = 45$$

Sol.101.(a) Let third number = 100

Then, two number = $100 \times 130\% = 130$
and $100 \times 160\% = 160$

Ratio of two numbers = $130 : 160$
 $= 13 : 16$

Sol.102.(b) Required ratio

$$= (640 - 400) : 400 = 240 : 400 = 3 : 5$$

Sol.103.(c) $15\% = \frac{3}{20}, 25\% = \frac{1}{4}$

Sonu : Raju : Ramu

$$20 : 23$$

$$4 : 5 \times 5$$

$$20 : 23 : 25$$

Required ratio = $23 : 25$

Sol.104.(b) Given :- $x : y = 5 : 6$

$$2x + 3y : 3x + 5y$$

$$= 2(5) + 3(6) : 3(5) + 5(6)$$

$$= 10 + 18 : 15 + 30 = 28 : 45$$

Sol.105.(b)

$$\frac{7}{4} = 1.7, \frac{8}{3} = 2.6, \frac{2}{3} = 0.6, \frac{1}{5} = 0.2$$

Clearly, we can see $\frac{8}{3}$ is the greatest of all the fractions given.

Sol.106.(d) No of red sweets in bag

$$= \frac{3}{4} \times 120 = 90$$

Sol.107.(a) $0.72 : x :: x : 2.85$

Product of extreme term = product of mean term

$$0.72 \times 2.85 = x \times x$$

$$x^2 = 2.052 \Rightarrow x = 1.4$$

Sol.108.(b) According to question,

$$\frac{5+x}{11+x} = \frac{3}{5}$$

$$25 + 5x = 33 + 3x \Rightarrow 2x = 8 \Rightarrow x = 4$$

Sol.109.(a) Let the no. be $3x$ and $7x$

According to question,

$$\frac{3x-11}{7x-11} = \frac{7}{20}$$

$$60x - 220 = 49x - 77 \Rightarrow 11x = 143 \Rightarrow x = 13$$

So, the smaller number = $3x = 3 \times 13 = 39$

Sol.110.(c)

$$40\% \text{ of } A = 60\% \text{ of } B = \frac{1}{6} \text{ of } C$$

$$\frac{2}{5} \text{ of } A = \frac{3}{5} \text{ of } B = \frac{1}{6} \text{ of } C = k \text{ (Let)}$$

$$A = \frac{5}{2}k, B = \frac{5}{3}k, C = 6k$$

$$A : B : C = \frac{5}{2}k : \frac{5}{3}k : 6k = 15 : 10 : 36$$

Sol.111.(c)

Ratio $\rightarrow A : B : C$

$$3 : 4$$

$$4 : 5$$

$$\text{Final} \rightarrow 3 : 4 : 5$$

$$C - A = 5 - 3 = 2 \text{ unit}$$

$$2 \text{ unit} = 30 \text{ marks}$$

$$\text{Marks of B (4 unit)} = \frac{30}{2} \times 4 = 60 \text{ marks}$$

Sol.112.(d)

Ratio $\rightarrow$ Jeevan : Praveen : Kanan

$$5 : 8 :$$

$$3 : 2$$

$$\text{Final} \rightarrow 15 : 24 : 16$$

Share of Jeevan

$$= \frac{15}{15+24+16} \times 17,600$$

$$= \frac{15}{55} \times 17,600 = 4,800$$

Sol.113.(d)

$$\text{Mean proportion} = \sqrt{75 \times 12} = 30$$

$$\text{Required natural number} = 30 - 13 = 17$$

Sol.114.(b) According to question,

$$a : b :: c : d \Rightarrow \frac{a}{b} = \frac{c}{d}$$

$$\text{Option (a)} :- b, a, d \text{ and } c, \Rightarrow \frac{b}{a} = \frac{d}{c}$$

$$\Rightarrow \frac{a}{b} = \frac{c}{d} \text{ (true)}$$

$$\text{Option (b)} :- c, d, b \text{ and } a, \Rightarrow \frac{c}{d} = \frac{b}{a}$$

$$\Rightarrow \frac{a}{b} \neq \frac{c}{d} \text{ (false)}$$

$$\text{Option (c)} :- d, c, b \text{ and } a, \Rightarrow \frac{d}{c} = \frac{b}{a}$$

$$\Rightarrow \frac{a}{b} = \frac{c}{d} \text{ (true)}$$

$$\text{Option (d)} :- a, c, b \text{ and } d, \Rightarrow \frac{a}{c} = \frac{b}{d}$$

$$\Rightarrow \frac{a}{b} = \frac{c}{d} \text{ (true)}$$

Sol.115.(c)

$$\text{Mean proportion} = \sqrt{ab} = \sqrt{a \times \frac{4}{a}} = 2$$

Sol.116.(d) $S = 75\% \text{ of } T$

$$\frac{S}{T} = \frac{75}{100} = \frac{3}{4} \Rightarrow S : T = 3 : 4$$

$$\text{Sol.117.(a)} 20\% = \frac{1}{5}$$

Girls = 5 unit, Boys = 6 unit

required ratio of Boy to Girls = $6 : 5$

Sol.118.(d) $42\% \text{ of Ranita's income} = 56\% \text{ of Bhaskar's income}$

$$\frac{\text{Income of Ranita}}{\text{Income of Bhaskar}} = \frac{56}{42} = \frac{4}{3}$$

According to question,

$$\frac{4x}{3x+200} = \frac{5}{4} \Rightarrow 16x = 15x + 1000$$

$$x = 1000$$

Hence, Ranita's Income

$$= 4 \times 1000 = 4000$$

$$\text{Sol.119.(a)} 25\% = \frac{1}{4}$$

Money left after saving

$$= \frac{3}{4} \times 40,000 = 30,000$$

Money spent on education

$$= \frac{2}{5} \times 30,000 = 12,000$$

$$\text{Required \%} = \frac{12000}{40000} \times 100 = 30\%$$

Sol.120.(a) No of students in high school

$$= \frac{324}{4} \times 3 = 243$$

Sol.121.(c) Ratio of boys and girls in class A = $11 : 18 = 0.61$

Ratio of boys and girls in class

$$B = 13 : 17 = 0.76$$

Ratio of boys and girls in class

$$C = 20 : 9 = 2.22$$

⇒ class C has the biggest ratio of boys to girls.

Sol.122.(b)

Ratio → cheese : butter
4 : 3

$$\Rightarrow 4 \text{ units} = 8$$

$$\Rightarrow 3 \text{ units} = \frac{8}{4} \times 3 = 6$$

So, she used 6 bowls of butter.

Sol.123.(c)

	S	:	F
Salaries →	2	:	3
Savings →	3	:	4

Since, total savings are equal to the salary of S.
Total savings = 3 + 4 = 7.

Now,

	S	:	F
Salaries →	2	:	3
			(× 7)
Savings →	3	:	4
			(× 2)

Hence,

	S	:	F
Salaries →	14	:	21
Savings →	6	:	8

Expenditure →	8	:	13
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Sol.124.(b) $S \propto (R + 7)$

$$S = k(R + 7)$$

Putting, $S = 42$ and $R = 17$,

$$42 = k(17 + 7) \Rightarrow 42 = 24k \Rightarrow k = \frac{42}{24} = \frac{7}{4}$$

Now, putting $R = 29$ in the above expression, we have ;

$$S = k(R + 7) \Rightarrow S = \frac{7}{4}(29 + 7)$$

$$\Rightarrow S = \frac{7}{4} \times 36 = 63$$

Sol.125.(c) $8 - 7 = 1 \text{ unit} = ₹4800$

$$7 \text{ unit} = 4800 \times 7 = ₹33,600$$

So, the price of the washing machine = ₹33,600

Sol.126.(c) Let

Rupees with Pooja = $x + 50$ Rs.

Rupees with Chameli = x Rs.

Rupees with Anju = $2(x + 50)$ Rs.

Total Rupees = $x + 50 + x + 2x + 100$
= 450 Rs.

$$\Rightarrow 4x + 150 = 450 \Rightarrow 4x = 300 \Rightarrow x = 75$$

Hence,

Pooja's Rupees = $75 + 50 = 125$ Rs.

Chameli's Rupees = 75 Rs.

Anju's Rupees = $2(75 + 50) = 250$ Rs.

Sol.127.(b) Let first number = $4x$

Second number = $5x$

According to question,

$$\frac{4x + 3}{5x - 7} = \frac{1}{1} \Rightarrow 4x + 3 = 5x - 7 \Rightarrow x = 10$$

Hence, First number = 40

And second number = 50

Sol.128.(c) Mean proportional

$$= \sqrt{45x^4 \times 5y^2} = 15x^2y$$

Sol.129.(a) According to the question,

$$\Rightarrow \frac{4x + 8}{7x + 8} = \frac{2}{3} \Rightarrow 12x + 24 = 14x + 16$$

$$\Rightarrow 2x = 8 \Rightarrow x = 4$$

Present age of A and B will be 16 yrs and 28 yrs respectively,

Now, sum of the age of A and B = $16 + 28$
= 44 years

Alternate method :-

	A	:	B
Present Age →	4	:	7×1
	4	:	7
Age after 8 yrs →	2	:	3×3
	6	:	9

Diff. 2

$$\Rightarrow 2 \text{ units} = 8 \Rightarrow 1 \text{ unit} = 4$$

$$\Rightarrow 11 \text{ units} = 44 \text{ years}$$

Sol.130.(d) Given: $\frac{3 + x}{11 + x} = \frac{4 + x}{14 + x}$

By putting option one by one, we get
 $x = 1$

$$\text{As } \frac{3 + 1}{11 + 1} = \frac{4 + 1}{14 + 1} \Rightarrow \frac{5}{12} = \frac{5}{17}$$

Sol.131.(a) Let first number = $4x$

2nd number = $5x$

$$\text{Given } \frac{4x - 17}{5x - 17} = \frac{11}{14}$$

$$\Rightarrow 56x - 238 = 55x - 187 \Rightarrow x = 51$$

$$\Rightarrow \text{original ratio} = \frac{51 \times 4}{51 \times 5} = \frac{204}{255}$$

Hence,

$$\text{Required ratio} = \frac{204 + 20}{255 + 20}$$

$$= \frac{224}{275} \text{ or } 224 : 275$$

Sol.132.(c)

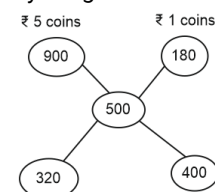
Let all coins are of denomination ₹5

$$\Rightarrow \text{Total value} = 180 \times 5 = 900 ₹$$

Let all coins are of denomination ₹1

$$\Rightarrow \text{Total value} = 180 \times 1 = 180$$

By allegation



$$\Rightarrow 4 : 5$$

$$\Rightarrow 9 \text{ unit} \rightarrow 180 \Rightarrow 1 \text{ unit} \rightarrow 20$$

Hence,

the number of ₹5 coins in the bag
= $4 \times 20 = 80$.

Sol.133.(d) Let the third proportional = x

$$\frac{9}{7} = \frac{x}{14} \Rightarrow 9 \times 14 = 7x \Rightarrow x = 18$$

Sol.134.(d) Fourth proportional of 42, 56,

$$63 = \frac{63 \times 56}{42} = 84$$

Fourth proportional of 189, 273, 153

$$= \frac{153 \times 273}{189} = 221$$

Required ratio = 84 : 221

Sol.135.(d) According to the question,

$$\frac{9}{16}R = \frac{3}{4}S \Rightarrow \frac{R}{S} = \frac{3}{4} \times \frac{16}{9} = \frac{4}{3}$$

Now, $4 + 3 = 7 \text{ unit} = ₹4,760$

$$\text{Then, } 3 \text{ unit} = \frac{4760}{7} \times 3 = ₹2040$$

Sol.136.(c) Cyrus Rohan Mishti

3	:	2	:	2
7	:	7	:	12

21	:	14	:	24
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According to the question,

$$24 - 21 = 3 \text{ unit} = ₹1800$$

$$\text{Then, } 14 \text{ unit} = \frac{1800}{3} \times 14 = ₹8400$$

Sol.137.(d) $(5^2 - 3^2) = 16, 5 - 3 = 2$

Third proportional to 16 and 2

$$= \frac{2 \times 2}{16} = \frac{1}{4}$$

Sol.138.(c) According to question,

$$b = 35 + a \text{ ----- (i)}$$

$$d = 60 + c \text{ ----- (ii)}$$

Since a, b, c, d are in proportion. Then we have ;

$$\frac{a}{b} = \frac{c}{d}$$

$$ad = bc$$

$$a(60 + c) = (35 + a)c \text{ [from eqn(i)&(ii)]}$$

$$60a + ac = 35c + ac \Rightarrow 60a = 35c$$

$$\frac{a}{c} = \frac{35}{60} = \frac{7}{12}$$

Let a and c be $7x$ and $12x$

$$\text{Then, } 7x \times 12x = 5376$$

$$84x^2 = 5376 \Rightarrow x^2 = \frac{5376}{84} = 64 \Rightarrow x = 8$$

$$\text{Then, } a = 7 \times 8 = 56, b = 35 + 56 = 91,$$

$$c = 12x = 12 \times 8 = 96, d = 60 + 96 = 156$$

$$\text{So, } a + b + c + d = 56 + 91 + 96 + 156 = 399$$

Sol.139.(d)

Types of coins ₹1 50P 25P

No of coins 3 : 4 : 6

$$\text{Total amount} = 3 \times 1 + 4 \times \frac{1}{2} + 6 \times \frac{1}{4}$$

$$= 3 + 2 + 1.5 = 6.5$$

$$6.5 \text{ unit} = ₹143$$

$$4 \text{ unit} = \frac{143}{6.5} \times 4 = 88$$

Sol.140.(c) According to question, we have given ratio;

A	B	C	D
1	: 2	: 2	: 2
3	: 3	: 1	: 1
2	: 2	: 2	: 3

$$6 : 12 : 4 : 6 = 3 : 6 : 2 : 3$$

$$\text{Value of C} = \frac{2}{3+6+2+3} \times 875$$

$$= \frac{1750}{14} = 125$$

Sol.141.(b) Ratio of expenditure and saving = 2 : 1, So income = 3 unit
Now,

$$\begin{aligned}\text{Income} &\rightarrow 3 \times 20\% = 60\% \\ \text{Expenditure} &\rightarrow 2 \times 15\% = 30\% \\ \text{Saving} &\rightarrow 1 \times ? = 30\%\end{aligned}$$

$$\therefore ? = \frac{30}{1} = 30\%$$

% increase in her saving = 30%

Sol.142.(a)

Expenditure = price $\times$ consumption

Ratio of price = 3 : 4

Ratio of quantity = 4 : 7

$\therefore$ Ratio of expenditure = 12 : 28 or 3 : 7

Sol.143.(c)

$$A : B = 3 : 4$$

$$B : C = 12 : 13$$

$$A : B : C = 9 : 12 : 13$$

Here total amount = 34 unit = ₹ 7786

$\therefore$ B's amount = 12 unit

$$= \frac{7786}{34} \times 12 = ₹ 2748$$

Sol.144.(b)

$$\text{Area of triangle} = \frac{1}{2} \times \text{base} \times \text{height}$$

$$\text{Area} \rightarrow 4 : 3$$

$$\text{Height} \rightarrow 6 : 5$$

$$\text{Then ratio of bases} \rightarrow \frac{4}{6} : \frac{3}{5} = 10 : 9$$

Sol.145.(c) According to the question,

$$\frac{X}{Y} = \frac{3}{1} \text{ (put the value in equation)}$$

$$\frac{2X + 3Y}{X - 2Y} = \frac{2 \times 3 + 3 \times 1}{3 - 2 \times 1} = 9$$

$$\text{Sol.146.(b)} P \propto \frac{1}{Q} \Rightarrow P = k \times \frac{1}{Q}$$

$$10 = k \times \frac{1}{6} \Rightarrow k = 60$$

$$\text{now, } 15 = 60 \times \frac{1}{Q} \Rightarrow Q = 4$$

Sol.147.(d) Mean proportion

$$= \sqrt{(3 + 2\sqrt{2})^2 \times (3 - 2\sqrt{2})^2}$$

$$= (3 + 2\sqrt{2}) \times (3 - 2\sqrt{2}) \Rightarrow 9 - 8 = 1$$

Sol.148.(c) According to the question,

$$\text{Ratio - } A : B : C$$

$$\text{Profit - } 15x : 6x : 5x$$

$$\text{B's share} = 29120 \times \frac{6x}{15x + 6x + 5x}$$

$$= ₹ 6720$$

Sol.149.(b)

Asha : Bhupinder : Gargi

$$\begin{array}{ccc} 9 & : & 5 \\ 2 & : & 2 \end{array}$$

$$18 : 10 : 5$$

Hence, shares of Asha

$$= \frac{66000}{(18 + 10 + 5)} \times 18 = ₹ 36,000$$

Sol.150.(a) According to the question,

$$5x - 2x = ₹ 150 \Rightarrow x = ₹ 50$$

$$\text{Hence, total sum} = (2x + 5x + 7x) = 14x$$

$$= ₹ 700$$

Sol.151.(d) $p : q = 5 : 2$,

$$\text{then } p^2 - q^2 : p^2 + q^2 = 25 - 4 : 25 + 4$$

$$= 21 : 29$$

Sol.152.(b)

Marks obtained by the student after

$$\text{studying for one hour} = \frac{60}{4} = 15 \text{ Marks}$$

So the time taken by him to study to get

$$75 \text{ marks} = \frac{75}{15} = 5 \text{ hours}$$

Sol.153.(d) According to the question

Let the numbers be $5x, 2x, 7x$

According to the question,

$$(5x)^2 + (2x)^2 + (7x)^2 = 3822$$

$$= 25x^2 + 4x^2 + 49x^2 = 3822$$

$$= 78x^2 = 3822 \Rightarrow x = 7$$

$$\text{So the largest no.} = 7x = 7 \times 7 = 49$$

Sol.154.(b) According to the question,

$$(19 + 6) - (7 + 15) \text{ units} = 72$$

$$3 \text{ units} = 72$$

Smallest number (6 units)

$$= \frac{72}{3} \times 6 = 144$$

Sol.155.(b) Let the savings of T = 1 unit

Then income of T = 8 units

Expenditure of T = $(8 - 1) = 7$ units

Hence, expenditure of T is 7 times of his savings

Sol.156.(d) Total students = 1650

Ratio $\rightarrow$ Boys : Girls

$$6 : 5$$

$$\text{Number of boys} = 1650 \times \frac{6}{11} = 900$$

$$\text{Number of girls} = 1650 \times \frac{5}{11} = 750$$

After Admission -

$$\text{Total boys} = 900 + 124 = 1024$$

New Ratio $\rightarrow$ Boys : Girls

$$4 : 3$$

$$4 \text{ unit} = 1024, 3 \text{ unit} = 768$$

$$\text{Newly admitted girls} = 768 - 750 = 18$$

$$\text{So, required ratio} = 18 : 124 = 9 : 62$$

Sol.157.(a) According to the question,

$$\frac{x + 20}{2x} = \frac{1}{1} \Rightarrow x = 20$$

$$\text{Number of student initially} = 3x = 60$$

$$\text{Sol.158.(c)} \sqrt{A^2 + B^2} : \sqrt{C^2 - B^2}$$

$$= \sqrt{9 + 16} : \sqrt{25 - 16} \Rightarrow \sqrt{25} : \sqrt{9} = 5 : 3$$

Sol.159.(a)

$$3C + 5P = 1080 ₹ \dots \text{e.q. (1)}$$

$$2C + 4P = 840 ₹ \dots \text{e.q. (2)}$$

$$C + P = 240 ₹$$

$$\Rightarrow 2C + 2P = 480 ₹ \dots \text{e.q. (3)}$$

On solving e.q. (3) and e.q. (2), we get

$$2P = 360 \Rightarrow P = 180 ₹ \text{ and } C = 60 ₹$$

$$\text{Required ratio} = 1 : 3$$

Sol.160.(c) Compound ratio :-

$$\frac{3 \times 6}{4 \times 7} = \frac{36}{x} \Rightarrow x = 56$$

Sol.161.(d) Mean proportional (25) = $\sqrt{ab}$

$$\Rightarrow a = \frac{625}{b}$$

$$\text{Third proportional (25)} = \frac{b^2}{a}$$

$$\Rightarrow b^2 = 25a = 25 \times \frac{625}{b}$$

$$\Rightarrow b = 25 \text{ and } a = 25$$

$$\text{Sol.162.(a)} \frac{2x + 8.5}{x + 8.5} = \frac{7}{4}$$

$$x = 8.5 \times 3 = 25.5 \text{ years}$$

Sol.163.(c) Required ratio

$$= \left(\frac{23 + 7}{2}\right) : \left(\frac{23 - 7}{2}\right) = 15 : 8$$

$$\text{Sol.164.(d)} \frac{x + 7}{2x + 7} = \frac{3}{5}$$

$$x = 7 \times 2 = 14$$

$$\text{Greatest number (2x)} = 28$$

Sol.165.(a)

$$\text{Mean proportional} = \sqrt{7 \times 63} = 21$$

$$\text{Required difference} = 24 - 21 = 3$$

Sol.166.(c)

$$\frac{(6a^2 - 2b^2)}{(b^2 - a^2)} = \frac{(6 \times 25 - 2 \times 49)}{(49 - 25)}$$

$$= \frac{(150 - 98)}{(24)} = \frac{(52)}{(24)} = \frac{(13)}{(6)} = 13 : 6$$

Sol.167.(d)

$$\text{Fourth proportional} = \frac{3\sqrt{10} \times 5\sqrt{8}}{3\sqrt{5}}$$

$$= \sqrt{2} \times 5\sqrt{8} = 5 \times 4 = 20$$

Sol.168.(a)

Let their age be $5x$ and $3x$, then

$$5x \times 3x = 960 \Rightarrow x = 8$$

Their present age = 40 years and 24 years

$$\text{Required ratio} = 40 + 6 : 24 + 6 = 23 : 15$$

Sol.169.(a) Given numbers are in

$$\text{proportional, so } \frac{8}{P} = \frac{16}{18} \Rightarrow P = 9$$

$$\text{Third proportional} = \frac{12 \times 12}{9} = 16$$

Sol.170.(d) Number of male worker
 $= 960 - 420 = 540$
 Required ratio $= 540 : 420 = 9 : 7$

Sol.171.(d) Given:- $x = 45$, $y = 9$, $x = Ky$
 $45 = K \times 9$, $K = 5$
 Now, $x = ?$, $y = 50$
 $x = 5 \times 50 = 250$

Sol.172.(d)
 Ratio - X : Y : Z
 $4 : 5 : 5$
 $7 : 7 : 6$

Final - 28 : 35 : 30

Share of Z $= \frac{1488}{(28 + 35 + 30)} \times 30 = ₹480$

Sol.173.(a) Ratio :- Boys : Girls = 11 : 15
 difference between the number of boys and girls (4 units) = 200

No. of boys (11 units) $= \frac{200}{4} \times 11$
 $= 550$ boys

Sol.174.(c)
 Ratio - Income = expenditure + savings
 Initial - 80 = 50 + 30
 Final - 96 = 65 + 31
 Change % in savings $= \frac{1}{30} \times 100$
 $= 3\frac{1}{3}\%$ increased

Sol.175.(d) 48% of male workers
 $= \frac{2}{3} \times \text{female worker}$
 $\frac{\text{Male workers}}{\text{Female workers}} = \frac{2}{3} \times \frac{100}{48}$
 $= \frac{100}{72} = \frac{25}{18}$

So, required ratio :-
 Male worker : female worker = 25 : 18

Sol.176.(b) Let the third number be 100.
 According to question,
 The two numbers = 117 and 150
 Required ratio = 117 : 150 = 39 : 50

Sol.177.(b)
 According to question, $\frac{96}{a} = \frac{c}{96}$
 Going through all the options, while putting option (b)
 $a = 32$ and $c = 288$
 $\frac{96}{32} = \frac{288}{96} \Rightarrow 3 = 3$ [satisfy]

Sol.178.(c)
 Ratio of no. $= \frac{4}{5} : \frac{5}{6} : \frac{9}{10} = 24 : 25 : 27$
 Difference between greatest and smallest $= (27 - 24)$ units $\Rightarrow 3$ units = 12
 Middle number (25 units) $= \frac{12}{3} \times 25 = 100$

Sol.179.(a) Let the length of colored

saffron, white and green be $6x$, $7x$ and $8x$
 According to question,
 $6x + 7x + 8x = 9030 \Rightarrow x = 430$
 Length of green part $= 8x = 8 \times 430$
 $= 3440$ cm

Sol.180.(d) Let the number of storey be x
 According to question,
 $\frac{28}{7} = \frac{48}{x} \Rightarrow x = 12$

Sol.181.(a)
 Let the number subtracted be x .
 $\frac{(2-x)}{(3-x)} = \frac{(3-x)}{4}$
 $\Rightarrow 8 - 4x = 9 + x^2 - 6x$
 $\Rightarrow x^2 - 2x + 1 = 0 \Rightarrow (x-1)(x-1) = 0$
 $\Rightarrow$ (smallest possible number) $x = 1$

Sol.182.(a) Third proportional to $3x^2$ and $4xy$
 $4xy = \frac{4xy \times 4xy}{3x^2} = \frac{16}{3}y^2$
 $\frac{16}{3}y^2 = 48 \Rightarrow y^2 = \frac{48}{16} \times 3 = 9 \Rightarrow y = 3$

Sol.183.(d)
 Ratio $\rightarrow$ Cheese : sauce = 2 : 1
 According to the question,
 Sauce used to make pizza (1 unit)
 $= 15$ cups
 Cheese to be used (2 unit) $= 30$ cups
Sol.184.(a) Let total Income = 100 units
 Then, Savings = 15 units
 Expenditure on health, education and food be 17 units, 34 units and 34 units respectively.
 According to the question,
 $3 \times (34 \text{ units}) - 100 \text{ units} = 300 \text{ Rs.}$
 $2 \text{ units} = 300 \text{ Rs.}$
 Required difference
 $= (17 \text{ units} - 15 \text{ units}) = 2 \text{ units} = 300 \text{ Rs.}$

Sol.185.(c) $M = \sqrt{16 \times 9} = 12$
 $N = \frac{15^2}{9} = 25$
 $5M + 3N = 5 \times 12 + 3 \times 25 = 60 + 75 = 135$

Sol.186.(d) Ratio of manufacturing $\rightarrow$
 Plate : Glasses = 2 : 1
 and Glasses : Bowl = 2 : 1
 Plate : Glasses : Bowl = 4 : 2 : 1
 According to the question,
 Total manufacturing (7 units) = 2380
 Required difference (3 units)
 $= \frac{2380}{7} \times 3 = 1020$ units

Sol.187.(a) Let the numbers be $7x$ and $6x$
 According to question,
 $(7x)^2 - (6x)^2 = 637$
 $= 49x^2 - 36x^2 = 637$
 $= x^2 = 49 \Rightarrow x = 7$
 Therefore, the numbers are 49 and 42.

Sol.188.(b) Ratio of Students $\rightarrow$

Initially = 2 : 3 : 5
 Finally = 4 : 5 : 7
 Increase in ratio (2 units) = 20 students
 Initially total students (10 units) = 100 students

Sol.189.(c) Let the marks of practical and theory be 300 and 700.
 According to question,
 Required ratio
 $= (700 \times \frac{60}{100}) : (300 \times \frac{80}{100})$
 $= 420 : 240 = 7 : 4$

Sol.190.(b) $a : b : c = 4 : 7 : 9 \times 4$
 $\Rightarrow a : b : c = 16 : 28 : 36$
 and $b : c : d = 28 : 36 : 21$
 or $a : b : c : d = 16 : 28 : 36 : 21$

Sol.191.(b) Let the number of boys and girls in the village be 500 and 300.
 According to question,
 Total number of boys and girls who appeared in the examination
 $\rightarrow (500 \times \frac{40}{100}) + (300 \times \frac{60}{100}) = 380$
 Total number of boys and girls who did not appear in the examination
 $\rightarrow 800 - 380 = 420$
 Required ratio = 380 : 420 = 19 : 21

Sol.192.(c) Volume of cylinder $= \pi r^2 h$
 Ratio $\rightarrow$ Initial : Final
 $(\text{radius})^2 \rightarrow (2)^2 : (1)^2$
 $\text{height} \rightarrow 5 : 2$
 Volume $\rightarrow 20 : 2 = 10 : 1$

Sol.193.(b) Let the numbers be $= 5x$, $9x$
 According to the question,
 $9x + 6 = 6(5x - 6)$
 $9x + 6 = 30x - 36 \Rightarrow 42 = 21x \Rightarrow x = 2$
 So the required numbers = $5 \times 2 = 10$,
 $9 \times 2 = 18$

Sol.194.(d) $\frac{J}{4} = \frac{K}{5} = \frac{L}{3}$
 $J = 4$, $K = 5$, $L = 3$
 According to the question,
 $(L + J)^2 : (J + K)^2 : (K + L)^2$
 $(3 + 4)^2 : (4 + 5)^2 : (5 + 3)^2$
 $49 : 81 : 64$

Sol.195.(d) 1st no. : 2nd no. : 3rd no.
 $3 : 4 : 4$
 $3 : 3 : 5$
 $9 : 12 : 20$

According to the question,
 $(9x \times 20x)$ unit = ₹4500
 $x^2 = \frac{450}{18} \Rightarrow x = \sqrt{\frac{225}{9}} = 5$
 Sum of three numbers $= (41x) = 41 \times 5$
 $= 205$

Sol.196.(b) Total amount = 50×80

$$\text{Number of children} = \frac{50 \times 80}{50} = 80$$

Sol.197.(c)

$$(x + y) : (y + z) : (z + x) = 3 : 5 : 7$$

$$\text{Let, } x + y = 3a, y + z = 5a, z + x = 7a$$

$$2(x + y + z) = 15a \text{ -----e.q.(1)}$$

$$45 = \frac{15a}{2} \Rightarrow a = 6$$

$$(x + y) = 3a = 18$$

$$\text{Now, } (x + y + z) = 45$$

$$18 + z = 45 \Rightarrow z = 27$$

Sol.198.(c) Let share of r be x.

$$\text{Then, share of q and p} = (x+6) \text{ and } (x+15)$$

$$\text{According to question,}$$

$$x + (x + 6) + (x + 15) = 120$$

$$3x = 99 \Rightarrow x = 33$$

$$\text{Required ratio} = 48 : 39 : 33 = 16 : 13 : 11$$

Sol.199.(c) According to the question,

$$\text{Ratio} \rightarrow \text{Income} : \text{expenditure} : \text{savings}$$

$$\text{Initial} \rightarrow 7 : 4 : 3$$

$$\text{Change} \rightarrow -1.4 : +1.4$$

$$\text{Final} \rightarrow 7 : 2.6 : 4.4$$

$$\text{His savings increased } \frac{1}{5} \text{th of their total}$$

$$\text{income and income remains same}$$

$$\text{Therefore, the new ratio of his expenditure and savings} = 2.6 : 4.4 = 13 : 22$$

Sol.200.(c) Let, weight of Ayush and Abhishek be $8x$ and $5x$ respectively.

$$\text{Increased weight of Abhishek}$$

$$= 5x \times \frac{140}{100} = 7x$$

$$\text{Overall increased weight}$$

$$\rightarrow (8x + 5x) \times \frac{160}{100} = 20.8x$$

$$\text{According to the question,}$$

$$20.8x = 104 \text{ kg.} \Rightarrow x = 5 \text{ kg.}$$

$$\text{Increased weight of Ayush}$$

$$\rightarrow 20.8x - 7x = 13.8x$$

$$13.8x = 13.8 \times 5 = 69 \text{ kg.}$$

Sol.201.(a) Let the income of Jitendra

$$\text{and Narendra} = 4x \text{ and } 7x$$

$$\text{And expenditure of Jitendra and Narendra} = 6y \text{ and } 5y$$

$$\text{According to the question,}$$

$$4x = 2800 \Rightarrow x = 700$$

$$\text{Savings of Jitendra} \Rightarrow 4x - 6y = 400$$

$$\Rightarrow y = 400$$

$$\text{Savings of Narendra} \Rightarrow 7x - 5y$$

$$= 4900 - 2000 = 2900 \text{ Rs.}$$

Sol.202.(b) Let income of P = x

$$\text{According to the question;}$$

$$\frac{1}{3}x + 4400 = \frac{1}{2}x \Rightarrow \frac{1}{6}x = 4400$$

$$x = \text{Rs. } 26,400$$

$$\frac{P}{Q} = \frac{1}{2} \Rightarrow Q = 26400 \times 2 = \text{Rs. } 52800$$

Sol.203.(d) Let the larger number be x

$$\text{and the smaller number be } y.$$

$$\text{According to question,}$$

$$x - y = \frac{20}{100} \times (x + y) \Rightarrow \frac{x - y}{x + y} = \frac{1}{5}$$

$$x - y = 1 \text{e.q.(1)}$$

$$x + y = 5 \text{e.q.(2)}$$

$$\text{On solving e.q.(1) and (2); we get}$$

$$x = 3 \text{ and } y = 2$$

$$\text{required ratio, } x : y = 3 : 2$$

Sol.204.(d) Total expenditure = 12000₹

$$\text{According to the question,}$$

$$\text{Expenditure of D} = \text{Saving of C} = x$$

$$\text{Then, Expenditure of C} = (12000 - x)$$

$$\text{Ratio} \rightarrow \text{C} : \text{D}$$

$$\text{Income} \rightarrow 3 : 4$$

$$\text{Expenditure} \rightarrow (12000 - x) : x$$

$$\text{Saving} \rightarrow x : (y)$$

$$\text{In 'C' case} \Rightarrow \text{Income} - \text{Expenditure}$$

$$= \text{Savings}$$

$$3 \text{ units} - (12000 - x) = x$$

$$\Rightarrow 1 \text{ unit} = 4000 \text{ ₹}$$

$$\text{Total Income (7 units)} = 28000 \text{ ₹}$$

$$\text{Now,}$$

$$\text{total saving} = 28000 - 12000 = 16000 \text{ ₹}$$

Short Trick :-

$$\text{Ratio} \rightarrow \text{C} : \text{D}$$

$$\text{Income} \rightarrow 3 : 4$$

$$\text{Expenditure} \rightarrow 2 : 1$$

$$\text{Saving} \rightarrow 1 : 3$$

$$\text{Total expenditure (3 units)} = 12000 \text{ ₹}$$

$$\text{Total Savings (4 units)} = 16000 \text{ ₹}$$

Sol.205.(a) Let current age of father be x and son be y

$$x + y = 94 \text{ -----e.q.(1)}$$

$$(8 - x) = 2(8 - y)$$

$$2y - x = 8 \text{ -----e.q.(2)}$$

$$\text{On adding e.q.(1) and e.q.(2); we get}$$

$$y = 34 \text{ and } x = 60$$

$$\text{Required ratio} = 70 : 44 = 35 : 22$$

Sol.206.(a) Ratio $\rightarrow$

$$P : Q + R = 4 : 7 \Rightarrow 7$$

$$Q : P + R = 2 : 5 \Rightarrow 11$$

$$\text{Total amount of P, Q, R} = 77 \text{ units}$$

$$\text{And R's unit} = 77 - (7 \times 4 + 11 \times 2)$$

$$= 27 \text{ units}$$

$$77 \text{ units} = ₹43120$$

$$(R) 27 \text{ units} = \frac{43120}{77} \times 27 \rightarrow ₹15120$$

Sol.207.(a)

$$\text{Ratio} \rightarrow A : B$$

$$\text{Income} \rightarrow 7 : 4 \Rightarrow 2$$

$$\text{Expenditure} \rightarrow 3 : 1 \Rightarrow 3$$

$$\text{Savings of A} = \{(7 \times 2) - (3 \times 3)\}$$

$$= 5 \text{ units} \Rightarrow 5 \text{ units} = 4800 \text{ ₹}$$

$$\text{Total Income of A and B}$$

$$= (7 + 4) \times 2 = 22 \text{ units}$$

$$22 \text{ units} \rightarrow \frac{4800}{5} \times 22 = 21120 \text{ ₹}$$

Sol.208.(b) Ratio of their marks $\rightarrow 2 : 4 : 9$

$$\text{According to the question,}$$

$$\text{Marks obtained by Rajesh (2 units)} = 30$$

$$\text{Marks obtained by Rakesh (4 units)} = 60$$

$$\text{Marks obtained by Ramesh (9 units)} = 135$$

Sol.209.(c)

$$\text{Ram} : \text{Ramesh} : \text{Naresh}$$

$$\frac{1}{3} : \frac{1}{4} : \frac{1}{6} \Rightarrow 4 : 3 : 2$$

$$\text{Ratio} \rightarrow \text{Ram} : \text{Ramesh} : \text{Naresh}$$

$$\text{Initial} \rightarrow 4 : 3 : 2 = 9 \text{ units} \times 13$$

$$\text{Final} \rightarrow 3 : 4 : 6 = 13 \text{ units} \times 9$$

$$\text{Make units equal as money divided is same}$$

$$\text{Ram} : \text{Ramesh} : \text{Naresh}$$

$$52 : 39 : 26 = 117 \text{ units}$$

$$27 : 36 : 54 = 117 \text{ units}$$

$$-25u \quad -3u \quad +28u$$

$$\text{According to question,}$$

$$117 \text{ units} = \text{Rs. } 702$$

$$\text{Then, } 28 \text{ units} = \text{Rs } 168$$

$$\text{Clearly, Naresh got Rs.168 More}$$

Sol.210.(d) If c is the third proportion of

$$\text{the number a and b, then } c = \frac{b^2}{a}$$

$$\text{Let, } x \text{ be the no. which is subtracted from 9 and 13 so that 18 is the third proportional to them.}$$

$$\text{According to question,}$$

$$\frac{(13 - x)(13 - x)}{(9 - x)} = 18$$

$$169 - 26x + x^2 = 162 - 18x$$

$$7 - 8x + x^2 = 0$$

$$x^2 - 7x - x + 7 = 0$$

$$\Rightarrow (x - 7)(x - 1) = 0 \Rightarrow x = 7 \text{ and } x = 1$$

$$\text{Since, smallest positive number to be subtracted. Hence, } x = 1$$

$$\text{Sol.211.(b)} A \propto \sqrt{B} \propto \frac{1}{C^3}$$

$$A = \frac{k \times \sqrt{B}}{C^3} \Rightarrow 15 = \frac{k \times \sqrt{27}}{2^3}$$

$$k = \frac{120}{3\sqrt{3}} \text{e.q.(1)}$$

$$\text{Again,}$$

$$A = \frac{k \times \sqrt{B}}{C^3} \Rightarrow 9 = \frac{120 \times \sqrt{B}}{3\sqrt{3} \times 2^3}$$

$$\sqrt{B} = \frac{9\sqrt{3} \times 8}{40} \Rightarrow B = \frac{243}{25}$$

Sol.212.(a) Mean proportion of 6 and 54

$$\Rightarrow \sqrt{6 \times 54} = 18$$

$$\text{so, } 18 \text{ is 3 more than } 15.$$

Sol.213.(d)

$$\text{Let the four no's be } x, 2x, 3x \text{ and } 5x$$

$$\text{According to the question,}$$

$$2(x + 2x + 3x + 5x) = 44$$

$$\Rightarrow 11x = 22 \Rightarrow x = \frac{22}{11} = 2$$

So, the no's are 2, 4, 6 and 10

Required ratio = $2^3 - 3 : 4^3 - 3 : 6^3 - 3 : 10^3 - 3$
 $= 5 : 61 : 213 : 997$

Sol.214.(d) LCM of (2, 5, 3) = 30

Then, the ratio of the salary of A, B and C

$$= \frac{3}{2} \times 30 : \frac{6}{5} \times 30 : \frac{4}{3} \times 30$$

$$= 45 : 36 : 40$$

$$\text{Required percentage} = \frac{45 - 40}{40} \times 100$$

$$= \frac{5}{40} \times 100 = 12.5\%$$

Sol.215.(d) Let the age of father and mother be $3x$ and $2x$ respectively

According to the question,

$$(3x) \times (2x) = 486$$

$$\Rightarrow 6x^2 = 486 \Rightarrow x^2 = \frac{486}{6} = 81$$

$$\Rightarrow x = \sqrt{81} = 9 \text{ yrs.}$$

Then, the age of father and mother

$$= 9 \times 3 = 27 \text{ yrs and } 2 \times 9 = 18 \text{ yrs.}$$

So, the ratio of the ages after 5 yrs.

$$= (27 + 5) : (18 + 5) = 32 : 23$$

Sol.216.(a)

Ratio $\rightarrow B : C : D : E$

Initial Salaries $\rightarrow 20 : 30 : 40 : 50$

Final salary $\rightarrow 24 : 39 : 56 : 75$

According to the question,

Increased salary of D = 56 unit

56 unit = ₹560

Sum of original salaries of B, C, D and E

$$= \frac{560}{56} \times 140 = ₹1,400$$

Sol.217.(c) It is given that, $CP \propto W^2$

Initial weight of a diamond = $(5 + 9)^2$

$$= 14^2 = 196 \text{ unit}$$

After break down, weight of a diamond

$$= 5^2 + 9^2 = 25 + 81 = 106 \text{ unit}$$

Required loss% in weight of diamond

$$= \frac{196 - 106}{196} \times 100$$

$$= \frac{90}{196} \times 100 = 45.918 \sim 45.92\%$$

Sol.218.(a) According to the question,

$$D = \frac{A}{2}, C = 1.2A$$

$$\text{Also, } B + D = \frac{14}{11} (A + C)$$

$$B + \frac{A}{2} = \frac{14}{11} (A + 1.2A)$$

$$B + \frac{A}{2} = \frac{14}{11} (2.2A)$$

$$B + 0.5A = 2.8A \Rightarrow B = 2.3A$$

So, ratio of sharing among A : B : C : D

$$= A : 2.3A : 1.2A : 0.5A = 1 : \frac{23}{10} : \frac{12}{10}$$

$$: \frac{5}{10} = 10 : 23 : 12 : 5$$

$$\text{Share of A} = \frac{10}{50} \times 1250 = ₹250$$

$$\text{Share of B} = \frac{23}{50} \times 1250 = ₹575$$

$$\text{Share of C} = \frac{12}{50} \times 1250 = ₹300$$

$$\text{Share of D} = \frac{5}{50} \times 1250 = ₹125$$

Hence, option (a) is the correct answer.

$$\text{Sol.219.(a)} \quad x = \sqrt{125 \times 5} = \sqrt{625} = 25$$

$$y = \sqrt{4^4 \times 6^2} = 4^2 \times 6 = 96$$

$$\text{So, } 2x : y = 25 \times 2 : 96 = 50 : 96 = 25 : 48$$

Sol.220.(d) According to question

$$\Rightarrow \sqrt{75x} = 5 \times \sqrt{15 \times 26}$$

$$\Rightarrow 5\sqrt{3x} = 5 \times \sqrt{15 \times 26}$$

$$\Rightarrow \sqrt{3x} = \sqrt{3} \times \sqrt{5 \times 26} \Rightarrow x = 5 \times 26 = 130$$

Sol.221.(a)

	No. of hens	No. of goats
Farm A	5	4 = 9) $\times 4$
Farm B	5	7 = 12) $\times 3$

No. of hens in farm A and B

$$= 5 \times 4 + 5 \times 3 = 20 + 15 = 35$$

No of goats in farm A and B

$$= 4 \times 4 + 7 \times 3 = 16 + 21 = 37$$

So, the required ratio = 35 : 37

Sol.222.(c) Let the no. of coins of ₹10, ₹5 and ₹2 be a, b, c respectively

According to question,

$$ab : bc : ca = 3 : 4 : 2$$

$$a : b : c = 3 \times 2 : 3 \times 4 : 4 \times 2$$

$$= 6 : 12 : 8 \text{ or } 3 : 6 : 4$$

So, the minimum amount of money that

$$\text{person has} = 3 \times 10 + 6 \times 5 + 4 \times 2$$

$$= 30 + 30 + 8 = ₹68$$

Sol.223.(b)

Ratio $\rightarrow$ 2-wheelers : 4-wheelers

$$11 : 9$$

$$\text{No. of petrol vehicle} = 9 \times \frac{1}{4} = 2.25 \text{ unit}$$

$$\text{No. of diesel vehicle} = 9 - 2.25 = 6.75 \text{ unit}$$

6.75 unit corresponds to 81

$$9 \text{ unit corresponds to } \frac{81}{6.75} \times 9 = 108$$

$$11 \text{ unit corresponds to } \frac{108}{9} \times 11 = 132$$

Number of 2-wheelers in the housing

$$\text{society} = 11 \text{ unit} = \frac{108}{9} \times 11 = 132$$

Sol.224.(d)

Ratio $\rightarrow$ Santosh : Ravi : Vipin

Valid Votes $\rightarrow 11 : 4 : 9$

According to question,

Winner - nearest rival = $(11 - 9)$ unit

$$= 2 \text{ unit}$$

$$2 \text{ unit} = 8350 \text{ votes}$$

$$\text{Total valid votes polled} = \frac{8350}{2} \times 24 = 1,00,200$$

Sol.225.(c)

$$A : B : C$$

$$\frac{8}{16} : \frac{4}{8} : \frac{3}{6} (\times 2)$$

$$16 : 8 : 6$$

A divides his parts

$$\begin{array}{c} A=(16) \\ \swarrow \quad \searrow \\ D=9 \quad E=7 \end{array} \quad \begin{array}{l} A=D+E \\ 16 \text{ unit} = 16 \text{ unit} \end{array}$$

Therefore,

$$C : D = 6 : 9 = 2 : 3$$

Sol.226.(b) let the first number = a

And second number = b

According to the question,

$$3b - a = \frac{3a}{4} \Rightarrow 3b = \frac{3a}{4} + a$$

$$\Rightarrow 3b = \frac{7a}{4} \Rightarrow \frac{a}{b} = \frac{12}{7}$$

$$\text{therefore ratio} \Rightarrow a : b = 12 : 7$$

Sol.227.(d)

Denomination of coin : ₹5 : ₹10 : ₹20

Ratio of coin $\rightarrow 2x : 3x : 5x$

$$\text{Value} \rightarrow 10x : 30x : 100x$$

$$10x = 240 \Rightarrow x = 24$$

Now total amount of money in the bag

$$= 10x + 30x + 100x$$

$$= 140x = 140 \times 24 = \text{Rs } 3360$$

Sol.228.(a)

Let x be added to each numbers

$$\Rightarrow (1 + x) : (7 + x) :: (7 + x) : (25 + x)$$

$$\Rightarrow (1 + x)(25 + x) = (7 + x)(7 + x)$$

$$\Rightarrow 25 + 26x = 49 + 14x$$

$$\Rightarrow 12x = 24 \Rightarrow x = 2$$

Sol.229.(d)

Let the cat be x and dogs be 2x.

$$\Rightarrow (4 \times x) + (4 \times 2x) + 3 \times 2 = 42$$

$$\Rightarrow 12x = 36 \Rightarrow x = 3$$

Total number of cats = 3

Sol.230.(b) Let the numerator be x and

$$\text{denominator be } y \Rightarrow \frac{x+8}{y+12} = \frac{1}{2}$$

$$\Rightarrow 2x - y = -4 \dots\dots\dots(i)$$

$$\Rightarrow \frac{x-2}{y-2} = \frac{1}{8}$$

$$8x - y = 14 \dots\dots\dots(ii)$$

On solving eq (i) and (ii), we get

$$x = 3, y = 10$$

$$\text{Original fraction} = \frac{3}{10}$$

$$\text{Sol.231.(c)} \quad \frac{a}{b} = \frac{b}{c} \Rightarrow ac = b^2$$

$$\text{Now, } a^6 : b^6 = a^6 : (b^2)^3$$

$$\Rightarrow a^6 : b^6 = a^6 : (ac)^3$$

$$\Rightarrow a^6 : b^6 = a^6 : a^3 c^3$$

$$\Rightarrow a^6 : b^6 = a^3 : c^3$$

Sol.232.(a)

Let the ₹5 coin be x and ₹10 coins be y

$$xy = 135 \text{ and } 5x + 10y = 165$$

$$\text{So, } 5\left(\frac{135}{y}\right) + 10y = 165$$

$$2y^2 - 33y + 135 = 0$$

$$y = 9 \text{ or } 7.5 \text{ (number of coins cannot be)}$$

in decimal)

So, amount = $10 \times 9 = ₹90$

Sol.233.(a) Total students = 60

B : G
4 : 1

$\Rightarrow 4 + 1 = 5 \rightarrow 60$ (given)

So number of boys = $4 \rightarrow \frac{60}{5} \times 4 = 48$

Number of non-vegetarian = $48 - 22 = 26$

Sol.234.(c) According to the question,
Ratio of steel (2018 : 2020) = $4 : 3 = 4 : 3$

	Steel	iron
2018	$\rightarrow (3 : 2) \times 4$	
2020	$\rightarrow (1 : 1) \times 3 \times 3$	

Here,

	Steel	iron
2018	$\rightarrow 12 : 8$	
2020	$\rightarrow 9 : 9$	

Therefore ratio of iron = 2018 : 2020
 $8 : 9$

Sol.235.(c)

Let the parrots be $7x$ and pigeon be $9x$.

According to the question,

$$\frac{7x-8}{9x+6} = \frac{1}{4} \Rightarrow 28x - 32 = 9x + 6$$

$$\Rightarrow 19x = 38 \Rightarrow x = 2$$

Initial Numbers of parrots = $7 \times 2 = 14$

Remaining parrots = $14 - 8 = 6$

Therefore ratio of parrots and sparrows

$$\Rightarrow 6 : 10 = 3 : 5$$

Sol.236.(d)

Let the age of Yuvansh be x year.

Ratio $\rightarrow$ Garvit : Yuvansh

4 years ago $\rightarrow 3x : x$

1 year ago $\rightarrow 3x + 3 : x + 3$

Present age $\rightarrow 3x + 4 : x + 4$

According to question,

$$3x + 3 = 2 \times (x + 3) \Rightarrow x = 3$$

Difference in their age =

$$(3x + 4) - (x + 4) = 13 - 7 = 6 \text{ years.}$$

Sol.237.(a) Let the number of boys be $2x$ and girls be $5x$

$$\text{Now, } \frac{2x-10}{5x+20} = \frac{4}{25} \Rightarrow 50x - 20x = 330$$

$$\Rightarrow 30x = 330 \Rightarrow x = 11$$

Number of boys now = $11 \times 2 = 22$

$$= 22 - 10 = 12$$

Sol.238.(a) Let Karan gets = x

Raj gets = $x - 9000$

As per question,

$$\Rightarrow x + x - 9000 = 21000 \Rightarrow x = 15,000$$

Ratio of Raj : Karan = $6000 : 15000 = 2 : 5$

Sol.239.(c) 4 years ago, ratio of age of Shyam : Ram = $3 : 4$

4 years hence, ratio of age of

Shyam : Ram = $5 : 6$

According to question,

$$\Rightarrow \frac{3x+8}{4x+8} = \frac{5}{6} \Rightarrow 18x + 48 = 20x + 40$$

$$2x = 8 \Rightarrow x = 4$$

Present age of Shyam = $3x + 4 = 3 \times 4 + 4 = 16$ years

Sol.240.(b) Let the expenditure and savings of Soham be 500 and 100 units

Income = Expenditure + Saving

Original $\rightarrow 600 = 500 + 100$

New $\rightarrow 750 = 630 + 120$

According to question,

Current expenditure (630 units) $\rightarrow ₹4347$

So, his Initial income (600 units) $\rightarrow$

$$\frac{4347}{630} \times 600 = ₹4140$$

Sol.241.(b) According to question,

$$10\% \text{ of } A + 8\% \text{ of } B = \frac{3}{5} \text{ (12\% of } A + 16\% \text{ of } B)$$

$$50\% \text{ of } A + 40\% \text{ of } B = 36\% \text{ of } A + 48\% \text{ of } B$$

$$14\% \times A = 8\% \times B \Rightarrow \frac{A}{B} = \frac{4}{7}$$

$$\text{Sol.242.(c)} (18x + 8y)\% = \frac{1}{3} (22x + 36y)\%$$

$$54x + 24y = 22x + 36y$$

$$32x = 12y \Rightarrow \frac{x}{y} = \frac{12}{32} = \frac{3}{8}$$

So, the ratio of x and $y = 3 : 8$

Sol.243.(d) Let the number be x and y .

According to question,

$$60\% \times x = \frac{3}{4} \times y$$

$$\frac{3}{5} \times x = \frac{3}{4} \times y \Rightarrow \frac{x}{y} = \frac{5}{4}$$

So, the required ratio = $5 : 4$

Sol.244.(c) Ratio $\rightarrow P : Q : R$

$$16 : 17 \quad) \times 15$$

$$15 : 16 \quad) \times 17$$

$$P : Q : R = 240 : 255 : 272$$

Total run scored by all batsmen =

$$(240 + 255 + 272) \text{ units} = 767 \text{ units}$$

767 units $\rightarrow$ 956 Run

Run scored by R (272 units)

$$= \frac{956}{767} \times 272 = \frac{260032}{767} \approx 339 \text{ Run}$$

Sol.245.(b) According to the question,

A varies with B and C

$$A = B \times C \Rightarrow 6 = 3 \times 2$$

$$\text{So, } A = 5 \times 7 = 35$$

Sol.246.(c) Ratio $\rightarrow A : B$

Income $\rightarrow 11 : 13$

Expenses $\rightarrow 9 : 11$

Savings $\rightarrow 2 : 2$

According to question,

Saving (2 units) = 4000 Rs.

Difference b/w their income $(13 - 11)$ units

$\rightarrow 2$ units = 4000 Rs.

Sol.247.(b) Let the present ages of A and B be a and b years respectively.

According to the question,

$$\frac{a-4}{b-4} = \frac{1}{2}$$

$$2a - 8 = b - 4 \Rightarrow 2a - b = 4 \quad \dots (1)$$

$$\text{Also, } a = b - 8 \Rightarrow a - b = -8 \quad \dots (2)$$

On solving equations (1) and (2), we get

$$a = 12 \text{ and } b = 20$$

So present age of B = 20 years

Sol.248.(c) Given,

$$(a^2 + b^2) : (b^2 + c^2) : (c^2 + a^2) = 34 : 61 : 45$$

Adding all we get, $2(a^2 + b^2 + c^2) = 140$

$$(a^2 + b^2 + c^2) = 70.$$

$$c^2 = (a^2 + b^2 + c^2) - (a^2 + b^2) = 70 - 34 = 36.$$

$$\text{Similarly, } b^2 = 25 \text{ and } a^2 = 9$$

$$\text{So, } a : b : c = \sqrt{9} : \sqrt{25} : \sqrt{36} = 3 : 5 : 6$$

$$b - a : c - b : c - a$$

$$= (5 - 3) : (6 - 5) : (6 - 3) \Rightarrow 2 : 1 : 3$$

Sol.249.(b) A : B

Income $2x : 3x$

Expenditure $3y : 4y$

Savings $(2x - 3y) : (3x - 4y)$

According to the question,

$$\frac{1}{3} \times 3x = 3y \Rightarrow x : y = 3 : 1$$

Savings ratio = $(2x - 3y) : (3x - 4y)$

$$\Rightarrow (2 \times 3 - 3 \times 1) : (3 \times 3 - 4 \times 1) = 3 : 5$$

Short Trick :-

Ratio $\rightarrow A : B$

Income $\rightarrow 2 : 3 \times 3$

Expenses $\rightarrow 3 : 4 \times 1$

$$\text{Savings of A} = (2 \times 3) - (3 \times 1) = 3$$

$$\text{Savings of B} = (3 \times 3) - (4 \times 1) = 5$$

$$\text{Required Ratio} \rightarrow 3 : 5$$

Sol.250.(d)

Ratio $\rightarrow x : y = \text{Total}$

$$\text{Alloy A} \rightarrow 5 : 2 = 7 \times 4$$

$$\text{Alloy B} \rightarrow 3 : 4 = 7 \times 5$$

Quantity of metal 'x' in Alloy C $\rightarrow$

$$(5 \times 4) + (3 \times 5) = 35 \text{ units}$$

Total quantity of Alloy C $\rightarrow$

$$(7 \times 4) + (7 \times 5) = 63 \text{ units}$$

$$\text{Required percentage} = \frac{35}{63} \times 100 = 55\frac{5}{9}\%$$

Sol.251.(a)

Ratio $\rightarrow$ Juice : water = total

$$\text{Cup 1} \rightarrow 5 : 2 = 7 \times 11$$

$$\text{Cup 2} \rightarrow 7 : 4 = 11 \times 7$$

Cup 3 $\Rightarrow$ Juice : Water

$$(5 \times 11) + (7 \times 7) : (2 \times 11) + (4 \times 7)$$

$$104 : 50 = 52 : 25$$

$$\text{Ratio} \rightarrow \text{water} : \text{juice} = 25 : 52$$

Sol.252.(a) Ratio $\rightarrow$ Last year Income

$$A : B = 4 : 3$$

Now,

Last Year : Present Year

$$A \rightarrow \times 4 \times 5 \quad (3 : 4)$$

$$B \rightarrow \times 3 \times 3 \quad (5 : 6)$$

Last Year : Present Year

$$A \rightarrow 60 : 80$$

$$B \rightarrow 45 : 54$$

According to the question,

$$(80 + 54) \text{ units} = 24.12 \text{ lakh}$$

$$1 \text{ unit} = 0.18 \text{ lakh}$$

Last year income of A + Present year

$$\text{income of B} = (60 + 54) \text{ units} = 114 \text{ units}$$

$$114 \text{ units} = 114 \times 0.18 = 20.52 \text{ Lakh}$$

$$\text{Sol.253.(a)} \quad \frac{z}{x} = \frac{5}{2} \Rightarrow z = \frac{5x}{2}$$

According to the question,

$$\frac{5}{2} = y + \frac{7}{4} \Rightarrow y = \frac{5}{2} - \frac{7}{4} = \frac{3}{4}$$

$$\text{Also, } x + y + z = 1 \frac{11}{12}$$

Putting the values of z and y as obtained in above equation, we get

$$x + \frac{3}{4} + \frac{5x}{2} = \frac{23}{12}$$

$$\frac{4x + 3 + 10x}{4} = \frac{23}{12} \Rightarrow 14x + 3 = \frac{23}{3}$$

$$42x = 23 - 9 \Rightarrow x = \frac{1}{3}$$

$$z = \frac{5 \times \frac{1}{3}}{2} = \frac{5}{6}, \text{ So, } (z - x) : (y - x)$$

$$= \left(\frac{5}{6} - \frac{1}{3}\right) : \left(\frac{3}{4} - \frac{1}{3}\right)$$

$$= \frac{3}{6} : \frac{5}{12} = 6 : 5$$

Sol.254.(b) Let, initial passed students be $25x$ and initial failed be $4x$.

According to the question,

$$\frac{25x + 1 + 3}{4x - 3} = \frac{22}{3}$$

$$75x + 12 = 88x - 66$$

$$13x = 78 \Rightarrow x = 6$$

$$\text{Initial passed} - \text{initial failed} = 25x - 4x$$

$$21x = 21 \times 6 = 126$$

Sol.255.(d) Let the total sum = ₹ x

Old ratio of share $\rightarrow 2 : 3 : 4$

New ratio of share $\rightarrow 6 : 4 : 3$

According to question,

$$\frac{6x}{13} - \frac{2x}{9} = 3500 \text{ Rs.}$$

$$\frac{28x}{117} = 3500 \Rightarrow x = 14,625 \text{ Rs.}$$

Sol.256.(b) Let number of Mathematics books = M, number of Physics books = P and number of Chemistry books = C

According to question,

$$\frac{M}{P} = \frac{P}{C} \Rightarrow P^2 = MC = 144 \times 100 = 14400$$

$$\text{So, } P = 120$$

$$\text{So, required ratio} \rightarrow (M + P) : (C + P)$$

$$= (144 + 120) : (100 + 120)$$

$$= 264 : 220 = 6 : 5$$

Sol.257.(c) Let x males and x females are added in the recruitment drive.

According to the question,

$$\frac{70 + x}{100 + x} = \frac{3}{4}$$

$$280 + 4x = 300 + 3x \Rightarrow x = 20$$

$$\text{Total males now} = 70 + 20 = 90,$$

$$\text{Total females now} = 100 + 20 = 120$$

$$\text{Total working population} = 90 + 120 = 210$$

Sol.258.(c)

Ratio $\rightarrow$ 1st year : 2nd year

$$\text{Income} \rightarrow 2x : 5x$$

$$\text{Expenses} \rightarrow 4y : 7y$$

According to question,

$$5x = 75000 \Rightarrow x = 15000$$

$$4y = 20000 \Rightarrow y = 5000$$

Savings of both years combined

$$= (2x - 4y) + (5x - 7y)$$

$$= (2 \times 15000 - 4 \times 5000) + (5 \times 15000 - 7 \times 5000) = 10000 + 40000 = 50000 \text{ Rs.}$$

Sol.259.(c) Let the shares of A, B and C be $5x + 130$, $2x + 150$ and $4x + 200$ respectively.

According to the question,

$$5x + 130 + 2x + 150 + 4x + 200 = 3780$$

$$11x + 480 = 3780 \Rightarrow 11x = 3300 \Rightarrow x = 300$$

$$\text{Share of C } (4x + 200) \rightarrow 4 \times 300 + 200 = 1400 \text{ Rs.}$$

Sol.260.(d)

$$\text{Marbles in Box A} : \frac{3}{15} \times 75 = 15,$$

$$\text{Marbles in Box B} : \frac{5}{15} \times 75 = 25,$$

$$\text{Marbles in Box C} : \frac{7}{15} \times 75 = 35$$

On transferring the marbles as per directions given in question, we get

$$\text{Total marbles in Box A} = 15 + 3 = 18$$

$$\text{Total marbles in Box B} = 25 - 3 + 5 = 27$$

$$\text{Total marbles in Box C} = 35 - 5 = 30$$

$$\text{So, required ratio} = 18 : 27 : 30 = 6 : 9 : 10$$

Sol.261.(b)

$$\text{Let } (a + b) = 7x, (b + c) = 4x, (c + a) = 5x$$

So, according to question

$$7x + 4x + 5x = 2(a + b + c)$$

$$7x + 4x + 5x = 2(16)$$

$$16x = 32 \Rightarrow x = 2$$

$$\text{So, } a + b = 7x = 14,$$

$$b + c = 4x = 8 \text{ and } c + a = 5x = 10$$

$$\text{So, } a = 8, b = 6, c = 2 \text{ Now,}$$

$$(a^2 + b^2 + c^2) : (ab + bc + ca)$$

$$(8^2 + 6^2 + 2^2) : (8 \times 6 + 6 \times 2 + 8 \times 2)$$

$$= (64 + 36 + 4) : (48 + 12 + 16)$$

$$= (104) : (76) = 26 : 19$$

Sol.262.(b)

$$\text{Ratio} \rightarrow \text{Maths} : \text{English} : \text{Science}$$

$$3 : 2 : 1 \times 2$$

$$2 : 3 : 1 \times 3$$

$$6 : 4 : 9$$

$$\text{Total marks } (6 + 4 + 9) \text{ units} = 190$$

$$\text{Marks in Science } (9 \text{ units}) = 90$$

Sol.263.(d) Let the number of chemistry books = x and let the price of each physics book and each chemistry book be 1 and 2 respectively.

So, total correct amount

$$= 10 \times 1 + x \times 2 = 10 + 2x$$

Total incorrect amount

$$= 10 \times 2 + x \times 1 = 20 + x$$

According to the question,

$$12.5 = \frac{(10 + 2x) - (20 + x)}{10 + 2x} \times 100$$

$$\frac{1}{8} = \frac{(10 + 2x) - (20 + x)}{10 + 2x}$$

$$\frac{1}{8} = \frac{x - 10}{10 + 2x} \Rightarrow 10 + 2x = 8x - 80$$

$$6x = 90 \Rightarrow x = 15$$

$$\text{So, required ratio} = 10 : x = 10 : 15 = 2 : 3$$

Sol.264.(b)

$$\text{Income} = \text{Expenditure} + \text{Savings}$$

$$17 = 14 + 3$$

According to question,

$$\text{Saving } (3 \text{ units}) = 12000 \text{ Rs.}$$

$$\Rightarrow 1 \text{ unit} = 4000$$

$$\text{Income } (17 \text{ units}) = 68000 \text{ Rs.}$$

$$\text{Expenditure } (14 \text{ units}) = 56000 \text{ Rs.}$$

Now,

$$\text{New income} = 68000 + 10000 = ₹78,000$$

$$\text{New expenditure} = 56000 - 2000$$

$$= ₹54,000$$

$$\text{Required ratio} \rightarrow 78000 : 54000 = 13 : 9$$

Sol.265.(d)

$$\text{Ratio} \rightarrow A : B : C : D$$

$$3 : 1$$

$$5 : 2$$

$$2 : 1$$

$$\text{Final} \rightarrow 30 : 10 : 4 : 2$$

$$\text{Difference between B and D} = (10 - 2) \text{ units}$$

$$8 \text{ units} = ₹1600 \rightarrow 1 \text{ unit} = 200 \text{ Rs.}$$

$$\text{Total sum } (30 + 10 + 4 + 2) =$$

$$46 \text{ units} = 46 \times 200 = 9200 \text{ Rs.}$$

Sol.266.(c) Here we need to check through options, and by doing so we can see that option (c) satisfies the conditions given in question.

$$\text{Ratio of marbles} \rightarrow A : B : C = 2 : 3 : 4$$

$$\text{Total marbles } (9 \text{ units}) = 360$$

$$1 \text{ unit} = 40$$

$$\text{Box A} = 80, \text{ Box B} = 120, \text{ Box C} = 160$$

So on applying option (c), we get

$$(80 - 20) : (120 + 20 + 20) : (160 - 20)$$

$$\Rightarrow 60 : 160 : 140 = 3 : 8 : 7$$

Sol.267.(c)

If we observe the given ratio we get

